

TRACKCSPC: A SMART CAMPUS VEHICLE TRACKING AND RESERVATION PLATFORM

A Capstone Project
presented to the Faculty of
College of Computer Studies
Camarines Sur Polytechnic Colleges

In Partial Fulfillment of the Requirements
for the degree Bachelor of Science in Information Technology

By

Raxine D. Beriña
Debie S. Layron
Ronan Gabriel Paat

December, 2025

DEDICATION

This research study is lovingly dedicated to our Almighty God, for the wisdom, strength, and guidance that made this work possible.

We also dedicate this work to our parents and families, whose endless love, sacrifices, encouragement, and support inspired us to persevere throughout this research journey.

To our instructors and advisers, thank you for your patience, guidance, and valuable knowledge that helped shape this study.

To our friends and classmates, thank you for the motivation, cooperation, and shared experiences that made this research meaningful.

Above all, this work is dedicated to everyone who believes in the value of education and continuous learning. May this study serve as a source of knowledge and inspiration for future researchers.

ACKNOWLEDGMENTS

The researchers would like to express their sincere and profound gratitude to all individuals who contributed to the successful completion of this research study.

Foremost, the researchers extended their deepest appreciation to **Rosel O. Onesa, MIT**, the Capstone Adviser, for her invaluable guidance, expertise, patience, and continuous support throughout the conduct of this study.

They also expressed their gratitude to **Dr. Jocelle B. Monreal**, their Capstone Subject Adviser, for her endless consideration, encouragement, and consistent support throughout the entire capstone project journey.

The researchers were also grateful to **Vincent B. Cortez, MIT**, the Consultant, for his professional insights, constructive suggestions, and technical guidance, which greatly helped in improving the quality of this research.

Special acknowledgment was given to **Ms. Ella Mae S. Alamo**, the Grammarian, for her time, dedication, and expertise in reviewing and refining the language of this manuscript, ensuring clarity, coherence, and correctness.

Sincere appreciation was likewise extended to the Panel Members, **Ian P. Benitez, DIT**; **Philip Alger M. Serrano, MIT**; and **Maricris L. Ramizares, MIT**, for their valuable time, insightful comments, and constructive recommendations, which significantly contributed to the refinement and improvement of this study.

The researchers likewise acknowledged their instructors, fellow students, and classmates for their cooperation, encouragement, and moral support throughout the research process. Finally, heartfelt gratitude was extended to the researchers' families for their unwavering support, understanding, and motivation throughout this academic endeavor.

ABSTRACT

Title:	TRACKCSPC: A Smart Campus Vehicle Tracking and Reservation Platform
Authors:	Raxine D. Beriña Debie S. Layron Ronan Gabriel V. Paat
Number of Pages:	197
School:	Camarines Sur Polytechnic Colleges
Degree Conferred:	Bachelor of Science in Information Technology
Keywords:	GPS Tracking, Vehicle Reservation System, Campus Vehicle Tracking

This study presents TrackCSPC: A smart campus vehicle tracking and reservations platform developed for the Camarines Sur Polytechnic Colleges–General Services Unit (GSU) to address inefficiencies in manual vehicle reservation processes, including delays, inconsistent documentation, and limited monitoring. The system integrates a mobile application for submitting vehicle requests and tracking reservations in real time with a web-based administrative dashboard for approval, driver assignment, and GPS-based vehicle monitoring. A mixed-method research design was employed using surveys, interviews, and observations, while system evaluation involved functional testing, load testing, and User Acceptance Testing (UAT). Evaluation results yielded an overall mean score of 1.00, interpreted as Completely Functional, indicating high reliability, usability, and operational efficiency. The findings demonstrate that TrackCSPC improves workflow management, communication, and centralized vehicle monitoring, with only minor refinements identified for specific user groups. The study concludes that TrackCSPC effectively enhances campus transportation management through automated reservation workflows and real-time tracking and recommends continuous system maintenance, user training, and future enhancements such as AI-based scheduling and online payment integration. The study employed a mixed-method research design and evaluated the system through surveys, interviews, and direct observations. System performance was assessed using functional testing, load testing, and User Acceptance Testing (UAT). The evaluation results yielded an overall mean score of 1.00, interpreted as Completely Functional, demonstrating the system’s high reliability, usability, and operational efficiency.

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CHAPTER 1

INTRODUCTION

This chapter provides the project context, purpose, objectives, significance of the study, scope and limitations of the study, and the project dictionary of the capstone project TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform.

1.1 Project Context

In today's digital era, technology is rapidly transforming various industries, including transportation [12]. Many institutions and universities worldwide integrated smart systems and GPS tracking technologies to enhance vehicle management, improve efficiency, and ensure safety. These advancements contributed to the reduction of delays, optimized routes, and provided real-time updates on vehicle locations, which benefited both passengers and administrators.

The implementation of GPS-based vehicle tracking and automated reservation systems became crucial for educational institutions. Universities with large campuses relied on school buses to transport students, faculty, and staff. However, the absence of an efficient system led to challenges, such as delayed bus arrivals, inefficient reservations, and safety concerns. In the Philippines, many universities still use manual, paper-based reservation systems, resulting in mismanagement, scheduling conflicts, and a lack of real-time tracking. This outdated approach created uncertainty for students and faculty, affecting bus availability and scheduling efficiency [14]. At Camarines Sur Polytechnic Colleges (CSPC), transportation management remained a significant challenge. The lack of an automated reservation system affected students, particularly nursing students with early hospital duties, participants in inter-school competitions, tourism tours, and other academic activities required CSPC buses. Without a digital system, confirming reservations often resulted in schedule conflicts, miscommunication, and delays.

Mohammad A. Al-Khedher [2011], developed a GPS-GSM-based vehicle tracking system that utilized Google Earth for real-time location monitoring [13]. The system installed a GPS

module on the vehicle to capture its current position, which is then transmitted via GSM as SMS along with data from the car's onboard system. A Kalman filter improved the accuracy of the GPS data before it was displayed on Google Earth. This study supported TrackCSPC by showcasing the effectiveness of GPS-GSM integration for accurate, real-time vehicle tracking and fleet management.

Shamsul Kamal Ahmad Khalid, Nurul Syahadah Mohd Salleh, & Noor Azah Samsudin, [2017]. examined the impact of real-time GPS tracking on reducing travel uncertainties in their study. They introduced the University Bus Arrival Information System (UBAIS), which leveraged mobile internet and consumer-grade GPS to provide continuous location updates. Implemented at Universiti Tun Hussein Onn Malaysia (UTHM), the system demonstrated high accuracy and increased passenger satisfaction by enabling real-time bus tracking via mobile devices [8]. This study served as a relevant reference for the development of TrackCSPC, as both systems aimed to enhance transportation efficiency through GPS tracking of vehicles and automated updates, ensuring optimized vehicle usage and improved user experience.

To address these issues, the present study proposed TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform. This web-based application will integrate automated bus reservations, real-time GPS tracking, and a user-friendly interface for students, faculty, and administrators. The system will also enable parents to monitor their children's travel status for enhanced security. By digitizing the reservation process, CSPC eliminated inefficiencies, reduced scheduling conflicts, and improved overall transportation management.

1.2 Purpose and Description of the Project

TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform aimed to design and develop a mobile and web-based application for the General Services Unit of the Camarines Sur Polytechnic Colleges (CSPC). This project aimed to minimize inefficiencies associated with the current manual vehicle reservation system, which frequently resulted in prolonged waiting

times, scheduling conflicts, and sub optimal vehicle allocation. The platform was designed with the primary objective of streamlining campus transportation operations by utilizing technological advancements to automate processes, improve user accessibility, and optimize resource management. Through the integration of real-time GPS tracking, cloud-based booking, and administrator scheduling, TrackCSPC introduced an innovative solution that facilitates automated vehicle reservations, thereby promoting efficiency, accessibility, and user convenience.

The prevailing manual reservation process often led to operational inefficiencies, causing delays and mismanagement of transportation services. Consequently, faculty, staff, and students experienced last-minute cancellations, overlapping schedules, and a lack of real-time visibility into vehicle availability. To address these concerns, TrackCSPC provided a centralized, automated, and digital platform for efficient vehicle reservations and monitoring. Its user-friendly interface allowed staff and administrators to secure available vehicles, track real-time locations, and receive instant notifications regarding and fostered a structured and efficient approach to campus transportation.

One of the system's primary advantages was its potential to improve time efficiency, reduce administrative errors, and provide greater user convenience. TrackCSPC automated the reservation process and provided real-time availability information, eliminating the need for manual inquiries and paper-based requests. As a result, staff members no longer had to deal with double bookings or ineffective reservation tracking. Furthermore, the platform enabled equitable and optimal vehicle deployment, reducing resource underutilization, scheduling conflicts, and inefficiencies in transportation management.

Real-time notifications and reminders kept users informed about their reservations, significantly reducing last-minute disruptions and scheduling inconsistencies. This system introduced a proactive approach to campus transportation management, offering an enhanced level of service reliability and operational efficiency.

Research underscored the need for an efficient transportation management system. Hsu,

Lu, and Chen [2022] highlighted how automation improved scheduling accuracy, reduced administrative workload, and enhanced user satisfaction [9]. Similarly, Zhang et al. [2018], demonstrated how real-time data, such as Wi-Fi signal strength, could optimize vehicle tracking and reduce waiting times [10]. Inspired by these advancements, TrackCSPC integrated similar technologies to improve campus transportation. Seniwoliba and Awuni [2023] emphasized the financial dangers associated with poorly managed vehicle fleets on campuses. They pointed out that issues like frequent vehicle malfunctions, delays in maintenance, and neglected repairs could significantly increase expenses and lead to the inefficient use of an institution's financial resources [1]. TrackCSPC addressed this by promoting accountability, optimizing vehicle usage, and minimizing inefficiencies, ensuring a transparent and effective transportation framework.

Through the adoption of advanced technological solutions, TrackCSPC aspired to revolutionize campus transportation operations, alleviate administrative challenges, and cultivate a more structured and efficient system. This initiative was projected to significantly enhance transportation reliability, accessibility, and overall efficiency, fostering seamless experience for CSPC faculty, staff, and administrators. The long-term advantages of TrackCSPC encompassed cost reduction, service quality enhancement, and improved user satisfaction, ultimately contributing to a sustainable and well-regulated transportation infrastructure for CSPC.

1.3 Objectives of the Project

This study aims to develop TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform, a mobile and web-based system designed to optimize campus transportation at Camarines Sur Polytechnic Colleges (CSPC). The platform will integrate real-time GPS tracking and automated reservations, to enhance efficiency, accessibility, and security in campus vehicle management. Ultimately, the system aims to make campus transportation more convenient and reliable for students, staff, and drivers by simplifying daily transport coordination and reducing unnecessary delays.

To achieve this, the study will:

1. Analyze the current vehicle reservation and tracking system at CSPC using interviews and questionnaires.
2. Develop a mobile and web-based application that provides real-time GPS tracking based on the challenges encountered.
3. Evaluate the system's usability and performance by conducting functional testing, user acceptance testing (UAT), and load testing through survey questionnaires and interviews.
4. Deploy the TrackCSPC mobile application and web-based platform within Camarines Sur Polytechnic Colleges (CSPC).

1.4 Significance of the Study

The study was significant to various stakeholders as it aimed to enhance transportation efficiency, security, and accessibility within the campus.

Camarines Sur Polytechnic Colleges (CSPC). This research project assisted CSPC in managing campus transportation more efficiently through the shift from manual vehicle management to a web-based system.

Students, Faculty and Staff. These stakeholders benefited greatly from the system, as it provided an easy and convenient way to locate and reserve campus vehicles in real-time. This reduced waiting times, eliminated scheduling conflicts, and ensured a smoother transportation experience within CSPC. With TrackCSPC, users were able to check vehicle availability and secure reservations without unnecessary delays.

CSPC's General Services Unit. Responsible for managing campus vehicles experienced greater efficiency in their work with the computerized system. They had real-time access to vehicle status, managed reservations more effectively, and allocated vehicles properly, reducing cases of misuse and delays. This led to better productivity and improved vehicle utilization.

CSPC Administrators. Administrators were equipped with a more efficient tool to oversee vehicle usage, monitor transportation schedules, and coordinate vehicle assignments. Through the TrackCSPC platform, they tracked vehicles in real-time, enhanced decision-making, and ensured that the transportation system ran smoothly and fairly for all users.

IT and Software Developers. This study contributed to the knowledge base of system developers and IT professionals interested in designing or improving vehicle tracking and reservation systems. Developers leveraged this research to understand the technical requirements, user needs, and challenges of implementing a smart transportation platform in an academic setting.

Future Researchers. Future studies on smart transportation systems within educational institutions were able to use this research as a valuable reference. It provides insights into the application of technology in managing campus vehicle services and can serve as a guide for similar projects in other schools or universities looking to improve their transportation management.

Researchers. As researchers, we aimed for this study to support further exploration into smart transportation systems in educational settings. Our work demonstrated the potential of technology to enhance campus vehicle services and offered insights that can inform both academic research and practical implementation. We invite future researchers to expand on our findings, explore new technological solutions, and address areas beyond the scope of our study.

1.5 Scope and Limitations of the Project

This study focused on developing TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform, a mobile and web-based system intended to optimize campus transportation at Camarines Sur Polytechnic Colleges (CSPC). The scope of the project included analyzing the existing manual reservation and tracking process to identify inefficiencies such as delays, miscommunication, and scheduling conflicts, which served as the basis for designing a more streamlined solution. To address these challenges, the system integrated real-time GPS tracking and an interactive map that allowed authorized users to monitor vehicle locations, while a mobile

application supported on-the-go access and a web-based platform facilitated administrative functions. The system also incorporated a reservation module for administrators and department heads, enabling them to request and manage vehicles for institution-related activities, supported by notifications and alerts to keep users informed of reservation updates. Functional testing, Load Testing, and User Acceptance Testing (UAT) were conducted to evaluate the system's usability, reliability, and operational performance, after which the platform was deployed in a controlled environment within CSPC to validate its readiness for broader institutional adoption.

However, this study has certain limitations. The system shall track designated campus vehicles on or off the premises of Camarines Sur Polytechnic Colleges such as GPS tracking functionality is dependent on network connectivity and device compatibility, which may affect real-time updates. The reservation feature is strictly limited to administrators and department heads, ensuring controlled access and proper vehicle management. Moreover, access to the application will be restricted to registered student, staff, and parent accounts, ensuring that only verified users can utilize its features.

1.6 Project Dictionary

The following technical terms were operationally and conceptually defined for clearer understanding in the context of TrackCSPC: A Smart Campus Vehicle Tracking and Reservation.

Automated Scheduling. A system function that optimizes vehicle allocation to reduce scheduling conflicts and ensure efficient vehicle usage [10]. In this research, automated scheduling refers to the process of using digital systems to optimize resource allocation, reduce scheduling conflicts, and enhance operational efficiency.

Load Testing. Load testing determines the behavior of the application when multiple users use it at the same time. It is the response of the system measured under varying load conditions.[19]. In this research, load testing will assess the system's performance by determining how the TrackCSPC platform responds to increasing levels of user activity.

Functional Testing. A software testing process that verifies whether each application feature behaves according to specified functional requirements, ensuring correct input/output and user workflow behavior [15]. In this research, functional testing refers to the process of verifying TrackCSPC's features such as login, vehicle tracking, and reservation processing to ensure each function performs correctly.

GPS Tracking. A system that uses Global Positioning System technology to provide real-time location updates for campus vehicles [7]. In this research, GPS tracking is a system that utilizes Global Positioning System (GPS) technology to determine and transmit the real-time location of objects or individuals.

Mobile Application. The user-friendly app is designed for students, staff, and administrators to monitor campus vehicles and check reservation availability [12]. In this research, a mobile application is a software program designed to run on smartphones, tablets, or other mobile devices, providing various functionalities.

Notification System. Alerts and reminders sent to users regarding their vehicle reservations and estimated arrival times [9]. In this research, a notification system is digital mechanism that delivers automated alerts or messages to users regarding specific updates or reminders.

Real-Time Tracking. The capability of the system to display live updates on vehicle locations and movements [8]. In this research, real-time tracking refers to the ability of a system to continuously monitor and display live updates on the location and movement of an entity.

Reservation Platform. Cloud or web-based software enabling users to book and schedule assets (e.g., rooms, vehicles) online, often with real-time availability and optional billing features [17]. In this research, the reservation platform refers to the web-based system that allows authorized users at CSPC to reserve campus vehicles for academic and official use. The system also helps streamline vehicle allocation, reduce scheduling conflicts, and improve overall resource.

Reservation System. A digital feature that allows authorized users to book campus vehicles in advance for official purposes [6]. In this research, a reservation system is a digital platform that facilitates booking and scheduling services for specific resources or activities.

Smart Campus. A smart campus, also known as a digital campus, is an educational environment enhanced with advanced technology to optimize facility management and costs, enrich learning experiences and streamline administrative operations [20]. In this research, Smart Campus refers to CSPC's initiative of adopting TrackCSPC to digitally manage and optimize campus transportation and resource utilization.

TrackCSPC. TrackCSPC is a smart and integrated platform developed for Camarines Sur Polytechnic Colleges (CSPC), combining vehicle tracking and reservation capabilities. It aims to enhance the management and utilization of campus vehicles.

User Acceptance Testing. The final testing phase where real end-users validate that software meets their requirements and works correctly in real-world scenarios [16]. In this research, user acceptance testing refers to the validation process conducted by CSPC students, staff, and administrators to assess the usability and readiness of TrackCSPC before deployment.

Vehicle Tracking. A vehicle tracking system allows business owners and fleet managers to track and monitor real-time data on the location, behavior, health, and security of their vehicles, all in a centralized and easy-to-use platform [18]. In this research, vehicle tracking refers to the integration of GPS-enabled modules to gather and transmit campus vehicle coordinates to the system for location monitoring and route visibility.

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CHAPTER 2

REVIEW OF RELATED LITERATURE AND SYSTEMS

This chapter presents a review of existing literature and systems relevant to the current study. It includes discussions on related concepts, theories, and previous works that contribute to the foundation and framework of the study. Moreover, it highlights the similarities and differences between the proposed system and existing systems, identifies the research gaps, and shows how the current study aims to address them.

2.1 Related Literature and Systems

The selected literature provides insights into technologies such as GPS, GSM, Arduino, mobile applications, and reservation systems. It also discusses various approaches in system evaluation and integration, highlighting how similar systems have been developed, deployed, and assessed. By comparing these systems to the proposed TrackCSPC platform, this review identifies both the contributions and limitations of past studies, helping to define the research gap. Ultimately, this chapter establishes the foundation for designing a more efficient, responsive, and user-centered solution tailored to the transportation needs of Camarines Sur Polytechnic Colleges (CSPC).

2.1.1 Arduino GPS Tracker with Mobile App

This section explores existing studies and systems that analyze the current state of vehicle reservation and tracking, particularly in institutional and transportation settings. These works highlight common challenges such as manual processes, booking delays, poor data management, and lack of real-time monitoring. By examining these limitations, the section provides a foundation for understanding the gaps in the current vehicle management system at Camarines Sur Polytechnic Colleges (CSPC).

Amores et al. [2021]. Vehicle Maintenance and monitoring system with GPS Tracking Device [2]. The system automated vehicle management tasks, such as maintenance scheduling, fuel consumption monitoring, and GPS tracking. To address the inaccurate control caused by human

error such as manual record-keeping and lack of driver information, the system utilized GPS technology to track vehicle locations, which helped reduce fuel waste, theft, and inefficiencies. Furthermore, the system allowed different user roles —Administrator, Staff, and Driver that have specific functional requirements. The administrator generated reports including maintenance logs and fuel records kept track of vehicle activities, and managed employee and vehicle information. Although they could track vehicle operations and handle inventories, staff members had limited authority over system settings, despite having identical functionalities to the administrator. In order to ensure effective fuel tracking on every journey, drivers were responsible for recording and viewing fuel usage as well as vehicle travel history.

Bus companies often struggled with delays and customer complaints caused by manual booking errors and slow processes. To address this, Chavan, et al. [2021] conducted a study titled Online Bus Reservation System [3], where they proposed a computerized system to manage passenger records, seat availability, schedules, and billing more efficiently. This system reduced errors and sped up reservations by handling everything digitally. This is relevant to the current research on TrackCSPC, as both projects sought to improve transport services through technology, focusing on real-time management, operational efficiency, and better user experience in institutional transport systems.

Abayon et al. [2023], conducted the study titled APC Car Rental Management System with GPS Vehicle Tracking [11]. The research and development study addressed operational challenges in the car rental field, particularly those related to manual booking processes, inefficient vehicle tracking, aligning with the current study's interest in real-time monitoring systems. This system is a web-based application that uses a modified Waterfall SDLC model that integrated GPS tracking, geofencing, and reporting features. The study further showed that digital solutions, such as GPS tracking and automated booking systems, could greatly improve efficiency and reliability of car rental services.

According to Albino and Acebedo [2021], in their study Development of Car Rental Management System Scheduling Algorithm [19]. The car management system integrated with a scheduling algorithm is composed of several modules tailored for both system administrators and end-users. It allowed clients to reserve vehicles while enabling the company to efficiently schedule and manage customer bookings. The system also displayed available cars, facilitating transparency in vehicle availability. A transaction interface was included for administrators to verify vehicle status and access relevant information. To safeguard sensitive data, the system implemented strict security measures that restrict unauthorized access to the database, thereby upholding data privacy and integrity.

The study of Ugwu et al. [2021], titled A Bus Reservation System on Smartphones [6], introduced a web-based bus reservation system designed that handled passenger bookings, schedules, fares, and bus details for transport companies. By using a round-robin scheduling method, the system assigned buses more efficiently, and ensured smoother operations. This study was relevant to the current research on TrackCSPC, as both projects aimed to enhance transport services through technology, focusing on improving scheduling, operational efficiency, and user experience in institutional and public transport systems.

2.1.2 Vehicle Monitoring System

This section reviews literature focused on the development and implementation of GPS-enabled tracking systems integrated with mobile and web platforms. The studies examined here emphasize the use of Arduino, GSM, and cloud-based technologies to support real-time data transmission and user accessibility. These systems demonstrate how modern tracking solutions can enhance transportation management through live location monitoring, route visibility, and user-friendly interfaces. The insights gathered support the development of TrackCSPC's tracking component, ensuring it meets the technical requirements for reliable and accessible real-time vehicle tracking across both mobile and web environments.

Tracking vehicles played a key role in making transport services safer, more efficient, and more convenient for users. In line with this, Morallo et al. [2021] introduced a system that combines GPS and GSM technologies, powered by an Arduino Uno microcontroller [4]. The system gathered real-time location data from GPS satellites and sent it to users via SMS through a SIM900A GSM module. Users could then easily view the vehicle's location using Google Maps. This setup was considered practical because of its affordability and dependable performance, even under various weather conditions. The researchers also planned to enhance the system by adding an automatic map display feature for easier access. This study was closely related to the current research on TrackCSPC, as both aimed to improved transport management through real-time tracking, enhanced security, and more efficient operations—particularly within institutional and modern transport systems.

Mahajan et al. [2024] developed the GSM and GPS-Based Women Tracking System Using Arduino to maximize the security of women and protect their lives during emergency situations [15]. The system highlighted the importance of GSM and GPS technologies in locating the user's exact position, quickly sending alerts, and transmitting urgent messages. It provided individual protection through fast support and continuous monitoring, making it a reliable tool for ensuring women's safety. Moreover, the system was budget friendly, user-friendly, and easy to operate, thereby enhancing personal security.

Sandeep et al. [2023] conducted a study that introduced a Real-Time Object Tracking System Using Arduino with Spot-it Mobile Application to address the common issue of misplaced or stolen valuables [20]. The system integrated GPS technology with Arduino, allowing users to track objects such as wallets, keys, pets, and children in real time. The GPS module continuously provided the object's geographic coordinates, which were transmitted to the user's mobile device, for easy location monitoring. The system was designed to be compact, cost-effective, and user-friendly, offering continuous safety and surveillance. The experimental results demonstrated the

system's practicality and effectiveness in real-world scenarios, providing valuable insights for the development of similar tracking technologies, such as the TRACKCSPC: A Smart Campus Vehicle Tracking and Reservation Platform.

Sarosa et al. [2023] designed and implemented a school bus information and tracking system application that utilized GPS tracking in smartphones to locate school bus rides [12]. By providing real-time location and arrival schedules, the system made it easier to track the whereabouts of buses. Additional features allowed parents to monitor their children based on bus identity, while card tapping on mobile phones displayed passenger card information, improving safety and monitoring efficiency.

Bagad et al. [2024] conducted a study to improve the management and tracking of buses using the latest GPS technology by developing an online bus tracking system [14]. The system allowed users to monitor the real-time location of buses through Google Maps, manage schedules, and access bus details. It could also be adapted for other applications, such as accident detection and soldier tracking. This study was relevant to the current research on TrackCSPC, as both projects aimed to enhance transportation systems through mobile technology and GPS tracking for real-time vehicle monitoring, efficient scheduling, and improved safety within institutional transport services.

Based on the study of Velarde et al. [2020], UV Tracking and Booking with Accident Reporting and Speed Alert Logs Using Android Smartphone [1], the system provided efficient tracking of seat availability, managed bookings, and offered real-time location and travel time updates using smartphone-enabled GPS. As a result, the transportation service delivery was enhanced in terms of safety, efficiency, and accessibility via mobile devices. Moreover, it enabled direct communication between passengers and drivers and included features such as speed monitoring and automatic SMS-based accident alerts to notify management for quicker emergency response.

2.1.3 Vehicle Tracking and Reservation

Evaluate the System's Usability and Performance. This section reviews significant studies and applications that evaluate system usability, responsiveness, and overall performance using user feedback, testing frameworks, and real-time assessments. These studies provide essential insights for the third objective, which is to assess the usability and functionality of TrackCSPC through methods like functional testing, user acceptance testing (UAT), and feedback analysis. The findings from these works guide the evaluation criteria and testing process of the proposed system.

Payongayong and Palaoag [2024] conducted a system evaluation of the Smart Campus Driveway and Vehicle Asset Management System using the ISO/IEC 25010:2011 standard [13]. The evaluation focused on software quality characteristics such as functionality, interaction capability, reliability, security, and maintainability. The study involved students, employees, IT experts, and other campus stakeholders as respondents and found the system to be highly acceptable across the majority of evaluation criteria. This research highlighted the practicality of integrating emerging technologies such as IoT, blockchain, and GPS into campus systems and emphasized the importance of continuous system assessment to support innovation and sustainability in educational environments.

According to Respecia et al. [2024], Real-Time Vehicle Tracking Mobile Application for University Campus Mobility Aid [17], the researchers presented a cross platform mobile application designed to improve campus mobility by tracking e-jeep locations, estimated time of arrival (ETA) at each station, and seat availability in real time, utilizing Firebase, Google Maps, and React Native technologies. The system demonstrated high accuracy and responsiveness, offering an efficient transportation experience for campus users. This study aided in the development of TrackCSPC, which aimed to enhance mobility within CSPC by providing real-time vehicle location and reservation capabilities through a user-friendly mobile interface.

According to Paloy et al. [2022], the Online Bus Monitoring and Reservation Application (OBMRA) [8] was developed specifically for GL Transit Agencies operating in the Cordillera Region. The system was designed to modernize the traditional bus booking process by enabling real-time seat availability, ticket reservations, cancellations, and customer inquiry functions. The project addressed the shortcomings of GL Transit's offline system, such as long waiting times, manual record-keeping, and phone line downtimes. This study was relevant to the current research on TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform, as it reflected similar objectives—specifically the integration of GPS tracking and the modernization of manual operations within the General Services Unit (GSU) at Camarines Sur Polytechnic Colleges (CSPC).

Based on the study by Palconit et al. [2024] titled Bus Tracking, Seat Vacancy Monitoring, and Reservation Using an Android-Based Mobile Phone Application [16], the application was developed to address common challenges in Philippine bus commuting, such as uncertainty regarding vehicle arrival, seat availability, and estimated times of arrival. Commuters traveling from provinces to cities often faced the added inconvenience of waiting at bus terminals, leading to wasted time and frustration. To resolve these issues, the “Click N’ Book” mobile application was developed using Android Studio, Google Maps API, Firebase Cloud Services, and the Back4App Database. The app allowed passengers to track the real-time location of buses, monitor seat vacancies, check estimated times of arrival (ETA), request seat reservations, and pay fares digitally. Additionally, bus conductors were able to accept or deny reservation requests, update seat availability, and locate passengers for pick-up. Similarly, TrackCSPC shared corresponding objectives, focusing on campus vehicle tracking and reservation management to improve transportation efficiency within Camarines Sur Polytechnic Colleges (CSPC). In the same way, the proposed system seeks to reduce everyday waiting time and uncertainty for campus commuters by making transport services more predictable, organized, and user-friendly. More importantly, it aims to create a smoother daily experience for students, drivers, faculties, and staff.

2.1.4 Smart Vehicle and Transport System

Deploy the system with Integration, System, and User Acceptance Testing. This section discusses studies that involved the deployment of mobile and web-based systems to selected users for testing and evaluation. In line with this, the TrackCSPC mobile application and web platform will be deployed to respondents within Camarines Sur Polytechnic Colleges (CSPC) to assess the system's usability and effectiveness under actual usage conditions.

According to Andutan and Ucat [2022], the study titled Design and Development for a Vehicle Tracking System [7], focused on the design and advancement of vehicle tracking systems that utilized route deviation detection, an emergency SOS feature, and a status report. This study tracked the vehicle's current location and improved vehicle security by integrating GSM and GPS technologies that monitor vehicle movements in real-time. The researchers used an Arduino Mega 2560, SIM900 GSM module, and NEO-6M module, with a feature STATUS command for location monitoring. This method aligned with modern trends in vehicle monitoring and tracking, which is relevant and beneficial to the current study, TrackCSPC.

Lohar et al. [2023], in their study titled Design and Implementation of Vehicle Tracking System with GPS [18], developed a simple and affordable vehicle tracking system by combining a smartphone application with a microcontroller. The system used GPS to track the vehicle's location in real time and transmitted updates via GSM/GPRS to a centralized database. The microcontroller managed the overall operation, while the mobile application displayed the vehicle's exact position on Google Maps. This setup allowed users to easily monitor the vehicle's movement, check the distance traveled, and determine the estimated time of arrival.

Mamdya [2022], in the study titled GPS Tracking System [10], presented a GPS-based tracking system designed to monitor the movement of vehicles and employees efficiently. Utilizing satellite technology, the system provided accurate, real-time location and time data, ensuring effective supervision and control. Widely applied across military, civil, and commercial sectors, it

operated through the integration of satellite transmitters and GPS receivers. This study was relevant to the current research on TrackCSPC, as both emphasized the use of GPS technology for real-time monitoring, improved scheduling, and enhanced management of transportation services.

Hsu et al. [2022], in their study Integrated System for Official Vehicles with Online Reservation and Moving Path Monitoring [9], introduced a comprehensive vehicle management system that integrates GPS tracking, online reservations, and real-time movement monitoring through a set-top box, Line Bot, and cloud database. The system effectively prevents unauthorized vehicle use, double bookings, and mismanagement by providing accurate data on vehicle location and usage. This study supports the present research on TrackCSPC, as it similarly aims to enhance vehicle monitoring and reservation efficiency through technology-driven solutions.

Parents or guardians of children who ride the school bus often worry about whether their children have arrived at school safely. To address this concern, Chaturvedi et al. [2021], in their study titled TrackTot: Student Tracking Application for School Bus [5], developed a mobile application that allows parents to track their children in real time through their smartphones. The app also enables them to view the bus route and monitor its location. This study is relevant to the current research on TrackCSPC, as both systems aim to integrate mobile technology for real-time vehicle monitoring, passenger verification, and enhanced safety within institutional transport operations.

In conclusion, the reviewed literature on vehicle monitoring, tracking, and reservation systems emphasized their potential to enhance security, efficiency, and user convenience in campus transport management. For TrackCSPC, key considerations include the implementation of real-time GPS tracking, a user-friendly mobile reservation interface, integration with campus security protocols, and consistent system performance under varying network conditions. These insights are drawn from the collective successes and challenges identified in previous studies and serve as valuable foundations for the continued development and optimization of the TrackCSPC platform.

2.2 Synthesis of the State-of-the-Art

The related literature, studies and systems that the researchers gathered from the websites were closely associated with each other. The content of the information discussed evidence related to the researcher's topic.

Several recent studies on Arduino-based GPS tracking systems Talip et al. [2023], many studies on Arduino-based GPS tracking systems Talip et al. [2023] Sandeep et al. [2023] Mahajan et al. [2024] Motakabber et al. [2022] Pradeepkumar et al. [2022] share TrackCSPC's core objective of providing real-time vehicle location monitoring using GPS, GSM, and microcontroller-based technologies. These studies also emphasize the importance of accurate data transmission, remote access through mobile devices, and improved monitoring efficiency, all of which align with TrackCSPC's goal of enabling reliable, continuous tracking of campus vehicles.

Furthermore, research on bus tracking and reservation systems Jayaraam et al. [2020], Ayesha & Jambhulkar [2021], Respecia et al. [2024], Gogri et al. [2020], Hsu et al. [2022], directly parallels TrackCSPC's integration of digital reservation features, showing that reservation modules significantly reduce scheduling conflicts and improve administrative coordination. Studies focusing on smart vehicle and transport systems Agun et al. [2022], Payongayong & Palaoag [2024] also reflect similar goals of enhancing operational efficiency, user convenience, and transport security through the convergence of GPS, IoT, RFID, and mobile technologies—principles equally central to TrackCSPC's design.

Despite these similarities, there are notable differences in system context, scope, and implementation that distinguish TrackCSPC from the reviewed research. Most existing Arduino-based GPS tracking systems were designed for general-purpose vehicle tracking, such as fleet management, anti-theft applications, and public transport monitoring, rather than for a school-based transportation ecosystem like CSPC. Additionally, some studies implemented complex sensor networks, GSM modules, or RFID-based authentication systems that TrackCSPC does not fully

adopt, as the system primarily utilizes mobile-based GPS tracking or a simplified GPS module depending on deployment. Bus reservation systems in previous research typically cater to public commuters, whereas TrackCSPC's reservation module is restricted to authorized institutional users, such as administrators and department heads, reflecting its academic and organizational context. TrackCSPC focuses specifically on campus-bound mobility, simplified real-time tracking, and controlled reservation workflows tailored to CSPC's operational procedures.

These studies provide essential insights that inform the design of TrackCSPC, ensuring that the system is user-friendly, efficient, and equipped with strong security features. Collectively, the reviewed findings underscore the significant potential of TrackCSPC to enhance campus transportation through improved operational efficiency, real-time monitoring, and reliable vehicle management.

2.3 Gap Bridged by the Study

The related literature and studies presented and analyzed various technologies and systems implemented for TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform. Through the reviewed materials, the researchers identified numerous studies and innovations focused on GPS-based tracking, RFID integration, mobile reservations, and smart campus systems.

While numerous studies have explored vehicle tracking and reservation management systems, none have directly integrated both functionalities into a unified platform specifically tailored for school-based transportation. Moreover, previous research has not conducted a comprehensive analysis from the perspective of multiple user groups—such as students, faculty members, and the General Services Unit (GSU)—nor examined the system's implementation within a campus-based environment where vehicle availability and mobility efficiency are essential.

Existing studies on vehicle tracking and reservation systems often overlook the potential of merging both functionalities into a unified platform specifically designed for campus

communities such as CSPC. The integration of a real-time GPS-based tracking system with an intuitive reservation interface for students, faculty, and staff remains underexplored. Furthermore, there is insufficient in-depth examination of operational efficiency and user satisfaction within campus environments, emphasizing the need for an innovative solution like TRACKCSPC to address these gaps.

The present study is unique because, although numerous systems have been developed involving GPS tracking, RFID, mobile reservations, and smart campus solutions, this research does not merely focus on the technologies themselves. Instead, it emphasizes how these technologies can be integrated into a unified web and mobile platform that specifically addresses the operational needs of Camarines Sur Polytechnic Colleges (CSPC). Most importantly, none of the reviewed studies or systems were developed or implemented within the context of CSPC's campus operations, highlighting the distinct contribution of this research.

Therefore, the present study serves as a modest attempt to bridge this research gap by focusing on the development of a smart campus vehicle tracking and reservation platform specifically tailored to the operational needs and environment of Camarines Sur Polytechnic Colleges (CSPC).

Notes

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CHAPTER 3

TECHNICAL BACKGROUND

This chapter provides a comprehensive discussion of the technical framework supporting the development and implementation of the proposed system, TrackCSPC: A Campus Vehicle Tracking and Reservation Platform. It outlines the system architecture, hardware and software components, and the integrated technologies that enable the system's core functionalities. By detailing these technical elements, the chapter establishes the system's operational foundation, demonstrating its practicality, reliability, and alignment with the project's objectives.

3.1 Overview of Current Technologies Used in the System

The development of TrackCSPC: A Smart Campus Vehicle Tracking and Reservation System incorporates a range of modern technologies, frameworks, and development tools, each chosen for their efficiency, compatibility, and relevance to the system's functional requirements. This section presents a detailed discussion of each component, along with the rationale behind its selection, to demonstrate the technical viability and cohesion of the system.

The central microcontroller for the system's hardware-based GPS tracking component was the ESP32 DevKit V1. The ESP32 was chosen for its suitability in wireless communication and real-time location monitoring due to its dual-core processor, integrated Wi-Fi and Bluetooth, and low power consumption. Prototyping and deploying embedded solutions were made easier by its open-source development environment and strong community support [1]. The real-time geolocation data of campus vehicles were obtained using the NEO-6M GPS Module. Through UART communication, this GPS module provided precise coordinates with minimal latency to the ESP32. The coordinates were then transmitted to the cloud for real-time tracking. Because of its affordability and dependability, it was frequently utilized in GPS-based Internet of Things applications [7]. Overall, this hardware setup was designed to be both practical and reliable, ensuring that vehicle location data could be collected consistently.

The Flutter Framework was used to create a mobile application that displayed the real-time location of vehicles using the driver's phone GPS or data from an Arduino device. Flutter's cross-platform capabilities enabled development for both Android and iOS from a single codebase. It also integrated smoothly with Firebase to provide seamless real-time updates [4].

The Geolocator Package for Flutter was used to enable the mobile app to obtain the driver's current GPS location. This allowed the mobile device to serve as an alternate tracker when the Arduino-based tracker was unavailable or turned off. The software enabled background tracking and high-precision data collection [3].

The Firebase Realtime Database served as the system's central data repository, where real-time tracking and reservation information was stored. Firebase's cloud-based architecture enabled simultaneous access from multiple devices and facilitated easy synchronization across clients, eliminating the need for a traditional backend server (Google Firebase, 2023). Firebase Authentication secured the admin panel by providing authentication services for login. This ensured that only authorized individuals could access and manage vehicle reservations, as well as monitor vehicle positions [4].

These technologies were selected to ensure that the system was scalable, maintainable, and capable of delivering real-time location data with high availability. By leveraging modern embedded systems, mobile frameworks, and cloud-based services, TrackCSPC provided a reliable platform for campus vehicle tracking and scheduling.

3.2 Resources

This section outlines the essential resources necessary for the development and implementation of TrackCSPC: A Campus Vehicle Tracking and Reservation Platform. It details the required hardware, software, and other specifications that will support the system's core functionalities, performance, and integration. These resources are carefully selected to ensure the successful delivery of a reliable, efficient, and real-time tracking and reservation system.

3.2.1 Hardware Specifications

This section details the essential hardware components required for the development, testing, and deployment of the TrackCSPC system. It focuses on the embedded devices, GPS modules, GSM communication units, and developer workstations used in building and operating the tracking and reservation platform. These specifications are selected to ensure performance efficiency, system reliability, and compatibility with the technical requirements of the project.

Table 1 presents the hardware components utilized during the system's development and implementation phases. These include both general-purpose development tools, such as computers and peripheral devices, as well as embedded hardware essential for mobile tracking. The selection of these hardware elements was based on considerations of compatibility, performance efficiency, and reliability to support the system's requirements from initial prototyping to final deployment.

Table 1: Hardware Specifications

HARDWARE TOOL	SPECIFICATION
Processor	Intel Core i5 or AMD Ryzen 5
Memory	8GB RAM
Storage	256GB SSD
Input Device	Keyboard and Mouse (for development)
Output Device	Monitor (for development)
Mobile Device	Quad-core CPU, 2GB RAM, 32GB storage
ESP32 DevKit V1	Dual-core 32-bit processor, WiFi/Bluetooth, 3.3V logic level
NEO-6M GPS Module	Compatible with 3.3–5V input, UART output, 9600 baud rate
Breadboard and Jumper Wires	Used for prototyping and testing hardware interconnections

The hardware tools listed in Table 1 represent the essential physical components required for the development and operation of the TrackCSPC system. These components were carefully selected to ensure optimal system performance during both the development and deployment stages within the actual campus environment. For instance, the developer's laptop or PC should be equipped with at least an Intel Core i5 or AMD Ryzen 5 processor, 8 GB of RAM, and SSD storage to efficiently run necessary software such as Arduino IDE and Android Studio. This specification

minimizes performance slowdowns during coding, testing, and firmware uploads to the microcontroller.

On the tracking side, the system uses an ESP32 DevKit V1 as the brain of the device. It handles the GPS and GSM modules and sends data to Firebase.

As for the mobile side, the application was designed to run on Android devices, requiring at least Android 8.0, 2GB of RAM, and 32GB of storage. These specifications ensure that the mobile app operates smoothly, particularly when accessing GPS features or synchronizing data with Firebase. Overall, the hardware tools listed in the table collectively enable the practical implementation of the system's tracking and reservation functionalities in real-world conditions.

3.2.2 Software Specifications

This section enumerates the essential software platforms and development tools used in the implementation of the TrackCSPC system. These software tools are carefully chosen to ensure seamless development, integration, and deployment of the system's features for both web and mobile environments.

Table 2 presents the software tools and technologies utilized during the development and deployment of the system. It includes operating systems, integrated development environments (IDEs), software development kits (SDKs), libraries, and programming languages essential for both backend and frontend components. This set of software tools supports multi-platform functionality, enables real-time data transmission, and ensures seamless interoperability among the mobile, web, and embedded system components.

In addition, the selection of these tools was guided not only by technical capability but also by ease of use, accessibility, and long-term maintainability of the system. This approach helps ensure that future developers and system administrators can easily understand, update, and manage the platform without requiring highly specialized expertise. Ultimately, this makes TrackCSPC more sustainable and adaptable for continuous improvement within the campus environment.

Table 2: Software Specifications

SOFTWARE TOOL	SPECIFICATION
Operating System (Development)	Windows 10 or higher
Operating System (Deployment)	Android 8.0 (Oreo) or later,
Arduino IDE	Version 2.2.1 or higher
Visual Studio Code	Latest stable release
Flutter SDK	Version 3.10.0+
Firebase Realtime Database	Web Console
Firebase Authentication	Web Console
Geolocator (Flutter)	Version 10.0.1+
Firebase Database	Version 10.0.7+
Programming Language	Dart, C/C++
Scripting Language	HTML, CSS, JavaScript

The software tools listed in Table 2 represent the core applications and platforms used in developing the TrackCSPC system. These tools cover various aspects of system creation, including programming, device testing, application development, and online data management. The Arduino IDE is employed for the embedded system to program the ESP32 board, where the GPS and GSM modules are configured to collect and transmit data. It is a simple yet powerful development environment that supports a wide range of boards and libraries, making it ideal for hardware prototyping and integration.

Flutter is utilized for mobile application development because it allows deployment on both Android and iOS platforms, however, this project focuses solely on Android implementation. It provides high performance and a visually appealing user interface, making it suitable for real-time applications. The Geolocator plugin enables the app to retrieve the driver's current location and continuously update it in Firebase. On the backend, Firebase Realtime Database manages all tracking and reservation data in real time, while Firebase Authentication ensures secure access control by allowing only authorized administrators to manage system data and operations.

A user-friendly web dashboard was developed using core web technologies such as HTML, CSS, and JavaScript. was integrated to visualize the real-time location of campus vehicles on an interactive map. When more advanced mapping features are required, the Google Maps API may

be incorporated to enhance map detail and functionality.

3.2.3 Program Specifications

This section describes the TrackCSPC system's critical software components, including functional and non-functional requirements, as well as internal system design elements such as data flow, architectural layout, and user authentication mechanisms. These specs serve as the blueprint for the application's operation, ensuring that the platform is dependable, secure, and scalable for real-world use.

Figure 1 illustrates the system architecture of the TrackCSPC platform, highlighting the interaction between mobile and web applications, as well as the integration of GPS technology for real-time vehicle tracking and reservation management. The architecture depicts how various components including user interfaces, GPS-enabled tracking devices, and cloud-based databases work together to provide a seamless experience for end users, drivers, and system administrators. This architecture ensures efficient communication, real-time data synchronization, and centralized control, all of which are vital for the system's reliability and overall operational performance.

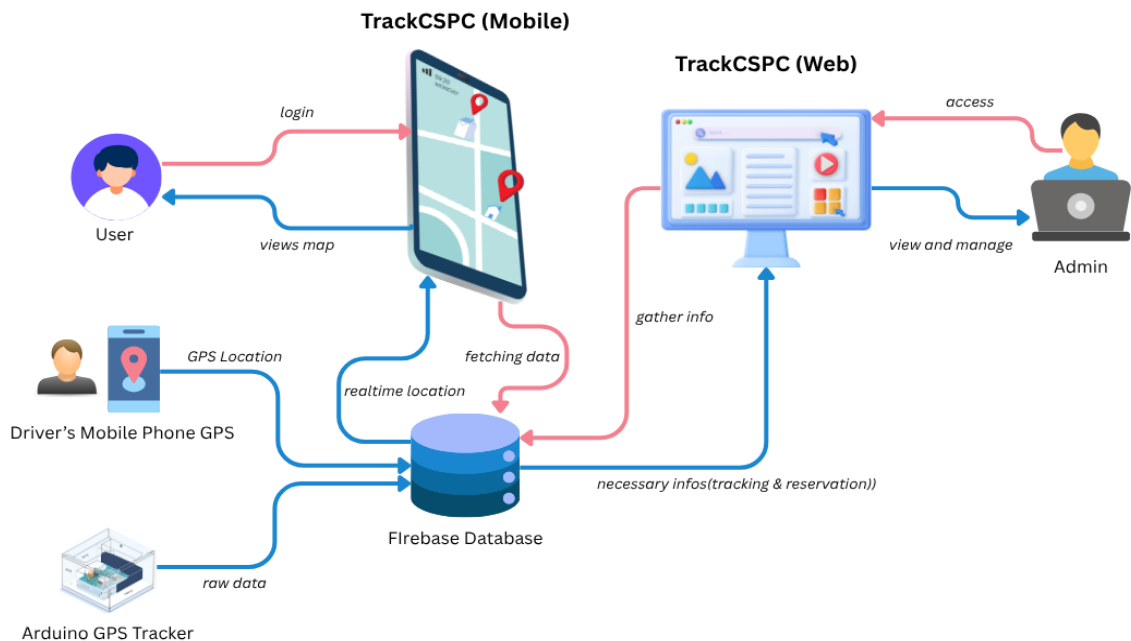


Figure 1. **TrackCSPC Mobile and Web Application Architecture**

Figure 1 presents the overall architecture of the TrackCSPC system, encompassing both the mobile and web application components. It highlights the core data flow and interactions among system users, tracking hardware, the Firebase cloud database, and the administrative web interface. This architectural overview demonstrates how each component communicates to support real-time tracking, reservation management, and centralized system control.

On the left side of the diagram, the system began with a GPS-enabled Arduino device and a driver's mobile phone, both of which served as sources of real-time location data. These devices continuously collected and transmitted raw coordinates to the Firebase Realtime Database. The mobile application, used by regular users such as students and faculty, fetched this live data from Firebase and visualized the vehicle's movement on an interactive map interface.

Meanwhile, the administrative web application accessed the same database to monitor and manage the tracking system. Admin users were able to view vehicle data, manage information, and access features such as tracking history or reservation requests. This centralized architecture ensured that both mobile and web clients retrieved synchronized, up-to-date information.

This diagram also reflected essential user interactions, such as mobile users logging in to view routes, while administrators accessed the system through a secure portal. This structure ensured seamless communication between components, efficient data handling, and real-time location visibility for campus vehicles.

Table 3 presents the functional requirements necessary for the system, detailing the system's primary features that support secure, convenient, and organized operations. These requirements serve as the foundation for system development, ensuring that each function aligns with the users' needs and overall project objectives. Furthermore, they guide the design and implementation process to achieve a reliable and user-friendly system performance. The web application functioned as the central administrative workspace of TrackCSPC. It included a complete vehicle reservation and scheduling module that enabled administrators to handle

booking requests smoothly Real-time tracking became possible through GPS data sent from both the driver's mobile device and the Arduino-GSM tracker, allowing accurate monitoring of vehicle movements at all times. The Firebase Realtime Database supported the system by keeping reservation records, tracking data, and user updates synchronized instantly across all interfaces.

Table 3: Functional Requirements

PLATFORM	FUNCTIONAL REQUIREMENTS	FUNCTIONAL REQUIREMENTS DESCRIPTION
Web Application	Vehicle Reservation and Scheduling	The system allowed an admin to reserve and schedule a vehicle via the web application.
	Real-time Tracking Dashboard	The system provided real-time vehicle tracking through two sources: mobile GPS (driver's smartphone) and an embedded GPS (Arduino with GSM).
	Data Storage and Synchronization	The system stored and synced vehicle data, location, and reservations through Firebase Realtime Database.
	Live Map Interface	The system displayed location updates on a live map in both the web and mobile interfaces.
	User Authentication	The system authenticated users and restricted access to admin functionalities.
Mobile Application	GPS Location Transmission	Sends real-time GPS coordinates from the driver's smartphone to Firebase.
	Mobile Notifications	Sends instant alerts about reservation confirmations, reminders, or changes through the mobile interface.
	User Authentication	Ensures that only authorized drivers and users can access mobile features securely.

The mobile application worked alongside the web platform by transmitting live GPS coordinates and allowing users to check vehicle availability and reservation status anytime. It also provided mobile notifications to keep users informed of important updates. Both the mobile and web applications implemented secure role-based authentication, ensuring user privacy and the protection of all system data.

Table 4: Non-Functional Requirements

NON-FUNCTIONAL REQUIREMENTS	NON-FUNCTIONAL REQUIREMENTS DESCRIPTION
Real-Time System Performance	The system maintained a location update interval of less than 5 seconds to ensure near real-time tracking.
Cross-Platform Compatibility	The application remained responsive and accessible across modern browsers and Android devices.
Security and Data Protection	All stored data were secured with Firebase Authentication and protected from unauthorized access.
System Reliability and Scalability	The system supported at least two active vehicle trackers without performance degradation.
Connectivity Efficiency	The application's lightweight GPS and GSM data packets provided reliable functionality in areas with limited mobile data connectivity.

The non-functional requirements ensured that TrackCSPC operated efficiently, securely, and consistently across different environments. The system delivered near real-time vehicle monitoring by maintaining location updates within five seconds, supporting reliable tracking throughout campus operations. Its compatibility with modern browsers and Android devices guaranteed that users could access the platform without performance issues. To protect sensitive reservation and location data, Firebase Authentication was implemented, securing all stored information against unauthorized access. The system also demonstrated strong reliability by handling multiple active vehicle trackers simultaneously without performance degradation. Additionally, the platform utilized lightweight GPS and GSM data packets, allowing stable functionality even in areas with limited mobile connectivity. Together, these requirements established the foundation for a dependable, secure, and user-friendly transportation management platform for CSPC.

3.2.4 Programming Environment

The programming environment for TrackCSPC: A Campus Vehicle Tracking and Reservation Platform integrates a combination of frontend and backend technologies, carefully chosen to ensure the system's efficiency, portability, and real-time performance. This environment

includes all essential programming tools, libraries, and frameworks used to build and maintain the web-based admin portal, the mobile tracking application, and the embedded GPS tracking system.

Front End

Flutter (Mobile Frontend Framework). Flutter is a cross-platform UI toolkit developed by Google and used for building the mobile application. It uses the Dart programming language and supports responsive layouts, fast development cycles, and smooth performance across a variety of Android devices.

Dart (Programming Language for Flutter). Dart is the primary language used within the Flutter framework. It allows developers to build efficient, high-performance mobile interfaces while maintaining clean and manageable code.

HTML, CSS, JavaScript (Web Admin Dashboard Technologies). The web-based admin dashboard is built using HTML for structure, CSS for styling, and JavaScript for interactivity styling, and JavaScript for interactivity. These standard web technologies ensure the system is compatible across modern web browsers and supports responsive design.

Back End

Firebase Realtime Database (Cloud Data Storage). Firebase Realtime Database is used as the main backend database. It supports real-time data syncing across users and devices, which is essential for updating vehicle location and reservation data instantly.

Firebase Authentication (User Management). This component handles user login and access control. It enables secure email-password authentication, restricting access to authorized administrators and ensuring data security.

Firebase SDKs and REST APIs (Data Interaction Tools). These tools enable mobile and web applications to interact with Firebase. They provide a reliable and secure communication layer between the client applications and the cloud database, ensuring consistent data synchronization across all platforms.

Embedded System

ESP32 Microcontroller (GPS Hardware Component); The ESP32 is a Wi-Fi-enabled microcontroller used in the embedded system. It is responsible for collecting GPS data and transmitting it to Firebase for real-time tracking.

Arduino IDE (Programming Tool for ESP32). The Arduino Integrated Development Environment is used to write and upload code in C/C++ to the ESP32 microcontroller. It enables the configuration of the GPS module and the transmission of data.

Development Tool

The development phase focused on building a functional, user-friendly, and reliable system for managing campus vehicle tracking and reservations. During this stage, the researchers selected tools and environments that would support efficient coding, testing, and integration of the system's front-end and back-end components. Emphasis was placed on using platforms that offered flexibility, debugging capabilities, and support for multiple programming languages to ensure smooth development and maintainability.

Visual Studio Code (Development Environment). VS Code is the primary IDE used for writing Dart, JavaScript, HTML, and CSS. It offers features such as syntax highlighting, extensions, and debugging tools, making it suitable for full-stack development.

GitHub is used for version control and collaboration. It tracks changes to the codebase, enables teamwork, and provides backup and deployment capabilities throughout the development lifecycle.

3.3 Testing Plan

This section outlines the strategies and methodologies to be used in testing the functionality, reliability, and performance of the TrackCSPC system. To ensure its overall success, a combination of test cases and appropriate tools per testing type is adopted. Testing will be done at several levels unit, integration, system, and user acceptance testing (UAT) each paired with its

suitable tools. Both the mobile and web components of the system will undergo rigorous testing to verify features such as location tracking accuracy, database synchronization, user interface behavior, and system responsiveness.

Gray-box testing was a mixed approach to software testing that incorporated aspects of both black-box and white-box testing. This testing method proved useful in uncovering specific bugs related to context, integration issues, and security vulnerabilities. During the study, gray-box testing enabled the researchers to identify areas where the system could be optimized for better performance. It also provided valuable insights into potential weaknesses that might affect the user experience, allowing improvements to be made before full deployment.

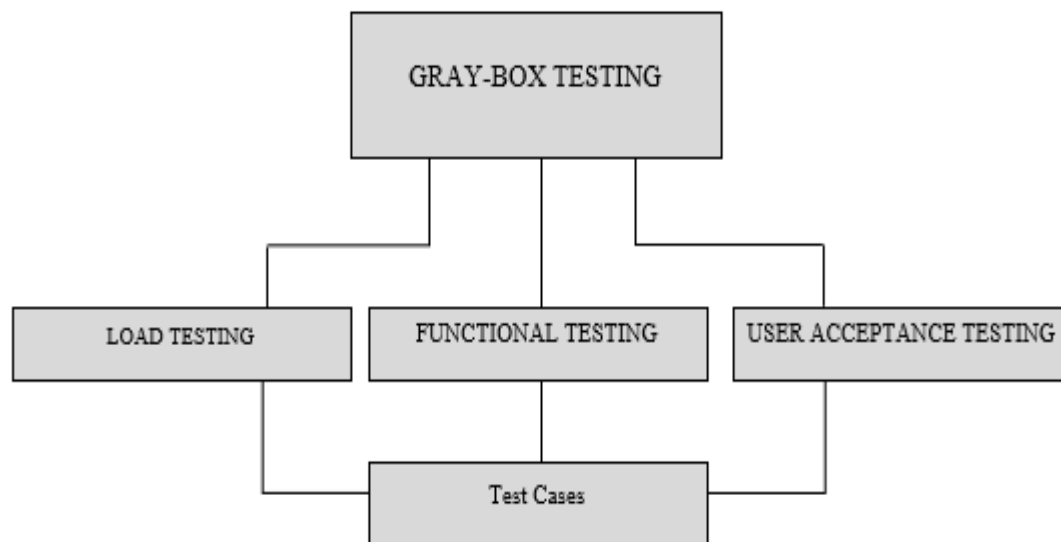


Figure 2. **Testing Plan**

Figure 2 illustrates the primary testing approach adopted in the development of the TrackCSPC system, which utilized gray-box testing. This approach allowed testers to evaluate the system with partial knowledge of its internal structure, including data flow, system architecture, and critical logic paths. Gray-box testing was particularly appropriate for TrackCSPC due to its integration of a mobile application, web-based administrative platform, GPS tracking modules, and a real-time Firebase database. As shown in the figure, gray-box testing was composed of three main

testing components: Load Testing, Functional Testing, and User Acceptance Testing (UAT). Load testing assessed the system's performance and stability under varying levels of user activity, such as simultaneous vehicle requests, GPS data transmission, and administrative actions. Functional testing focused on verifying that core system features—including vehicle reservation, real-time GPS tracking, authentication, and role-based access—operated according to specified requirements. User acceptance testing ensured that the system met end-user expectations by evaluating usability, accuracy of information, and overall system behavior under realistic operational conditions.

The results from these testing activities were consolidated into defined test cases, which served as the basis for validating system reliability and correctness. This layered approach enabled the identification of issues across both the front end and back end, offering insight into data flow, Firebase queries, and the rendering of live maps. Gray-box testing ensured a balance between functionality testing and structural validation, making it highly suitable for systems with integrated data processing and user interaction.

3.3.1 Types of Testing

This section details the system's reliability and effectiveness within real-world campus operations, employing structured testing methodologies. Evaluations validated system performance under both typical and peak usage scenarios, assessing user interaction, responsiveness, and workflow efficiency. The testing focused on the TrackCSPC system's management of vehicle reservation requests, real-time GPS tracking via Arduino-based modules, user authentication, notification delivery, and concurrent activities across mobile and web platforms. The objective was to ensure the system met expected standards of functionality, usability, and operational stability for campus-wide deployment.

Load Testing. This testing focuses on assessing how the system handles multiple simultaneous users, frequent GPS data transmissions from the ESP32 devices, and concurrent

vehicles. reservation requests, and real-time updates in the Firebase database.

Functional Testing. This type of testing ensures that each component performs its intended function correctly, both individually and as part of the overall system. Core functionalities such as user authentication, vehicle reservation processing, real-time GPS tracking, role-based access control, and map visualization are systematically tested.

User Acceptance Testing (UAT). Real users will test the system's usability and reliability. A UAT checklist and survey questionnaire will be used to collect feedback based on test cases such as tracking and reserving a vehicle.

3.3.2 Testing Tools and Framework

To ensure comprehensive and efficient testing, a combination of advanced tools and robust testing frameworks will be employed. These resources will help automate various aspects of the testing process, reduce manual effort, and improve the accuracy and consistency of results throughout the development lifecycle.

Firebase Emulator Suite. This built-in tool from Firebase will simulate real-time database behavior locally. It enables developers to test read/write operations without affecting live data, ensuring database logic works as intended.

Survey Questionnaire and UAT Checklist – Used during User Acceptance Testing to gather feedback on usability, design, and system satisfaction. Using these tools and frameworks not only improves testing efficiency but also provides better visibility into potential failures. According to Meszaros [2007], leveraging appropriate frameworks improves the reliability and maintainability of software projects.

Notes

- [1] Espressif Systems. n.d. ESP32 Wi-Fi & Bluetooth SoC. Retrieved April 16, 2025 from <https://www.espressif.com/en/products/socs/esp32>
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CHAPTER 4

METHODOLOGY, RESULTS AND DISCUSSION

This chapter discusses the system's Iterative Waterfall Model technique, which includes the phases of planning, analysis, design, implementation, and testing. It presents the information gathered, organized, and evaluated throughout the system's development. This process was carried out to ensure a systematic and continuous improvement of the project through multiple development cycles. Each iteration produced a working version of the system that was tested and refined based on user feedback. This approach provided a structured and flexible workflow for developing the "TrackCSPC: A Campus Vehicle Tracking and Reservation Platform" for Camarines Sur Polytechnic Colleges.

4.1 Planning & Requirements Gathering

The initial stage, Planning & Requirements Gathering, was crucial in the Iterative Waterfall Model. It involves defining the overall project plan, aligning objectives among team members, and identifying essential requirements that determine the project's success. This step ensures that every iteration remains focused on meeting clearly defined goals rather than deviating from the intended outcomes [1]. The project's scope was clearly defined to solve problems like manual scheduling, inefficient monitoring, and the lack of real-time tracking. A project timeline was created, along with a list of the necessary tools and technologies for both web and mobile platforms. This planning phase sets the groundwork for the entire project, ensuring that each succeeding iteration was guided by the defined goals. The utmost aim was to create a functional and reliable system, "TrackCSPC: A Campus Vehicle Tracking and Reservation Platform", to streamline the management and monitoring of campus vehicles at Camarines Sur Polytechnic Colleges (CSPC).

4.1.1 Research Design

The researchers used mixed-method research design that both quantitative and qualitative

approaches were mixed to critically assess the system developed in relation to the goals of the study.

A quantitative research design involves systematic gathering and analysis of numerical data for the purpose of determining trends and assessing performance [4]. The study used a mixed-method research design to evaluate the TrackCSPC platform. The quantitative method involved paper-based surveys where administrators, General Services Unit, drivers, faculty, and students rated each system module using a binary scale (1 = pass, 0 = fail). These numerical results were analyzed to measure the system's functionality, accuracy, and overall performance.

The qualitative method consisted of interviews and observations participants were encouraged to share their experiences using the TrackCSPC platform, including system usability, functionality, and suggestions for improvement. The responses were analyzed and interpreted to understand the system's usability and the users' level of satisfaction.

4.1.2 Research Method

This section described the methods that were used in the study, including the research instruments, data-gathering procedures, sampling technique, respondents, and statistical tools applied. These components were essential in collecting meaningful data to evaluate the functionality, usability, and overall performance of the TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform. The study employed a mixed-methods approach, which allowed the researchers to gather both quantitative and qualitative insights. Ethical considerations were also observed throughout the research to ensure the confidentiality and integrity of the respondents' information.

Instruments

This part describes the tools that the researchers used to collect the necessary information for the study. The instruments included an interview guide and questionnaire.

Interview Guide. The interview guide provided a structured set of questions used with the

General Services Unit staff, drivers, faculty, and students to gather qualitative insights about problems in CSPC's manual vehicle reservation and tracking process. Their responses helped the researchers understand the system's shortcomings and informed them about the design of TrackCSPC key features.

Questionnaire. The questionnaire was distributed through a paper-based format and served as the primary instrument for gathering quantitative data essential to the study, enabling the researchers to collect measurable feedback on usability, reliability, and user satisfaction across different stakeholder groups, including students, faculty, drivers, and General Services Unit staff. It was designed to support two specific objectives:

Objective 1: Gather demographic information of the respondents.

Objective 2: Evaluate the system in terms of functionality, User Acceptance Testing(UAT), and load testing, using Binary Scale (0–1)

Respondents included CSPC administrators, General Services Unit staff, Drivers, Faculty, IT Expert and students who participated in testing the TrackCSPC platform. Their responses enabled the researchers to assess user satisfaction and overall system performance.

Test Cases (via questionnaire). Test cases were carried out to examine whether each core feature of the TrackCSPC platform functioned as intended. These tests focused on checking the reliability of modules such as vehicle reservation, GPS tracking, authentication, and data synchronization. Each test case was evaluated using a simple binary scale, where a score of "1" indicated that the feature operated correctly, while a score of "0" signified that the feature failed or did not perform as expected. This method allowed the researchers to clearly determine the operational status of every module and identify areas that required adjustment or improvement.

Sampling Technique

The study used purposive sampling to select respondents who were directly involved in CSPC's vehicle reservation and monitoring process. This included General Services Unit,

drivers, faculty, IT Expert and students. These individuals were chosen because their experience with the manual system allowed them to provide reliable and relevant feedback on the usability and performance of the TrackCSPC platform. This sampling approach ensured that the evaluation results accurately reflected how the system functioned in actual campus operations.

To determine the appropriate number of student respondents, the researchers used arithmetic mean:

$$\text{Mean} = \frac{\sum X}{N}$$

Where:

$X = 1$ if Pass, 0 if Fail

N = number of respondents

Therefore, the computed sample size is 20 student respondents. In addition, 0 faculty, 5 General Services Unit staff, 5 drivers, and 1 IT expert participated, resulting in a total of 31 respondents for the study.

Table 5: Frequency Distribution Table of Respondents

RESPONDENTS	FREQUENCY	PERCENTAGE
Students	20	64.52%
Driver	5	16.13%
General Services Unit Staff	5	16.13%
IT Experts	1	3.22%
Faculty	0	0.00%
Total	31	100%

This composition ensured that the study gathered diverse and relevant perspectives from both system users and evaluators.

4.1.3 Statistical Analysis Tools

In analyzing the data gathered for the study, the researchers used different statistical tools to ensure that interpretations of results are accurate. The tools used serve as a guide to show the result of the study in a systematic approach and help identify patterns and relationships within the

data that support the study's objectives. These tools also aided in quantifying the system's performance and user responses, allowing for an objective evaluation of its effectiveness, which provided a factual basis for validating the developed system met the functional and operational requirements of the organization.

Arithmetic Mean. The arithmetic mean is a statistical measure that represents the central value of data set. It is obtained by adding all values in the data set and dividing the result by the total number of values [6]. In this study, the arithmetic mean is applied using a binary scale, where responses are limited to two possible values: 1 for pass and 0 for fail. This approach allows the researchers to determine the overall level of system functionality and acceptability based on the results of user acceptance and functional testing.

Frequency and Percentage. The researchers analyzed the respondents' demographic Profile using Frequency and Percentage, determining the proportion of each group in the sample [2]. It is computed by dividing the frequency by the total number of responses, Multiplying it by 100.

$$P = \frac{f}{n} \times 100$$

Where:

p = percentage

f = frequency of each group

n = total number of respondents

4.1.4 Identify Problems and Proposed Solutions

During the requirement analysis phase, several challenges in the existing manual vehicle reservation and tracking process at Camarines Sur Polytechnic Colleges (CSPC) were identified. These issues have long affected the efficiency of campus transportation services, leading to scheduling conflicts, communication delays, and limited visibility of vehicle availability. The manual system relied heavily on paperwork and face-to-face coordination, which often caused

confusion among users and administrators.

One of the major problems discovered was the lack of automation in vehicle scheduling and tracking. Requests for vehicle use were handled manually, resulting in frequent double bookings, overlapping schedules, and prolonged waiting times. Without a centralized system, the General Services Unit (GSU) encountered difficulties in monitoring which vehicles were available or currently in use, making it challenging to allocate resources efficiently.

Another significant issue was the absence of real-time tracking and monitoring. Administrators and staff had no direct visibility of vehicle locations once they were deployed, making it difficult to ensure accountability and respond quickly to emergencies or delays. This lack of transparency limited the institution's ability to manage its transportation resources effectively.

Furthermore, inefficient record-keeping and coordination contributed to delays in communication between requesters, drivers, and administrators. Vehicle logs were often handwritten, and information about trips was scattered across multiple documents, making it time-consuming to verify records or track usage history. This manual process increased the chances of human error and created gaps in the documentation of vehicle operations.

To resolve these challenges, the researchers developed TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform, a web and mobile-based system designed to digitalize and simplify campus transportation management. The platform automated the reservation process, allowing users to check vehicle availability, submit requests, and receive booking confirmations instantly. Real-time GPS tracking was integrated to provide administrators with live updates on vehicle locations, improving safety, accountability, and operational oversight.

The system also centralized all data through a cloud-based database, ensuring that records were accurate, synchronized, and easy to retrieve. Notifications and reminders were automatically sent to both drivers and requesters to reduce scheduling conflicts and missed reservations. By transforming the manual system into an automated digital platform, TrackCSPC improved

operational efficiency, enhanced data accuracy, and provided a more convenient and reliable transportation experience for CSPC faculty, staff, and students.

Table 6: Identified Problems and Proposed Solutions

IDENTIFIED PROBLEMS	DESCRIPTION	PROPOSED SOLUTION
Manual Vehicle Reservation Process	The reservation of campus vehicles was done manually, often resulting in scheduling conflicts, incomplete records, and delays in vehicle allocation.	Developed an automated reservation feature that allowed users to view availability, submit requests, and receive instant booking confirmation through the TrackCSPC platform.
Lack of Real-Time Vehicle Tracking	Vehicles cannot be monitored once deployed, causing difficulty in tracking locations, ensuring safety, and managing schedules effectively.	Integrated real-time GPS tracking using both mobile GPS and Arduino-based GSM modules to allow administrators to monitor active trips.
Poor Data Management and Accessibility	Records of reservations and vehicle usage were stored on paper, making retrieval time-consuming and prone to loss or misplacement.	Implemented a cloud-based database through Firebase Realtime Database for centralized, secure, and easily accessible data storage
Limited Communication Between Users and Administrators	Coordination between drivers, requesters, and administrators was slow, often leading to miscommunication and delays.	Added an in-app notification system and automated alerts to improve communication and ensure timely updates on reservations and trip schedules.
Inefficient Resource Utilization	Vehicles were sometimes underused or double-booked due to the absence of a structured monitoring system.	Designed a digital dashboard that displayed real-time vehicle status, helping administrators allocate resources efficiently and reduce idle time.

4.1.5 Operational Feasibility

Operational feasibility is dependent on the people available for the project and it involves assessing whether the system will be used once it is deployed and implemented. It measures how well the proposed system solves the problem and how it capitalized on the opportunities identified in the stage of determining the scope, and how it met the needs identified in the evaluation phase of the requirements in developing the system. It also examined the readiness of the organization to support the suggested system [8]. In this section, the researcher's study and examine how the system handles and resolves the identified problem at Camarines Sur Polytechnic Colleges (CSPC).

Fishbone Diagram

A fishbone diagram, also known as an Ishikawa diagram or a cause-and-effect diagram, is a graphical representation that helps to categorize the possible causes of a problem. Its fundamental role is to trace the main sources of the problem by depicting how various elements result in a particular effect [9]. In the context of this study, the Fishbone Diagram helped uncover both major and underlying factors that contributed to the delays, scheduling conflicts, and disorganized processes within CSPC's manual vehicle reservation and tracking system.

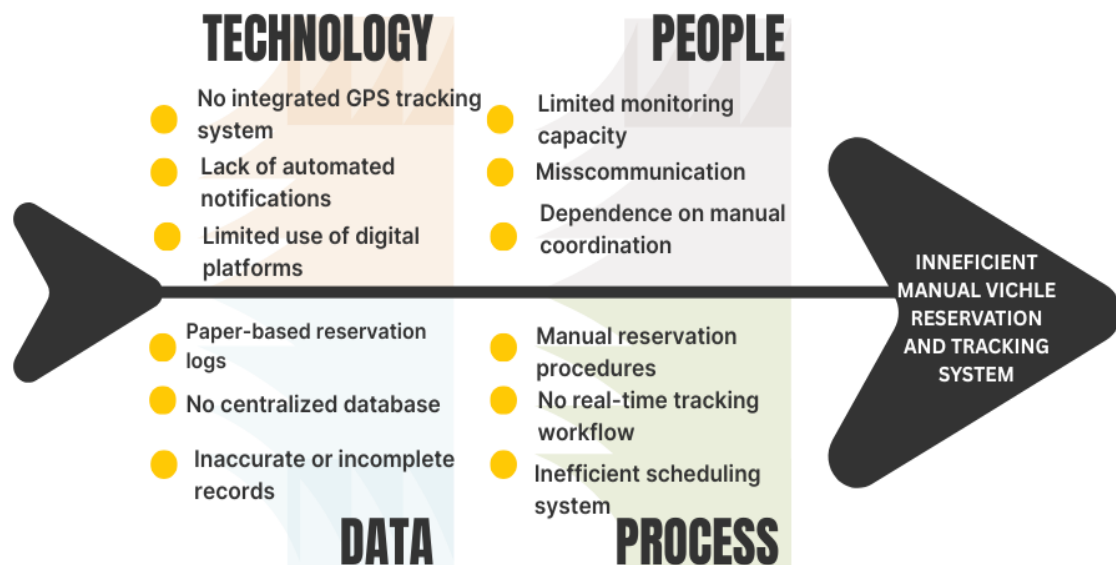


Figure 3. **Fishbone Diagram**

By understanding these root causes, the researchers were able to design the TrackCSPC platform, a smart campus vehicle tracking and reservation system that directly addresses these operational challenges.

Figure 3 displays the Fishbone Diagram used to determine the root causes of inefficient manual vehicle reservation and tracking system. It graphically depicts the most influential factors involved in the problems confronting the TrackCSPC, grouped into four broad categories: Technology, People, Data, and Process [7]. Under Technology, CSPC lacked digital tools that

could support modern transportation operations. There was no mobile or web-based platform for reserving vehicles, and the absence of a GPS-enabled tracking system made it impossible to monitor vehicle locations in real time. The campus also did not have an automated notification system, which made communication slow and prone to misunderstandings among users.

The People category reflected human-related challenges that contributed to the inefficiency of the manual system. Staff and drivers often depended on verbal coordination and handwritten logs, which led to miscommunication and forgotten reservations. Administrators also had limited capacity to monitor vehicle activities, making it difficult to manage multiple requests at the same time.

In terms of Data, the campus relied on scattered, paper-based reservation records that were easily misplaced, incomplete, or inaccurate. Without a centralized database, administrators struggled to verify vehicle availability, track previous bookings, or maintain organized transportation records. This often resulted in inconsistencies and overlapping reservations.

The Process category highlighted several operational inefficiencies in the existing workflow. All reservations were handled manually, which slowed down the approval process and increased the chances of double booking. The absence of a structured scheduling procedure and real-time tracking also made vehicle deployment harder to manage, leading to delays and uncoordinated transportation services.

Overall, the Fishbone Diagram showed that CSPC's transportation issues stemmed from interconnected problems involving technology, people, data, and processes. These combined weaknesses justified the development of TrackCSPC, a digital platform designed to automate reservations, provide real-time GPS tracking, organize vehicle data, and deliver a more reliable and efficient transportation system for the campus

4.1.6 Schedule Feasibility

Schedule feasibility analyzed the deadlines of the proposed project. This includes

estimating how long it takes for the team to finish the whole project. This has a big impact on an organization because the project's purpose or objective may fail if it is not finished on time [10]. Researchers established an organized schedule by specifying the main stage of the project and using Gantt chart, allocating amount of time for each task, and arranging activities based on their sequence.

Gantt Chart

A Gantt chart functions as a basic project management instrument which assists both planning and task monitoring. The chart displays tasks with duration bars which indicate both start and end dates for each activity [11]. The project progress becomes visible through this method while the upcoming tasks remain clear for execution.

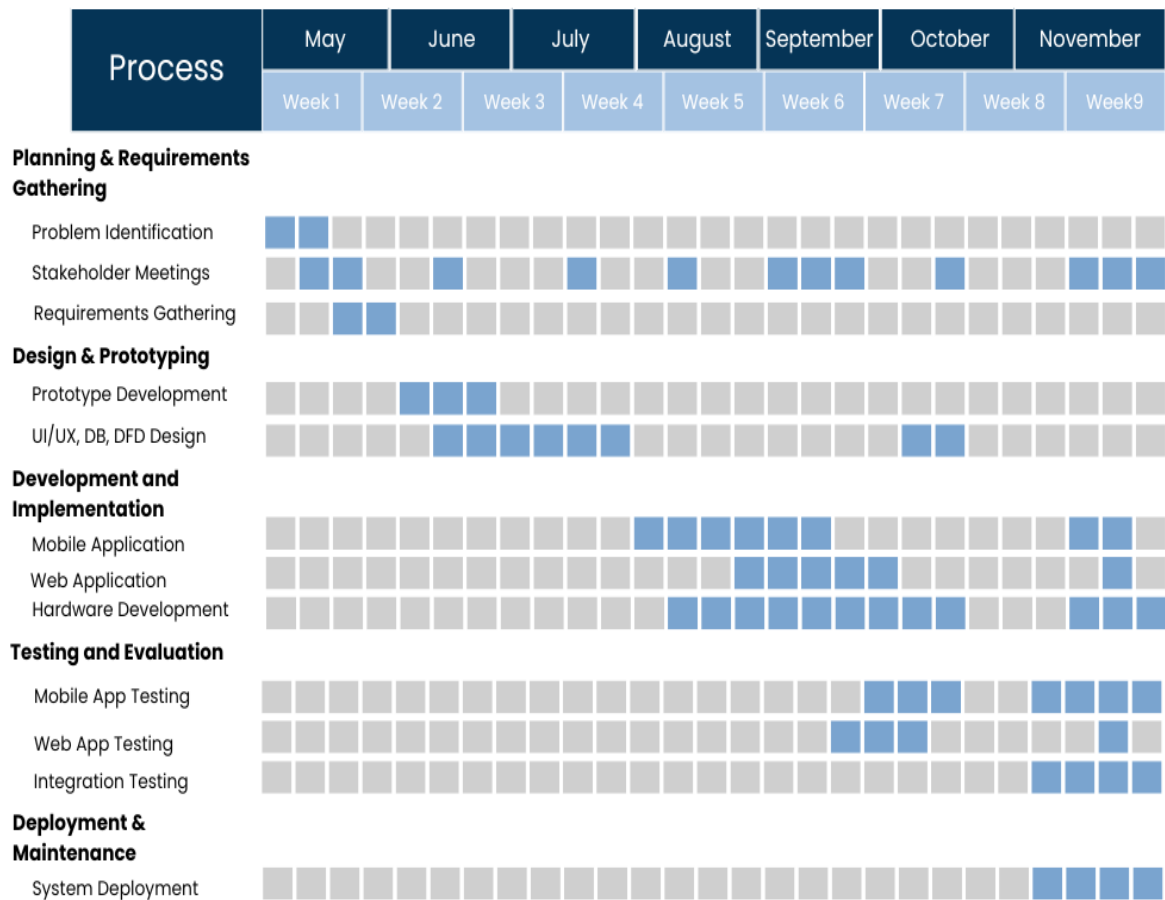


Figure 4. **Gantt Chart**

Figure 4 presents the Gantt Chart for the development of TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform. It is a graphical representation of the project timeline and progress through phases of development [5]. The chart outlines the sequence and duration of major tasks from May to November, covering the Planning & Requirements Gathering, Design & Prototyping, Development and Implementation, Testing and Evaluation, and Deployment & Maintenance phases.

In the Planning & Requirements Gathering Phase, the researchers conducted initial consultations with the General Services Unit (GSU), interviewed potential users, and assessed the existing vehicle reservation and tracking procedures.

The Design & Prototyping Phase involved producing early mock-ups, creating wireframes for the web and mobile interfaces, and drafting the database schema and data flow diagrams (DFDs). These prototypes were repeatedly reviewed and refined based on stakeholder feedback, ensuring that the visual structure and functional flow aligned with user expectations and system goals.

The Development & Implementation Phase was the most extensive stage, during which system components were coded, integrated, and enhanced across multiple iterations. Features such as the vehicle request form, reservation approval workflow, driver and vehicle management modules, and GPS tracking interface were gradually built and improved. Each cycle allowed the team to enhance system performance and address issues as development progressed.

The Testing and Evaluation Phase followed each iteration, where various testing activities such as functionality testing, integration testing, performance testing, and usability evaluation were conducted to verify system reliability and correctness.

After several cycles of refinement, the Deployment & Maintenance Phase marked the completion of the development process. At this point, the system was prepared for deployment within the institution, enabling pilot implementation and actual use by GSU staff and end-users.

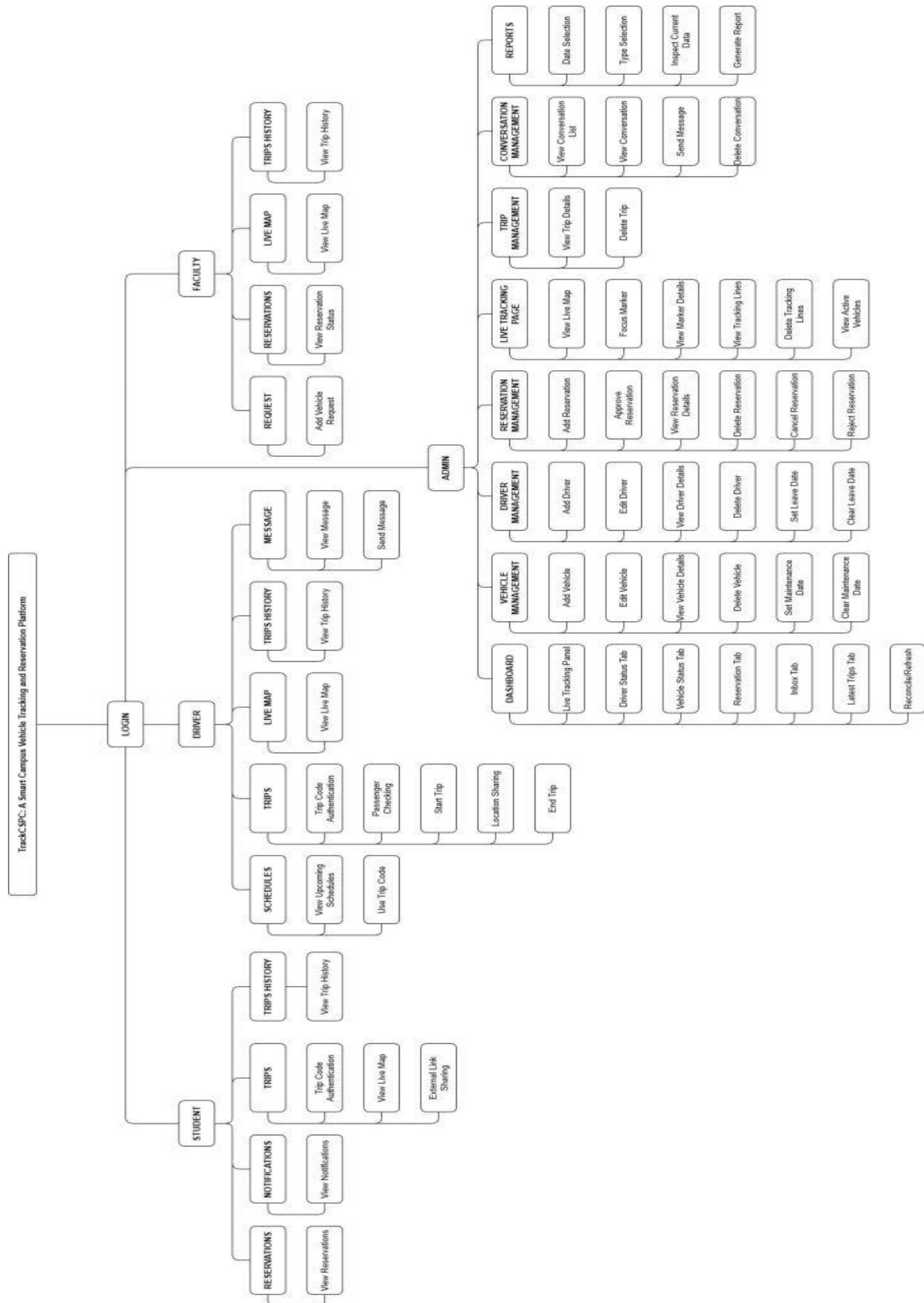


Figure 5. **Functional Decomposition Diagram**

Figure 5 illustrates the Functional Decomposition Diagram (FDD) of TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform, which breaks down the system into structured functional components to clearly show how each module supports the overall operation of the platform. The system begins with the Login Module, serving as the common entry point for all user roles—Student, Driver, Faculty, and Admin—where access is granted based on successful authentication and assigned privileges. Students are provided with features such as viewing reservations, notifications, trip details, live maps, trip history, trip code authentication, and shared trip links for convenient monitoring. Drivers are equipped with essential trip management functions, including viewing schedules, authenticating trips, checking passengers, starting and ending trips, sharing real-time location, accessing live maps and trip history, and exchanging messages. Faculty users can request vehicles, track reservation statuses, monitor live trips through maps, and review trip history to support campus-related travel needs. The Admin role encompasses comprehensive system control, allowing administrators to manage drivers, vehicles, reservations, trips, and communications; monitor active trips and live tracking; process requests; generate reports; set maintenance schedules; update vehicle status; and refresh dashboard data to ensure accurate and real-time system monitoring.

4.2 Design

The design phase for TrackCSPC was a crucial step in transforming the initial requirements into a functional and user-friendly vehicle tracking and reservation platform. The team meticulously crafted system models and user interface designs to illustrate the system's operation and user interactions. These models, such as Data Flow Diagrams and Use Case Diagrams were employed to visually represent internal processes, data flow, and user-system interactions, ensuring clear definition and validation of functional requirements. Role-based interface mockups were developed to provide tailored experiences for GSU administrators, drivers, and student/faculty requesters, enhancing security and streamlining workflows.

The design prioritized usability, accessibility, and operational efficiency, enabling administrators to manage requests, monitor availability, and track vehicles in real-time, while drivers accessed trip details and live tracking, and requesters submitted reservations and tracked their status. The accompanying diagrams provided detailed explanations of the system's architecture, functional components, and interface layouts, ultimately providing a clear structural foundation for building the TrackCSPC system.

4.2.1 Data Flow Diagram

The Data Flow Diagram (DFD) served as a fundamental analysis and design tool used to illustrate and describe the flow of data within the TrackCSPC system. It provided a visual representation of the major processes, the movement of data between them, and the interactions between the system and external entities such as users or hardware components. Through the use of DFDs, developers and analysts were able to clearly understand how information was processed, stored, and transmitted across the system, ensuring accuracy, efficiency, and consistency in the overall design. This method helped break down complex operations into smaller, manageable components, facilitated better communication among stakeholders, and supported the early identification of areas that required improvement.

Figure 6 showed the Context Diagram Data Flow Diagram of TrackCSPC and provided an overview of the system as a single main process interacting with four external entities: Admin, Faculty, Student, and Driver. This level illustrated the general flow of information, including reservation details, trip codes, driver updates, and real-time location data exchanged between users and the system. The diagram offered a high-level view of how TrackCSPC operated, providing a clear understanding of the system's overall functionality before presenting more detailed sub-processes in subsequent levels. It helped the researchers identify the key interactions and data exchanges necessary for efficient campus transportation management. These diagrams also made it easier for both technical and non-technical stakeholders to visualize how the system works.

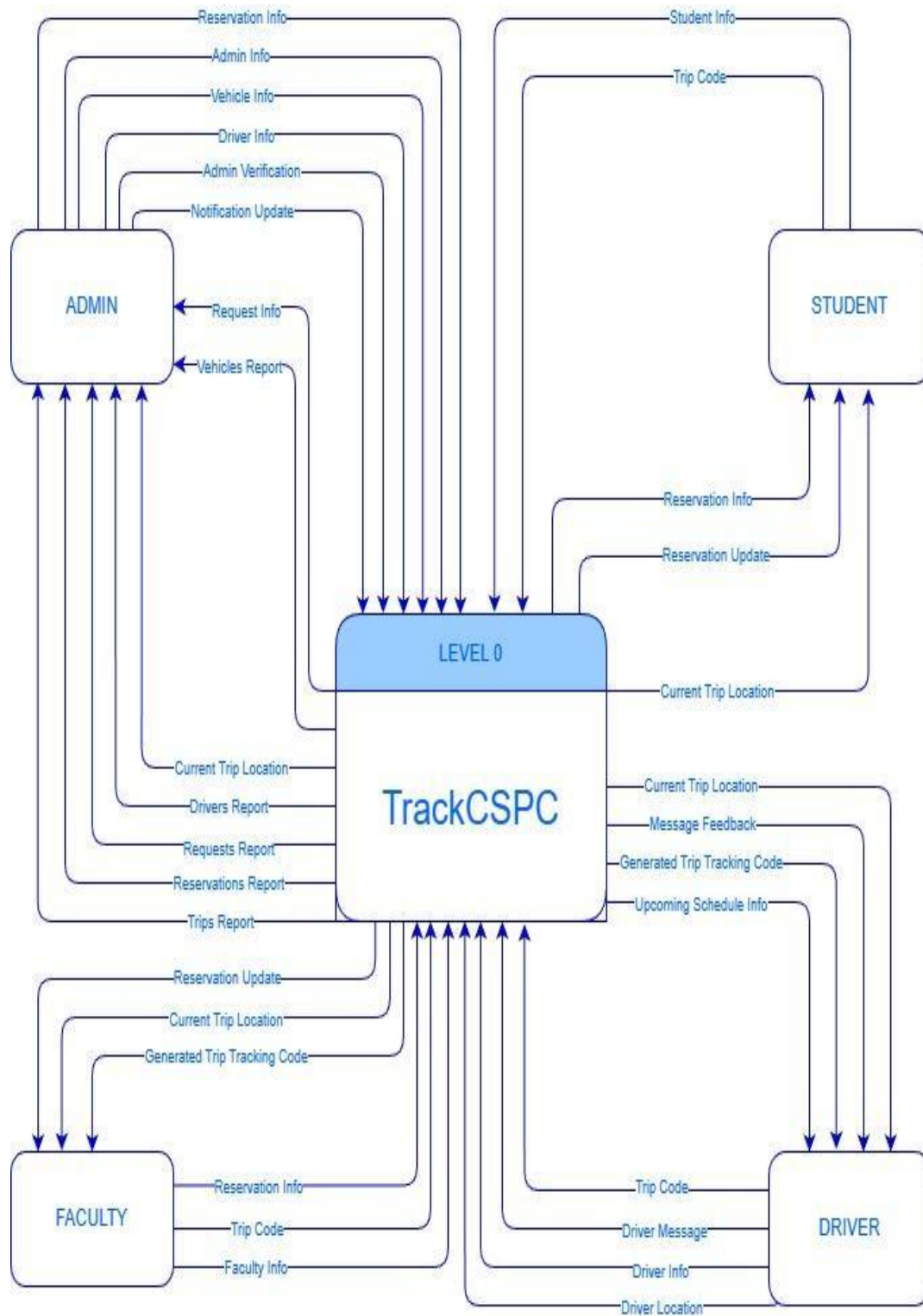


Figure 6. Context Diagram

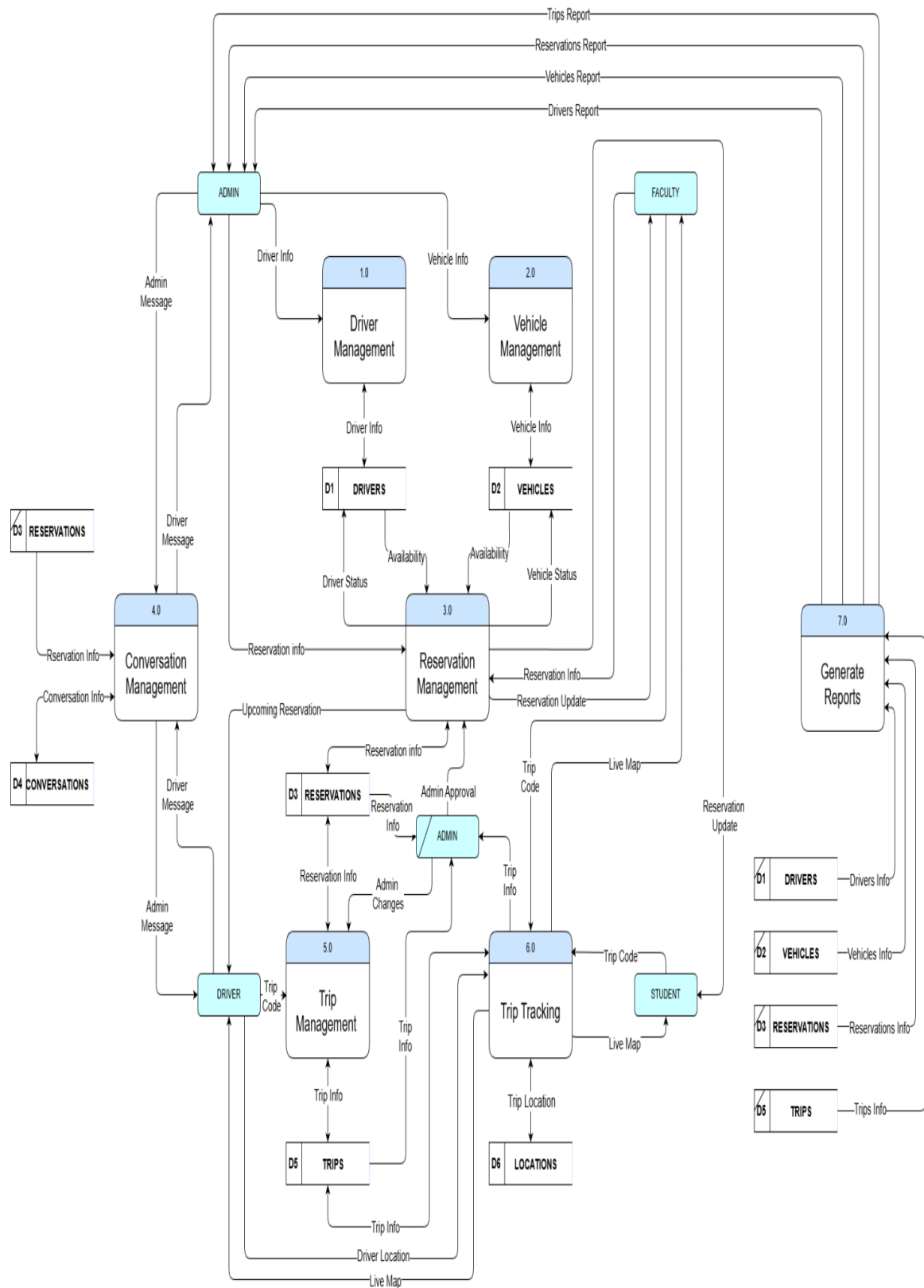


Figure 7. **Data Flow Diagram Level 0**

Figure 7 presents the Level 0 Data Flow Diagram of TrackCSPC, showing the breakdown of the system into its major interconnected processes. These include (1.0) Driver Management, (2.0) Vehicle Management, (3.0) Reservation Management, (4.0) Conversation Management, (5.0) Trip Management, (6.0) Trip Tracking, and (7.0) Generate Reports. Each module handles a specific function essential to the operation of the campus vehicle tracking and reservation platform.

Driver Management (1.0) handles the creation, updating, and monitoring of driver records. Vehicle Management (2.0) maintains vehicle information, availability, and status. Reservation Management (3.0) processes reservation requests from faculty and students, handles admin approval, and updates reservation records. Conversation Management (4.0) supports communication between the admin and drivers. Trip Management (5.0) generates and updates trip details, while Trip Tracking (6.0) monitors the live location of vehicles using the trip code. Lastly, Generate Reports (7.0) compiles data on drivers, vehicles, reservations, and trips for administrative evaluation.

This diagram illustrates how information flowed among users, processes, and data stores, providing a clearer understanding of the system's operations. It ensured transparency in how reservations were handled, trips were monitored, and reports were produced, supporting the overall goals of efficient vehicle management, real-time monitoring, and smooth coordination among all TrackCSPC users. Additionally, it allowed the researchers to identify potential bottlenecks or gaps in information flow. The diagram also helped validate the logical structure of the system by showing clear relationships between processes and data stores. It served as a useful reference during system development and testing to ensure consistency in data handling. It also helped ensure that each process was not working in isolation but was properly connected to other system components, creating a smooth and coordinated flow of operations. In this way, the diagram made the overall system easier to understand, manage, and improve throughout the development process. Moreover, it provided a practical guide for developers and administrators to anticipate real-world scenarios.

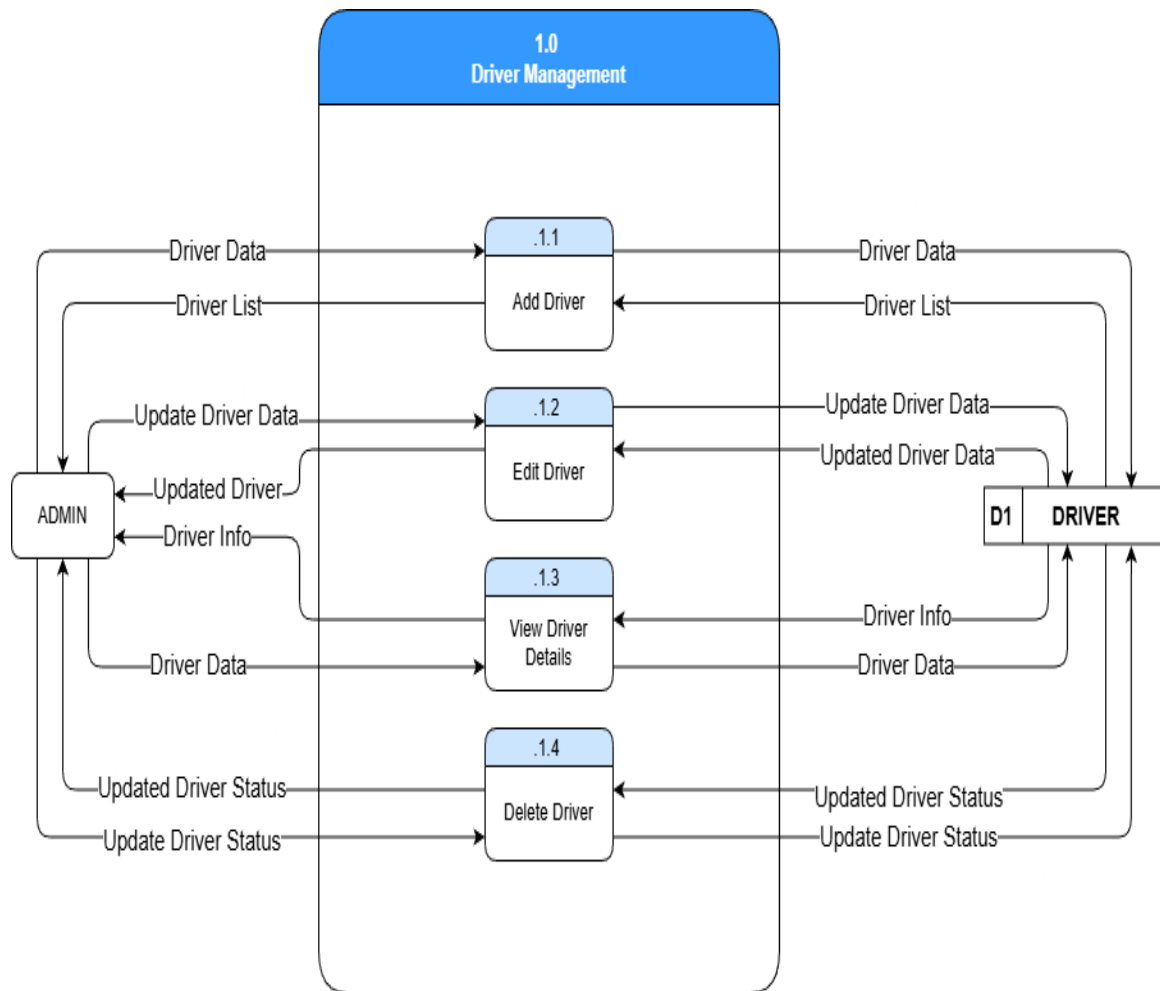


Figure 8. **DFD Level 1 Process 1.0 (Driver Management)**

Figure 8 presents the Driver Management module, which allows administrators to add, view, update, and delete driver records within the TrackCSPC system. This module ensures that all driver information including driver details, status, and availability is properly organized and updated whenever changes occur. Administrators can register new drivers, modify existing profiles, review driver data, and remove outdated records.

The module keeps driver information consistent and accessible, helping the system maintain accurate trip assignments and reliable fleet operations. By centralizing all driver-related processes, the Driver Management module supports efficient coordination, reduces administrative errors, and ensures that only verified drivers participate in campus transportation activities.

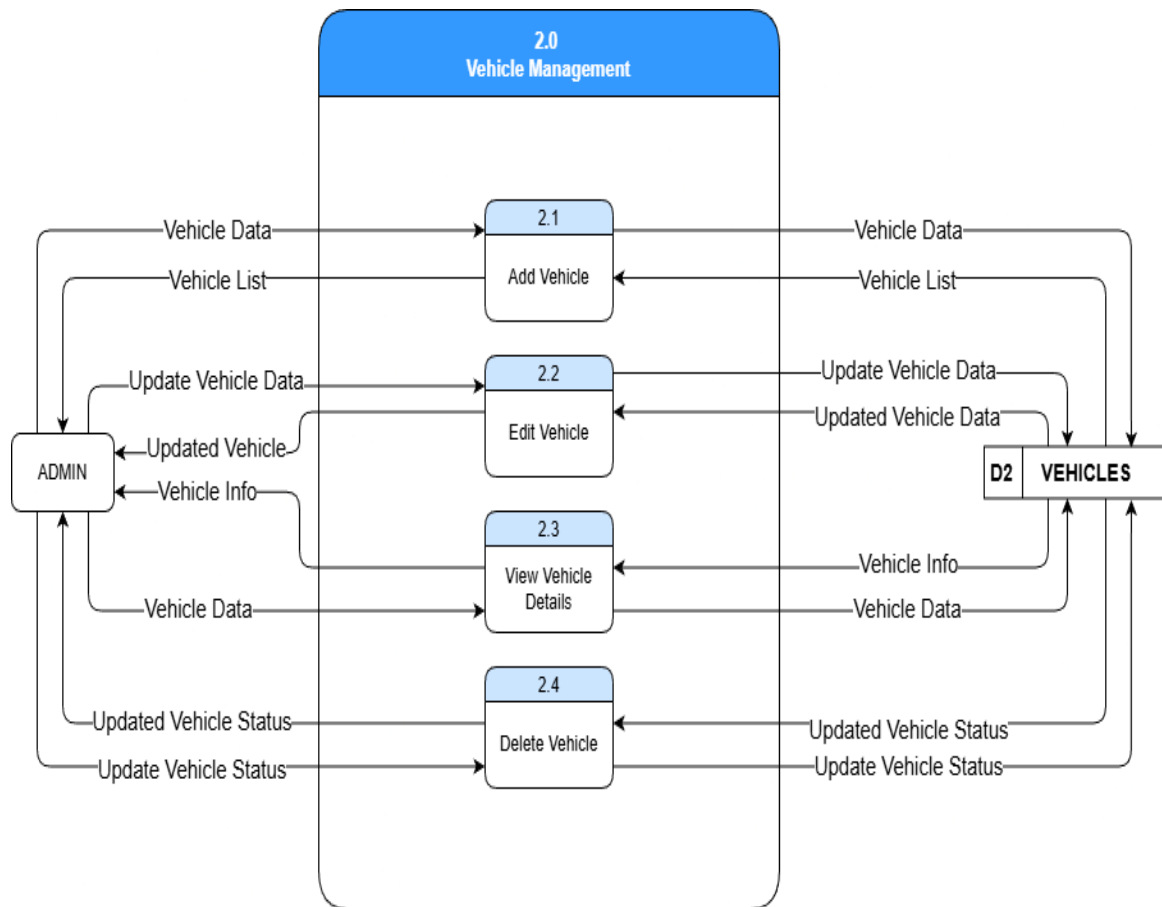


Figure 9. **DFD Level 1 Process 2.0 (Vehicle Management)**

As shown in Figure 9, the Vehicle Management module handles the organization and maintenance of all vehicle records in the system. It allows the admin to add new vehicles, update existing vehicle details, view complete vehicle information, and remove vehicles when necessary. Each process ensures that vehicle data remains accurate and up to date. By managing all vehicle information in one module, the system maintains efficiency, consistency, and proper tracking of all campus vehicles.

The module enables the admin to input essential vehicle information such as vehicle type, plate number, model, and assigned driver which is then stored and managed in the vehicle database. Whenever changes occur, such as updates to vehicle status or modifications to vehicle details, the admin can easily edit the information to keep the system current and reliable.

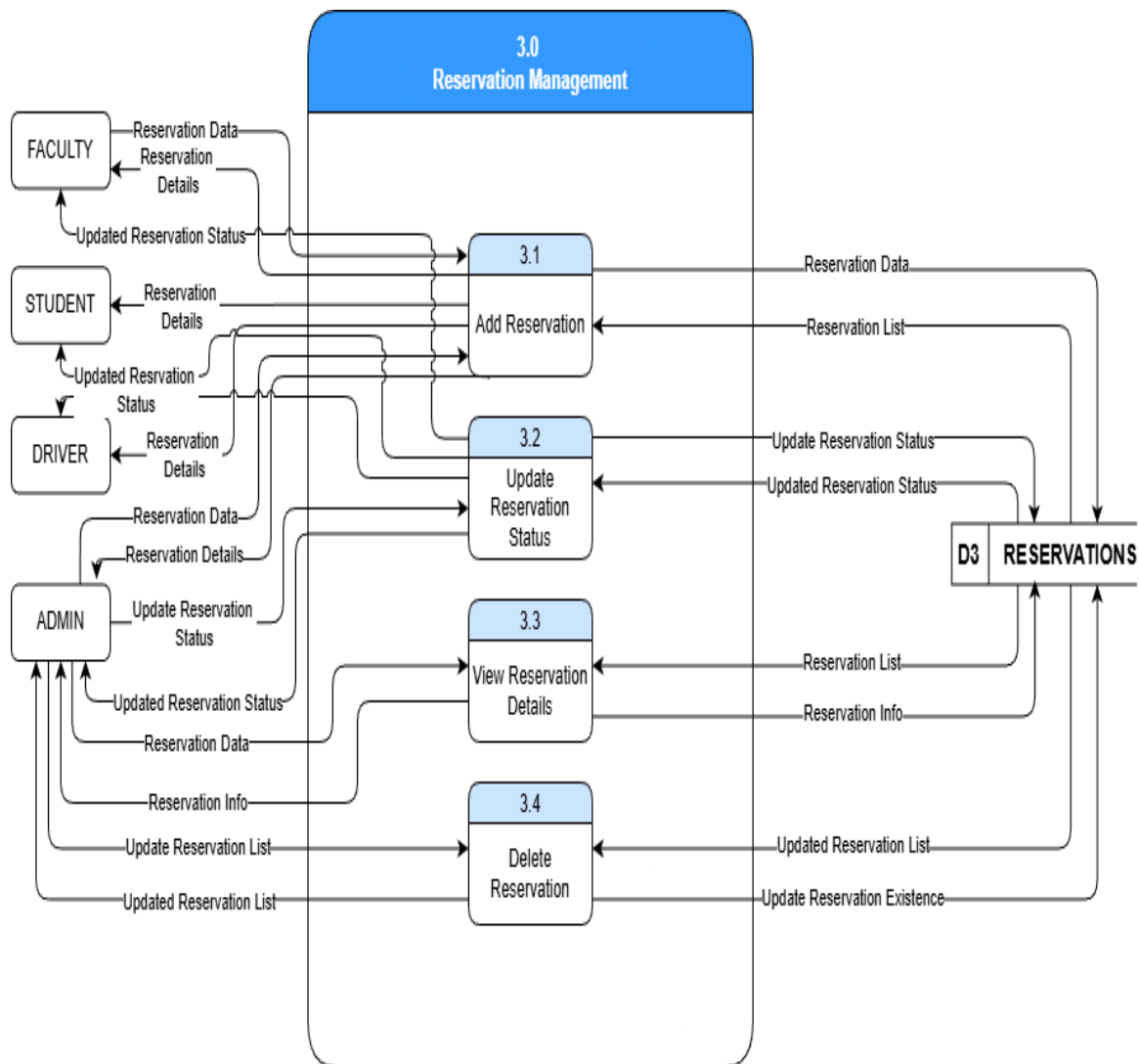


Figure 10. **DFD Level 1 Process 3.0 (Reservation Management)**

As shown in Figure 10, the Reservation Management module handles all processes related to booking and managing vehicle reservations. Through this module, faculty, students, drivers, and administrators can submit reservation requests, update reservation status, view reservation details, and delete reservations when necessary. All reservation data is stored and organized in a central database, allowing quick access to updated schedules and booking information. This centralized system helps prevent conflicts, ensures accurate tracking of reservation status, and supports efficient coordination between users and administrators.

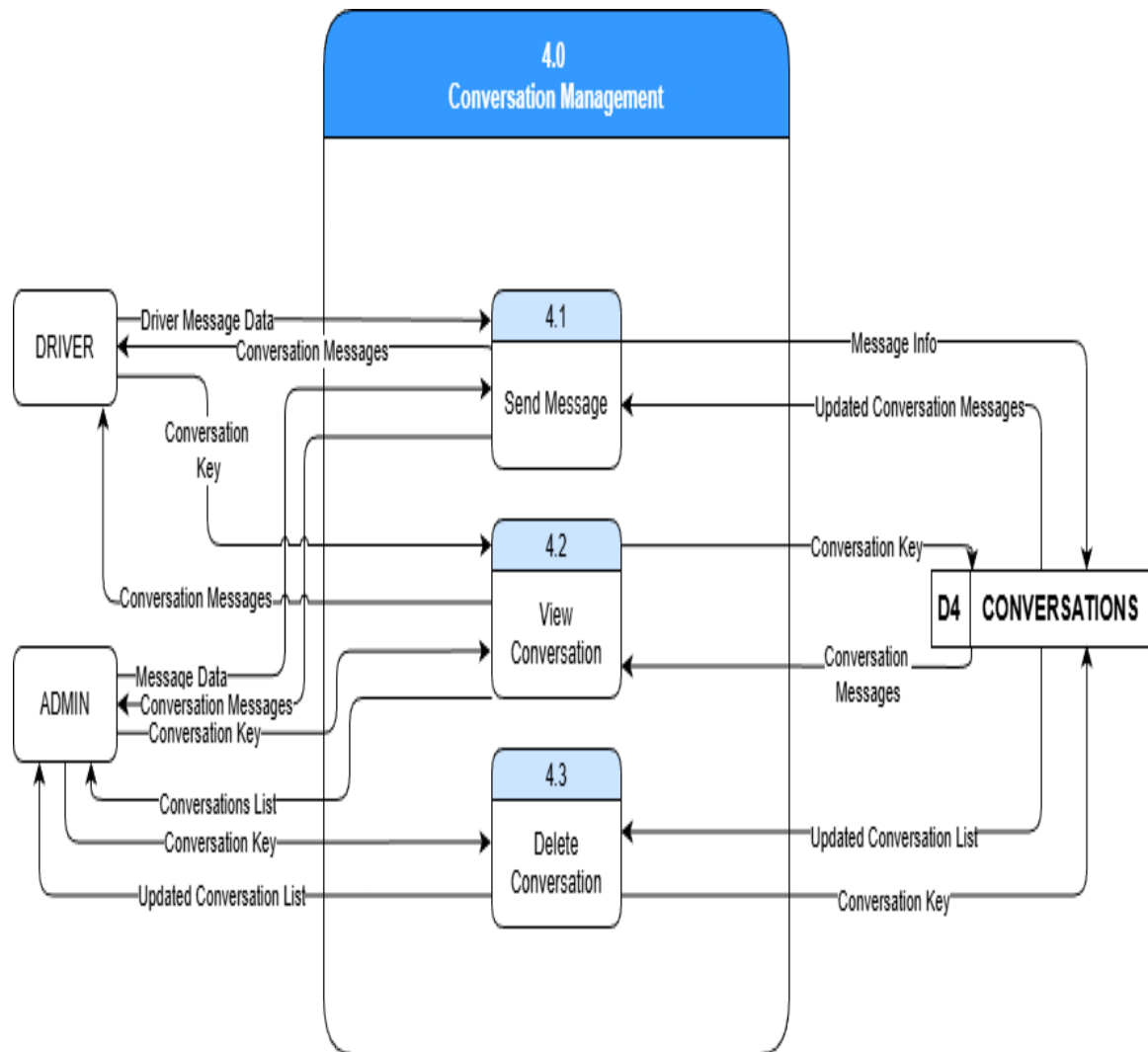


Figure 11. **DFD Level 1 Process 4.0 (Conversation Management)**

As shown in figure 11, the Conversation Management module enables both drivers and administrators to send, view, and manage conversation threads efficiently. Through this module, users can exchange messages in real time, ensuring clear communication between the driver and the admin. The system provides organized access to conversation histories, allowing users to review past messages using the assigned conversation key. Additionally, the module allows the deletion of entire conversation threads when needed, ensuring that only relevant and updated communication records are maintained. Every action sending, viewing, or deleting messages updates the centralized Conversations data store, guaranteeing consistency across the system.

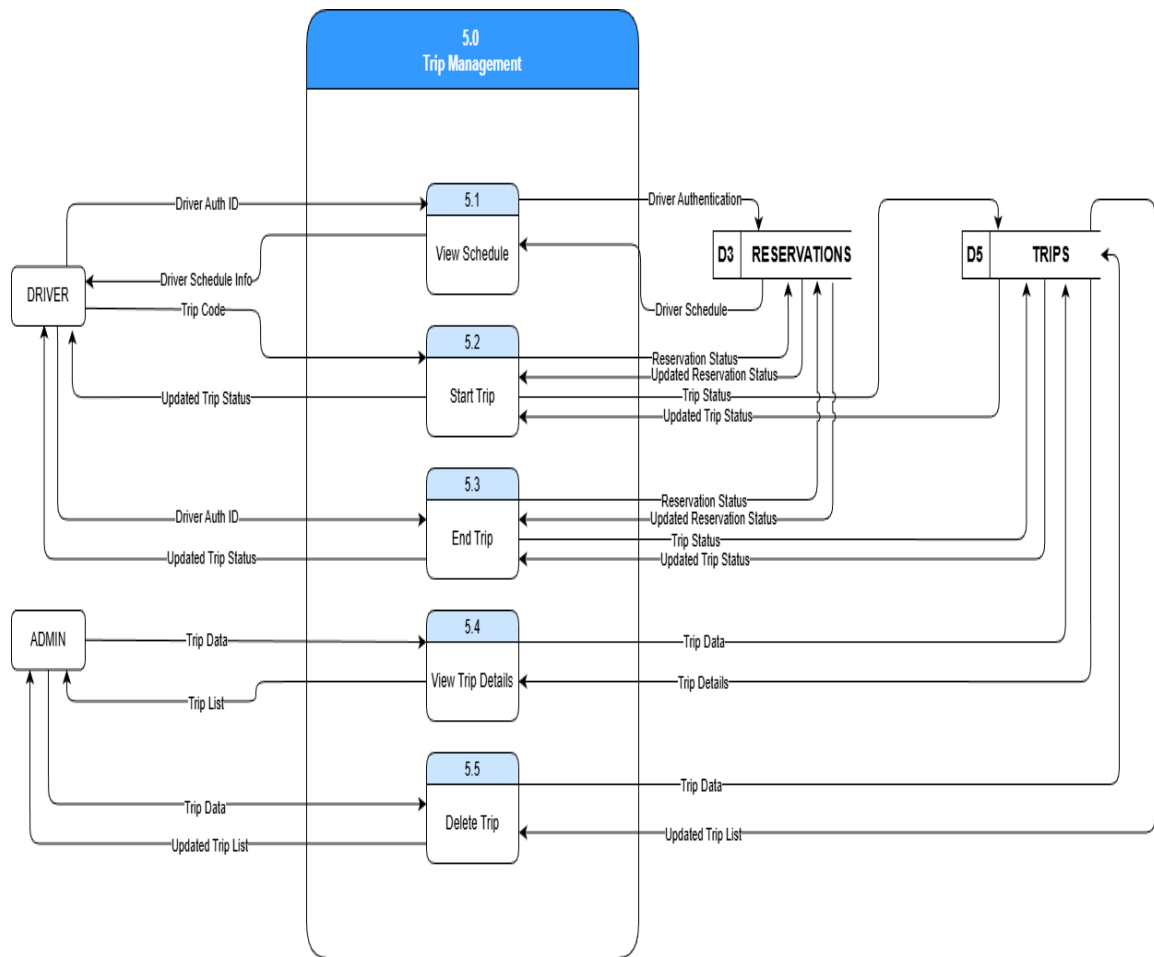


Figure 12. **DFD Level 1 Process 5.0 (Trip Management)**

Figure 12 illustrated the Trip Management module, which handled all trip-related activities for both drivers and administrators. This module allowed drivers to view their schedules, start assigned trips, and end trips once completed. It ensured that all trip records, including trip status, reservation status, and schedule information, were accurately updated in the system. Administrators were also able to view detailed trip information and delete trips when necessary, helping maintain an organized and up-to-date trip database. Through continuous synchronization with the Reservations and Trips data stores, the module kept the system accurate by reflecting changes such as updated trip statuses, reservation updates, and detailed trip data. Additionally, this process supported efficient coordination between drivers and administrators, contributing to a smoother and more reliable campus transportation experience.

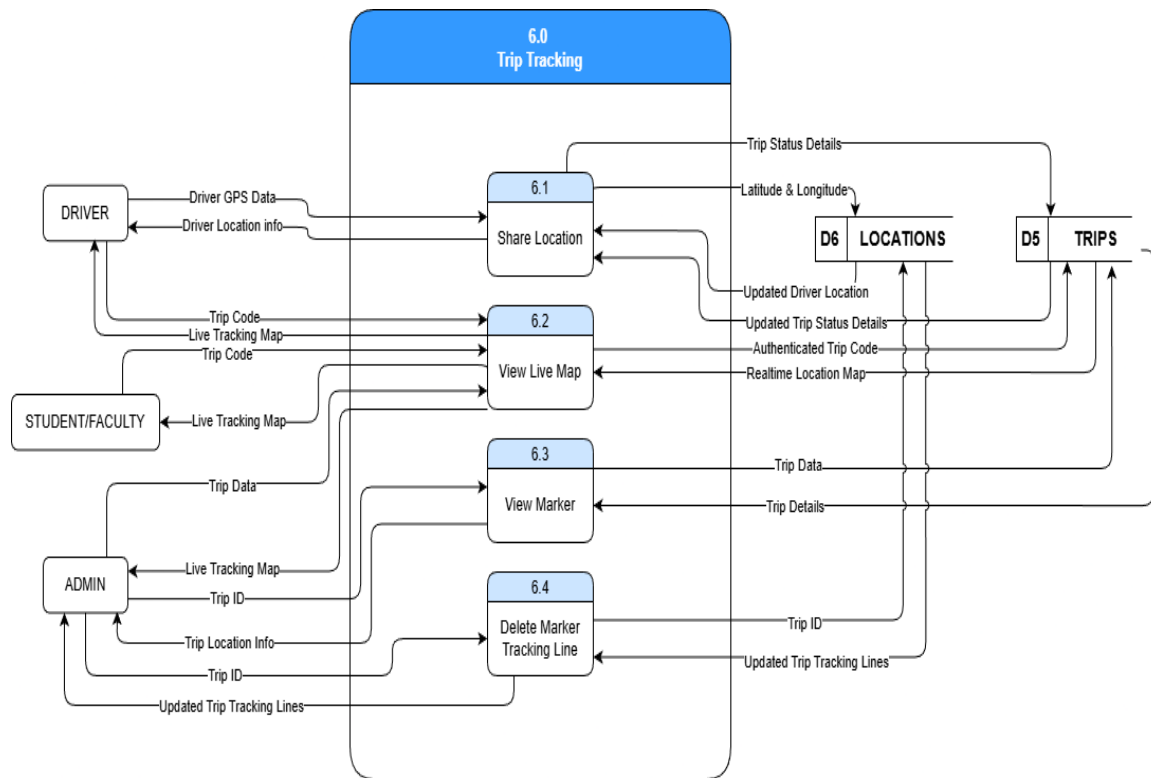


Figure 13. **DFD Level 1 Process 6.0 (Trip Tracking)**

Figure 13 illustrated the Trip Tracking module, which handled the real-time monitoring of trips for drivers, students, faculty, and administrators. The module captured the driver's live GPS location and continuously updated the system with accurate coordinates, allowing users to follow the vehicle's movement through a live map. Students and faculty were able to track their assigned trips as they occurred, while administrators monitored map markers and managed tracking lines to keep route information well organized. The module synchronized with the Locations and Trips data stores to ensure that trip statuses, validated trip codes, and location details remained current. It also reduced uncertainty by giving users timely visibility of vehicle movements throughout the trip. Administrators benefited from improved oversight, as they could quickly identify active, completed, or delayed trips. The clear presentation of routes and locations supported better decision-making during operations. Overall, this functionality enhanced efficiency and safety by providing accurate, transparent, and reliable trip monitoring.

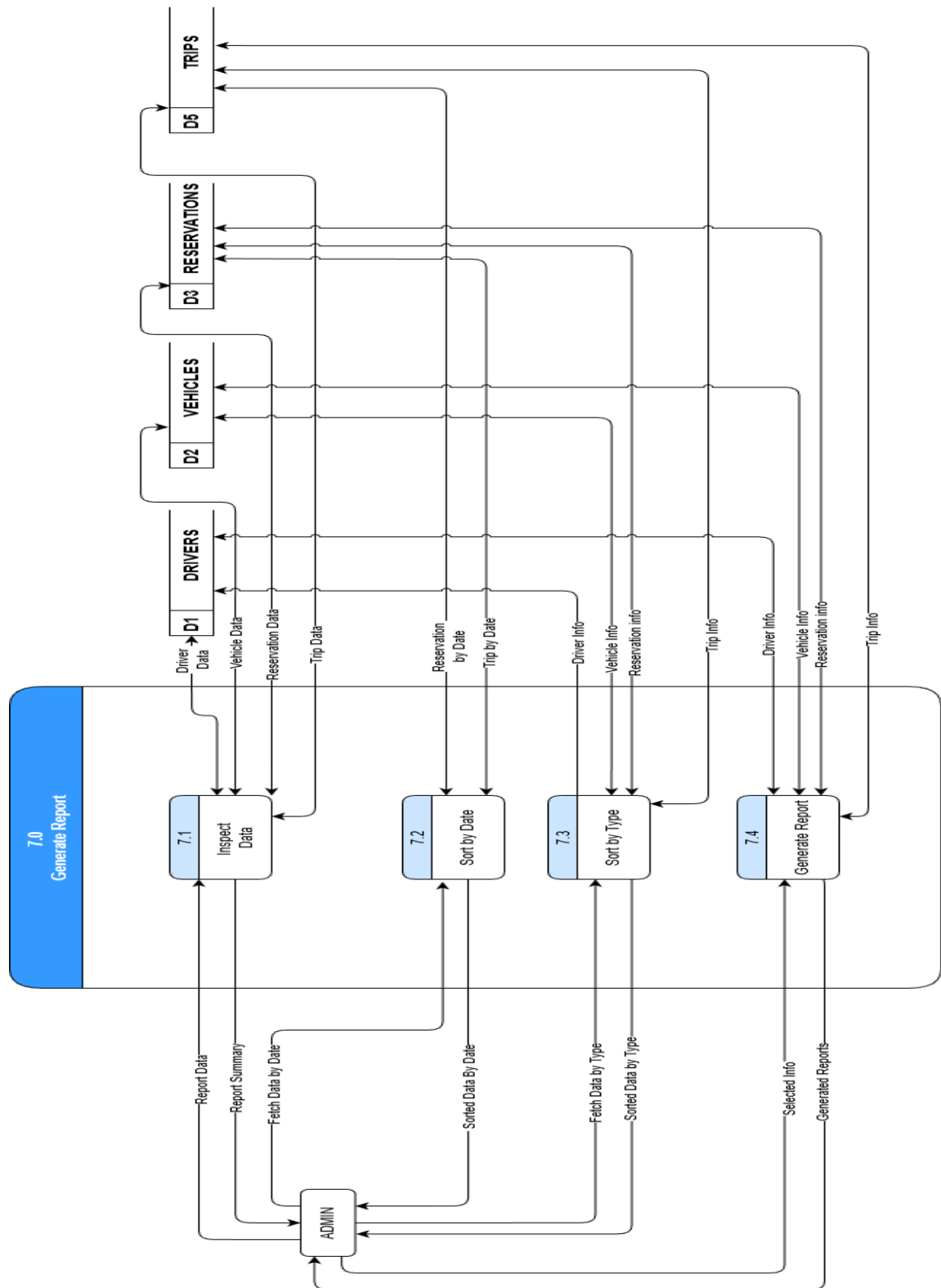


Figure 14. **DFD Level 1 Process 7.0 (Generate Report)**

Figure 14 illustrates the Generate Reports module of the TrackCSPC system, which allows administrators to inspect, sort, and generate detailed reports related to campus vehicle operations. This module retrieves data from the Drivers, Vehicles, Reservations, and Trips data stores, ensuring that all information used in the report is accurate and up to date. Within this process, the administrator can filter data by date, type, or specific operational details. The system organizes the information accordingly, allowing the admin to review sorted records before generating the final report. These reports may include driver activity, vehicle availability, reservation summaries, and trip histories. By digitizing the reporting process, TrackCSPC enhances administrative efficiency, reduces manual work, and provides faster access to essential operational insights.

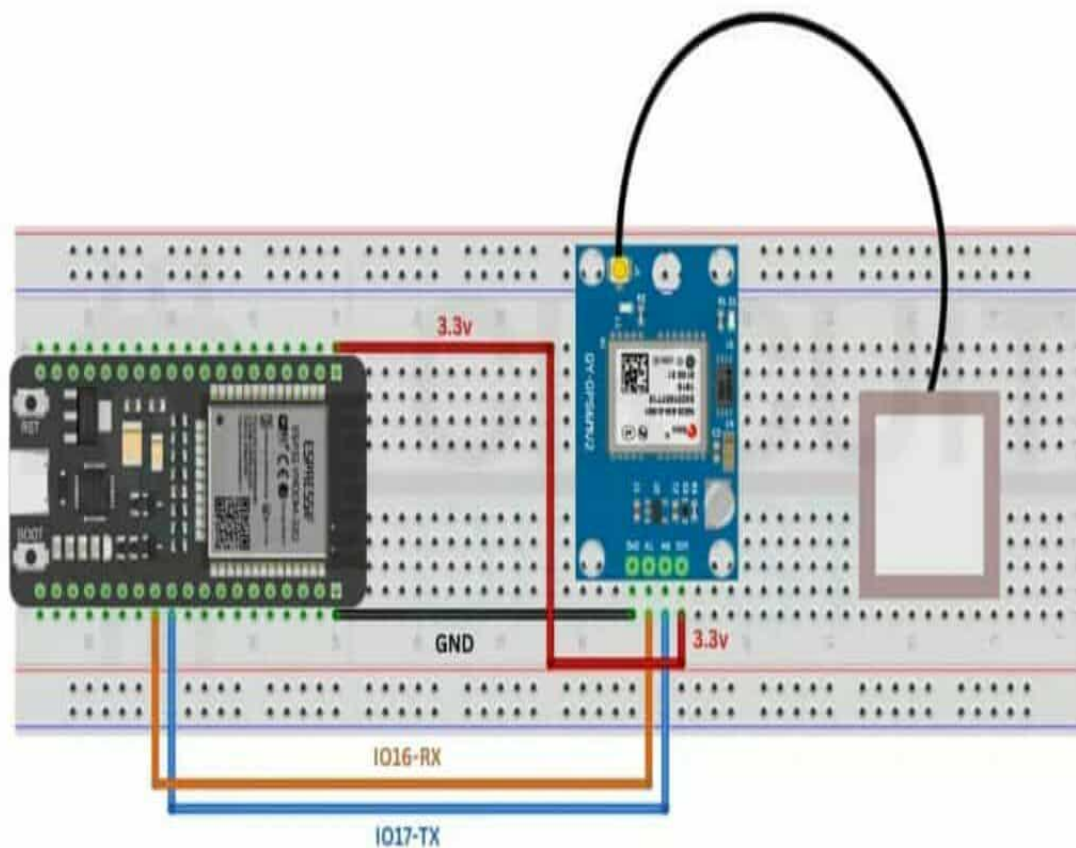


Figure 15. GPS Circuit

Figure 15 illustrates the hardware connection of the ESP32 microcontroller with a GPS module and an external GPS antenna mounted on a breadboard for prototyping purposes. The ESP32 serves as the main controller that receives, processes, and transmits GPS location data. The GPS module is powered through the 3.3V output of the ESP32 and shares a common ground connection to ensure stable operation. Serial communication is established using UART pins, where IO16 (RX) of the ESP32 is connected to the TX pin of the GPS module and IO17 (TX) is connected to the RX pin, allowing bidirectional data transmission. An external GPS antenna is attached to the module to enhance satellite signal reception and improve positioning accuracy. This circuit configuration enables continuous acquisition of geographic coordinates, which are transmitted to the TrackCSPC platform for real-time vehicle tracking and monitoring.

4.2.2 Use Case Diagram

A Use Case Diagram was an essential component of system design that graphically showed how various users interacted with the system. It was used to identify the functional requirements of the system by illustrating relationships between users (actors) and the tasks (use cases) they could perform. The diagram provided a clear and concise overview of user interactions, allowing developers and stakeholders to understand what the system needed to accomplish from the users' point of view. It also promoted effective communication throughout the development process, ensuring that each function was aligned with user needs, requirements, and the overall goals and objectives of the TRACKCSPC platform.

The Use Case Diagram played a crucial role in bridging the gap between the conceptual design and the actual implementation of the TRACKCSPC system by illustrating user interactions. This diagram allowed for the early identification of duplicate, missing, or ambiguous features, which was essential for refining the system before the coding process began. Its visual nature supported a user-centric approach by considering the roles and responsibilities of every user involved in the platform.

Figure 16 shows the Use Case Diagram for TRACKCSPC: A Smart Campus Vehicle Tracking and Reservation Platform. It provided a clear overview of the system's primary actors Students, Drivers, and General Services Unit (GSU) Administrators—along with their interactions with the system's core modules.

In the TrackCSPC system, the GSU Administrator held the highest level of access, enabling them to manage vehicles, drivers, reservations, and system users. Administrators also oversaw real-time vehicle monitoring, handled request approvals, and ensured the proper coordination of system operations across the campus. Drivers were able to update their trip status, send location data through the mobile app, and receive instructions from GSU. Meanwhile, the Student User could log in to submit vehicle reservation requests, view the status of their requests, check available vehicles, and receive notifications about approvals or updates. Each user role accessed only the features relevant to their responsibilities, ensuring security, efficiency, and smooth operation of the vehicle tracking and reservation process.

All the interaction between the users cautiously nurtures through the process of login, present role-based access control so that the sensitive information and operations of the office are not compromised. Comprehensively, the diagram presents the system's focus on accessibility, effectiveness, and safety in facilitating the vehicle tracking and reservation processes across the campus.

The Use Case Diagram served as a practical tool for both developers and stakeholders to visualize how each user interacts with the system, making it easier to identify potential gaps or overlaps in functionality. By clearly defining the roles and responsibilities of Students, Drivers, and GSU Administrators, the diagram helped ensure that every action within the system was purposeful and streamlined, reducing confusion and minimizing errors during daily operations. This level of clarity promoted better communication among the development team, campus staff, and end-users, as everyone could see how their tasks fit into the broader workflow of TrackCSPC.

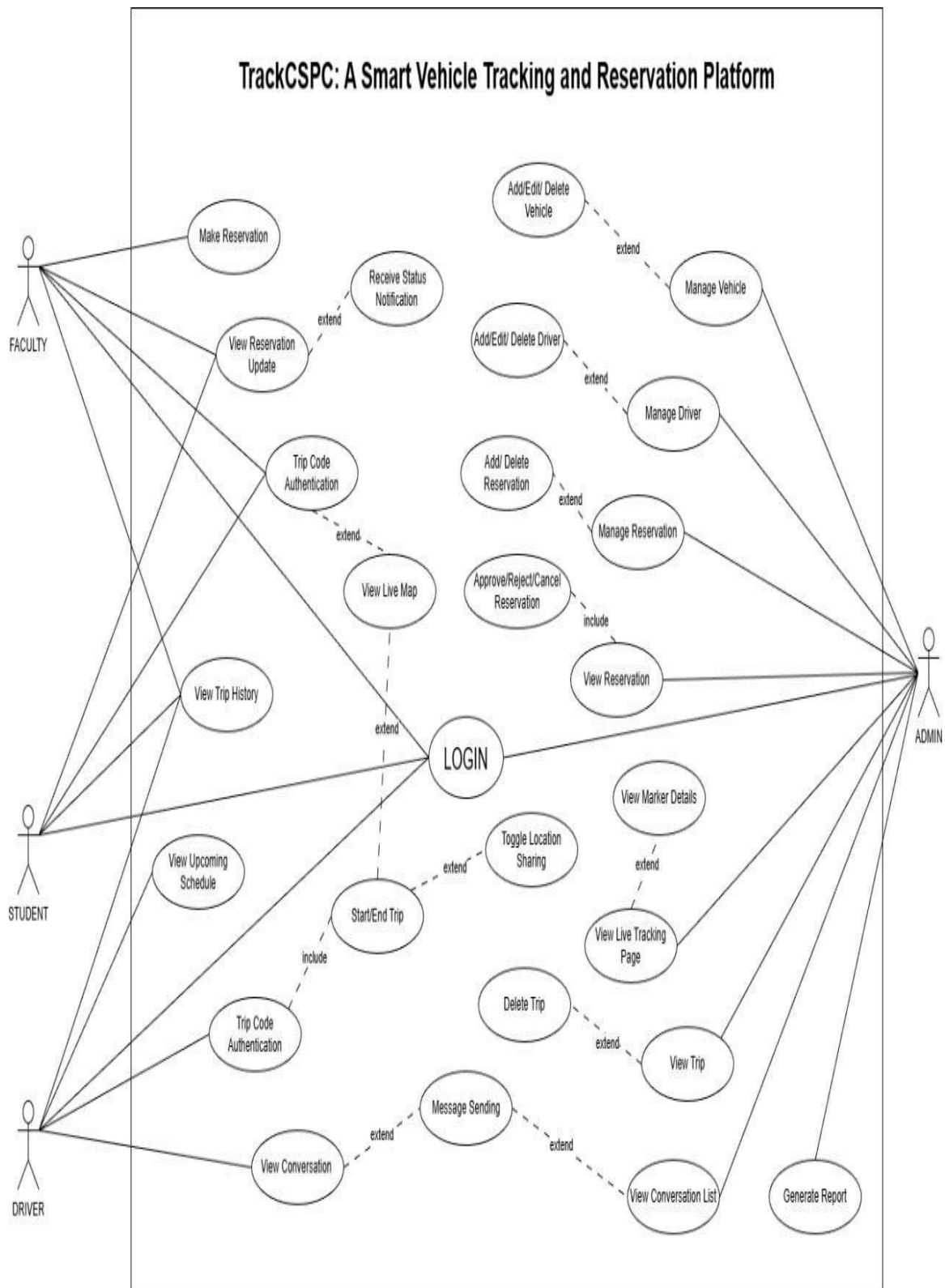


Figure 16. Use Case Diagram

Figure 16 shows the Use Case Diagram for TrackCSPC: A Smart Vehicle Tracking and Reservation Platform. It illustrates four primary actors—Admin, Faculty, Student, and Driver each with specific system interactions. All actors begin with the Login process, which enforces secure, role-based access.

The Faculty and Student actors share similar capabilities, including making a reservation, viewing reservation status updates, authenticating trips through a trip code, tracking vehicles through the live map, and viewing their trip history. They also receive automated status notifications.

The Driver plays a crucial operational role. Drivers can authenticate trips, start and end trips, perform passenger checking, share their live location, and view trip history. They can also send and receive messages through the conversation module, ensuring smooth communication with administrators during active trips.

The Admin actor has the most extensive system access. Admins can add, edit, and delete vehicles and drivers, manage reservations, approve, reject, or cancel trip requests, view detailed marker information on the live tracking page, remove trips, view and manage conversations, and generate system reports. These capabilities allow administrators to oversee the entire vehicle reservation process, ensuring accuracy, safety, and efficient tracking of all campus trips.

Overall, the Use Case Diagram demonstrates how TrackCSPC distributes responsibilities across its actors while ensuring secure and efficient system usage. By clearly defining each user's role and permitted interactions, the diagram supports the system's goals of improving reservation accuracy, enhancing communication, and enabling real-time vehicle tracking within the campus.

4.3 Development

Development tasks were performed iteratively throughout the creation of the TrackCSPC platform. Coding began based on the approved system specifications, ensuring that all features aligned with the project's goal of improving campus vehicle tracking and reservation management.

Collaboration among the developers helped maintain consistency and efficiency in every cycle. Each iteration ended with a review to confirm that the system met the required functionality and performance standards. This phase focused on integrating the web-based admin system with the mobile application, implementing key features such as real-time tracking, reservation processing, trip authentication, and notification services. Through this iterative approach, the system was developed into a cohesive, reliable, and user-friendly platform.

4.3.1 User Interface

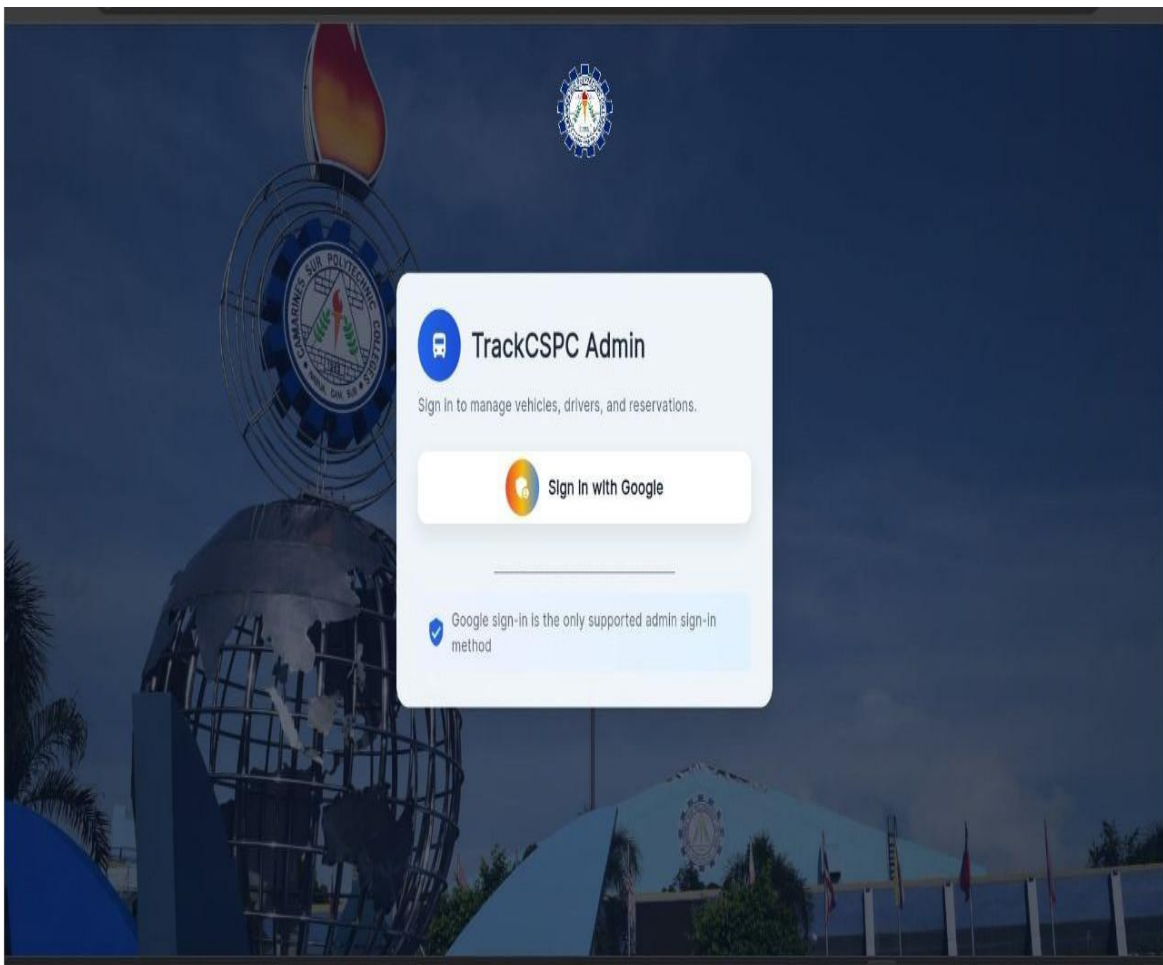


Figure 17. **User Interface**

The landing page serves as the system's primary access point for administrators, providing a clear and streamlined interface for secure authentication. As shown in the figure, the TrackCSPC Admin portal presents a minimalist design that highlights its main function—allowing authorized

personnel to sign in to manage vehicles, drivers, and reservation requests. The page uses Google Sign-In as the exclusive authentication method, ensuring both convenience and enhanced security. This entry interface directs users to the system's administrative features while maintaining a simple and user-friendly layout that supports efficient navigation and accessibility.

Figure 17, shows the landing page of the TrackCSPC Smart Campus Vehicle Tracking and Reservation System. The main function of this landing page is to provide administrators with secure access to the platform, where they can manage campus vehicles, assign drivers, and monitor reservation requests.

The interface prominently displays the system name, TrackCSPC Admin, along with a brief description indicating its purpose—managing vehicles, drivers, and reservations. A single action button, sign in with Google, is provided as the exclusive authentication method, ensuring a secure and convenient login process for authorized personnel. Once signed in, administrators gain access to the system's role-specific features, enabling them to oversee fleet activity, approve requests, and maintain operational efficiency within the campus transportation service.

4.3.2 Admin Dashboard and Features

The Admin Dashboard served as the central control panel of the TrackCSPC Smart Campus Vehicle Tracking and Reservation System, giving administrators full access to system functions, data records, and configuration settings. It was designed for intuitive navigation and real-time monitoring, allowing administrators to track vehicle locations, manage drivers, oversee reservation requests, and coordinate transportation activities. Unlike other users, administrators had unrestricted access to all modules as part of their role in system supervision and operational control.

The dashboard also provided a real-time summary of fleet status, showing how many vehicles were Available, In Use, or under Maintenance. At the center, an interactive map displayed the latest locations and routes of campus vehicles for effective monitoring. The Vehicle Status panel presented key details such as models and operational conditions. Through its organized top

and sidebar menus, administrators were able to access drivers, reservations, messages, trip histories, and reports, helping them stay informed and maintain efficient campus transportation management.

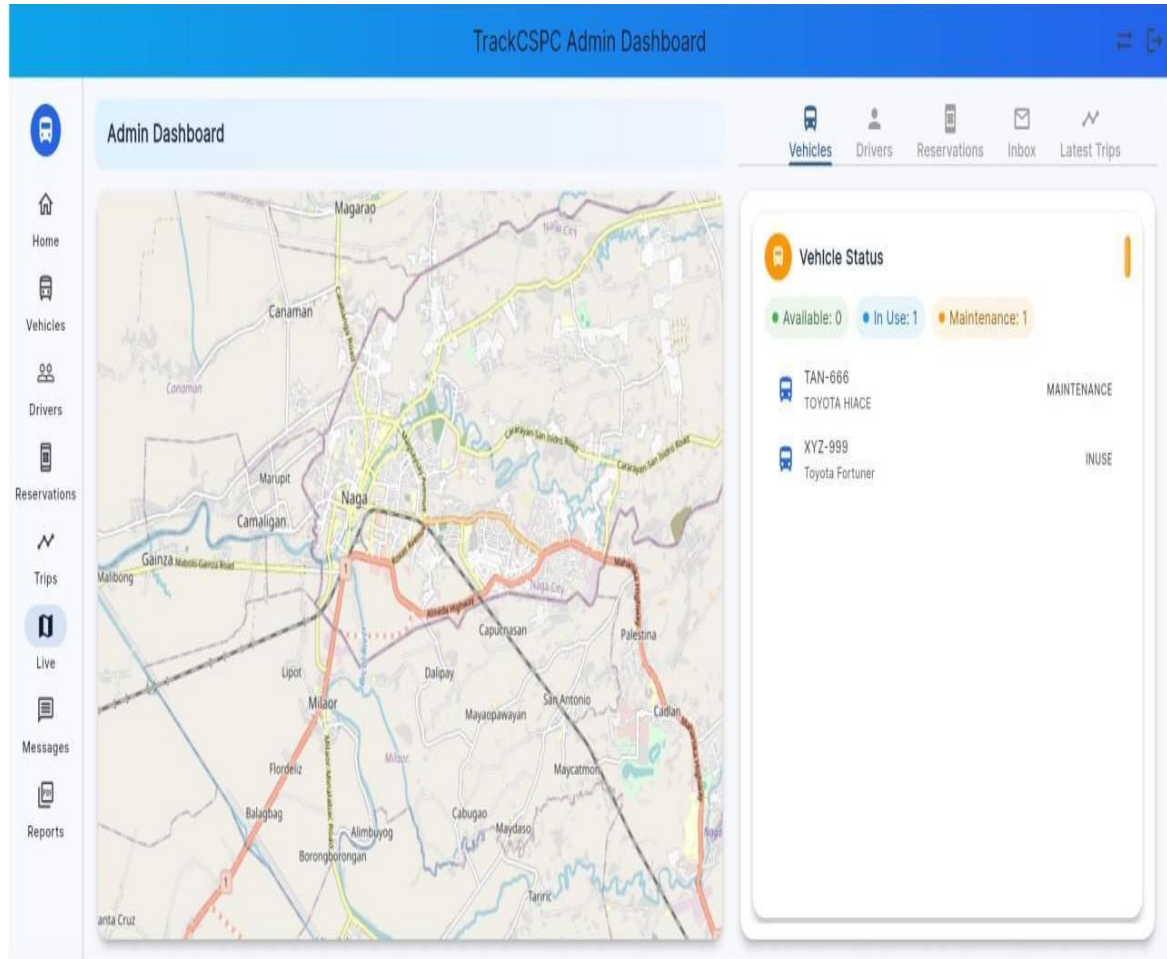


Figure 18. **Admin Dashboard**

Figure 18 showed the Admin Dashboard of the TrackCSPC Smart Campus Vehicle Tracking and Reservation System. This interface served as the central control panel for administrators, providing real-time access to vehicle locations, fleet status, and essential management functions. The dashboard featured an interactive map that visualized the current routes, locations, and movement patterns of campus vehicles, enabling efficient monitoring of transportation activities.

On the right side of the interface, the Vehicle Status panel categorized vehicles into Available, In Use, and Maintenance. Vehicle show current status, allowing administrators to quickly assess fleet availability. The upper navigation tabs Vehicles, Drivers, Reservations, Inbox, and Latest Trips—provided quick access to key system modules, while the left sidebar offered comprehensive navigation to features such as Home, Vehicles, Drivers, Reservations, Trips, Live Tracking, Messages, and Reports.

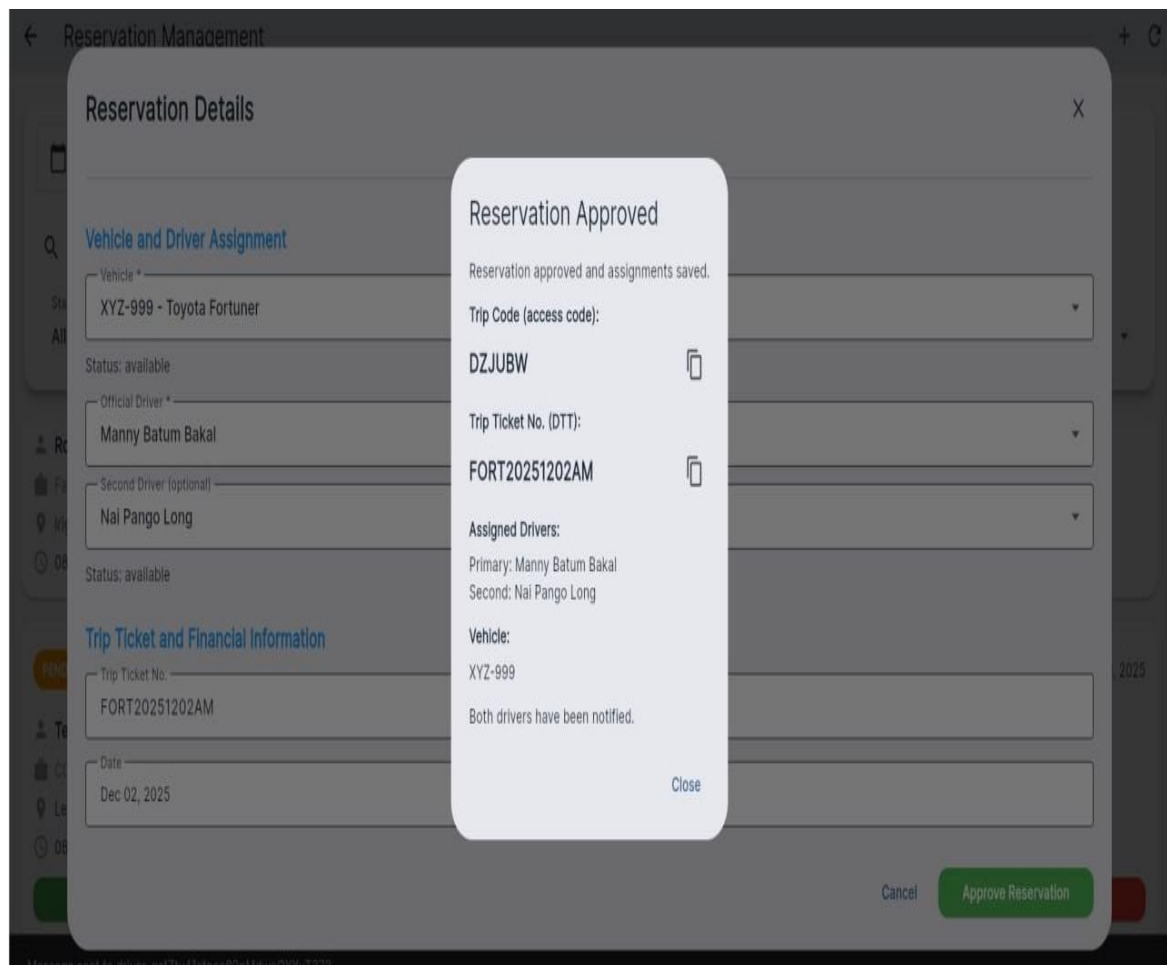
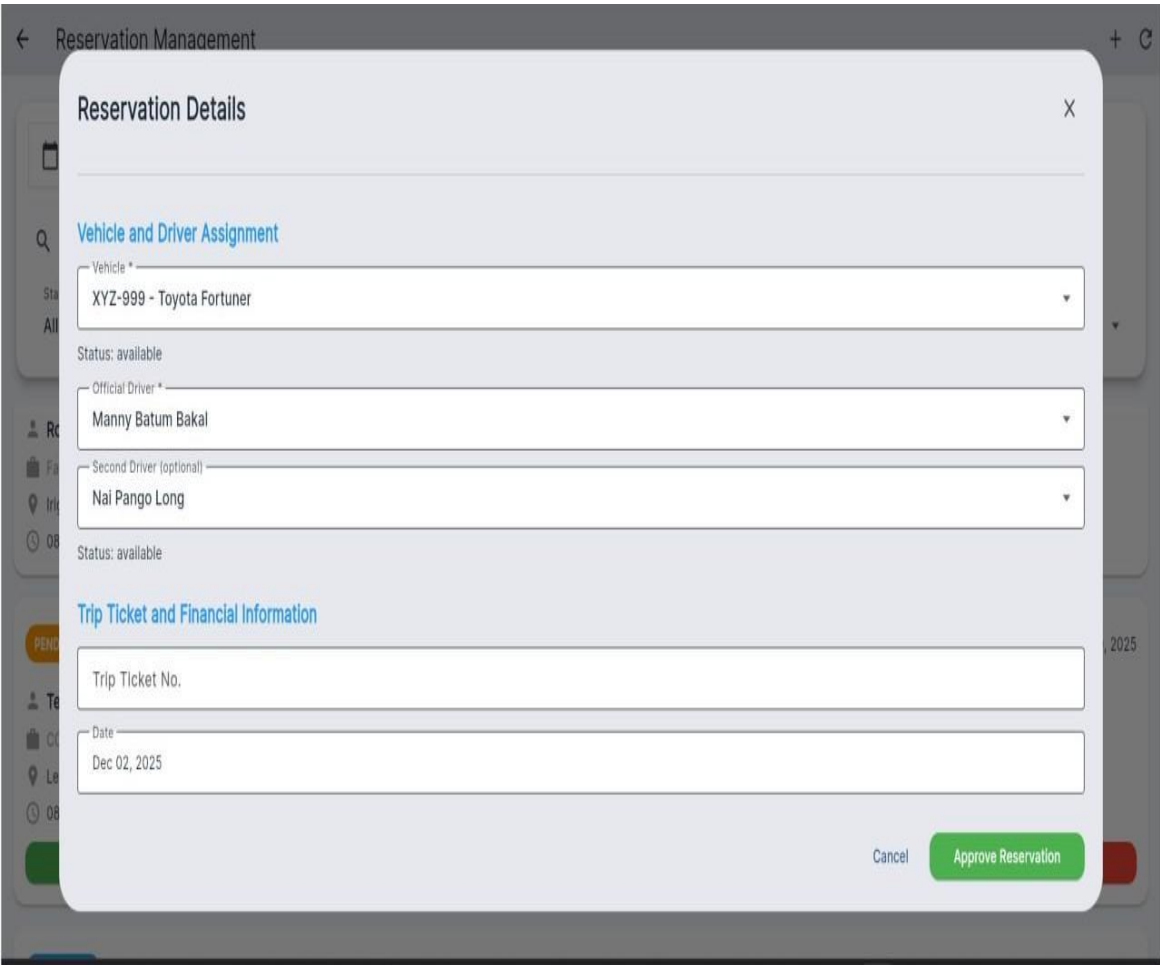


Figure 19. **Reservation Approved**

Efficient handling of vehicle reservations is essential for maintaining organized scheduling, driver coordination, and overall transport operational efficiency. The Reservation Approval Module confirms finalized booking details and ensures that all necessary assignments are

properly recorded. Figure shows the *Reservation Approved* dialog, which appears once an administrator successfully validates and approves a reservation request.

This confirmation panel displays important reservation information such as the Trip Code, Trip Ticket Number, and the assigned drivers for the trip, including both the primary and secondary drivers. It also identifies the vehicle assigned to the reservation, ensuring clarity and avoiding scheduling conflicts. Additionally, the system automatically notifies the designated drivers, promoting seamless coordination and timely communication regarding upcoming trips. This module enhances operational reliability by ensuring that every approved reservation is clearly documented, shared, and tracked.



Reservation Management

Reservation Details

Vehicle and Driver Assignment

Vehicle *
XYZ-999 - Toyota Fortuner

Status: available

Official Driver *
Manny Batum Bakal

Second Driver (optional)
Nai Pango Long

Status: available

Trip Ticket and Financial Information

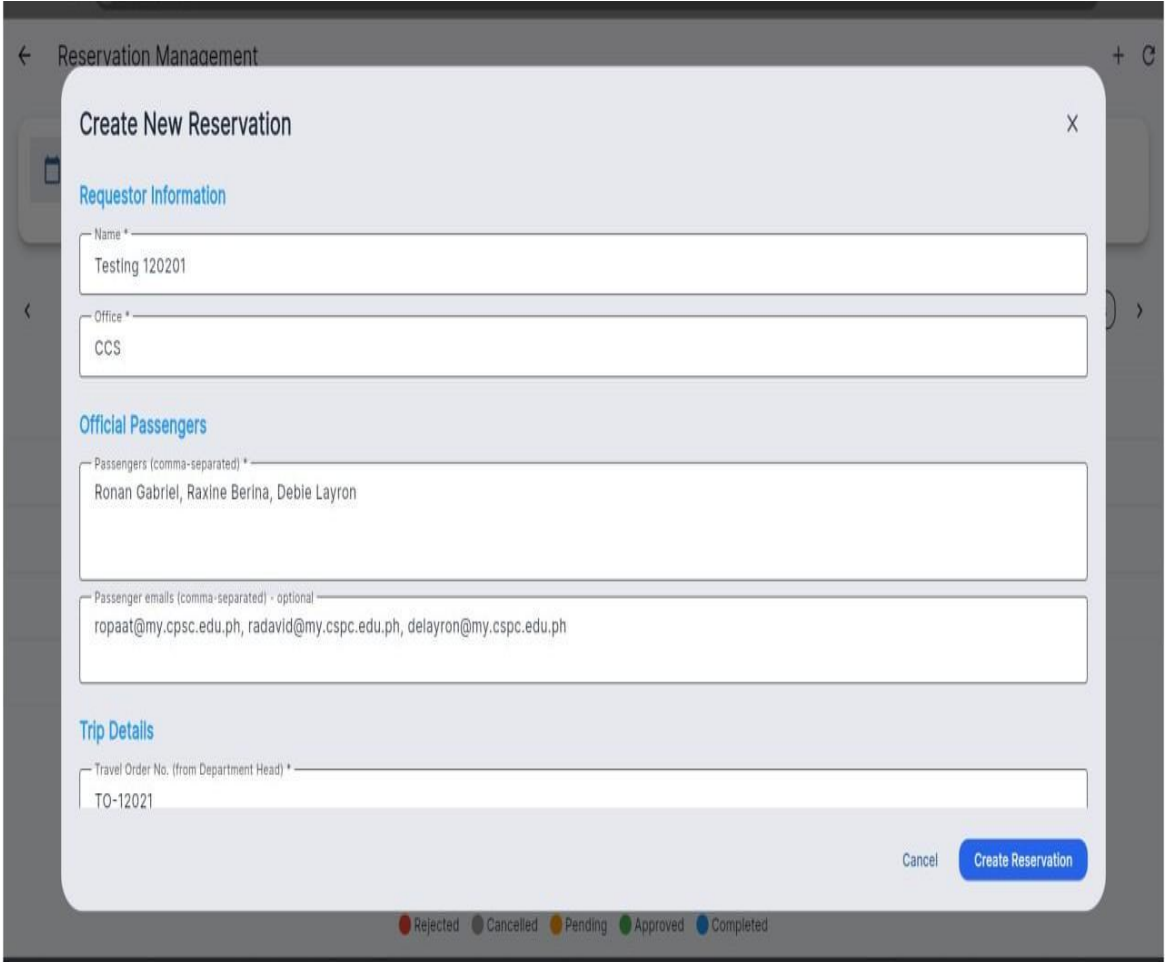
Trip Ticket No.

Date
Dec 02, 2025

Cancel Approve Reservation

Figure 20. **Reservation Details**

The completion of a resource allocation workflow within a digital Reservation Management system. The process transitions from the input of essential trip data to the formal confirmation of resource assignment. Specifically, the system confirms the booking of the vehicle and assigned drivers. Finalized upon approval, the reservation is officially recorded with the unique system identifiers of a Trip Code. The system verifies operational readiness by confirming that formal notifications have been successfully dispatched to both assigned drivers.



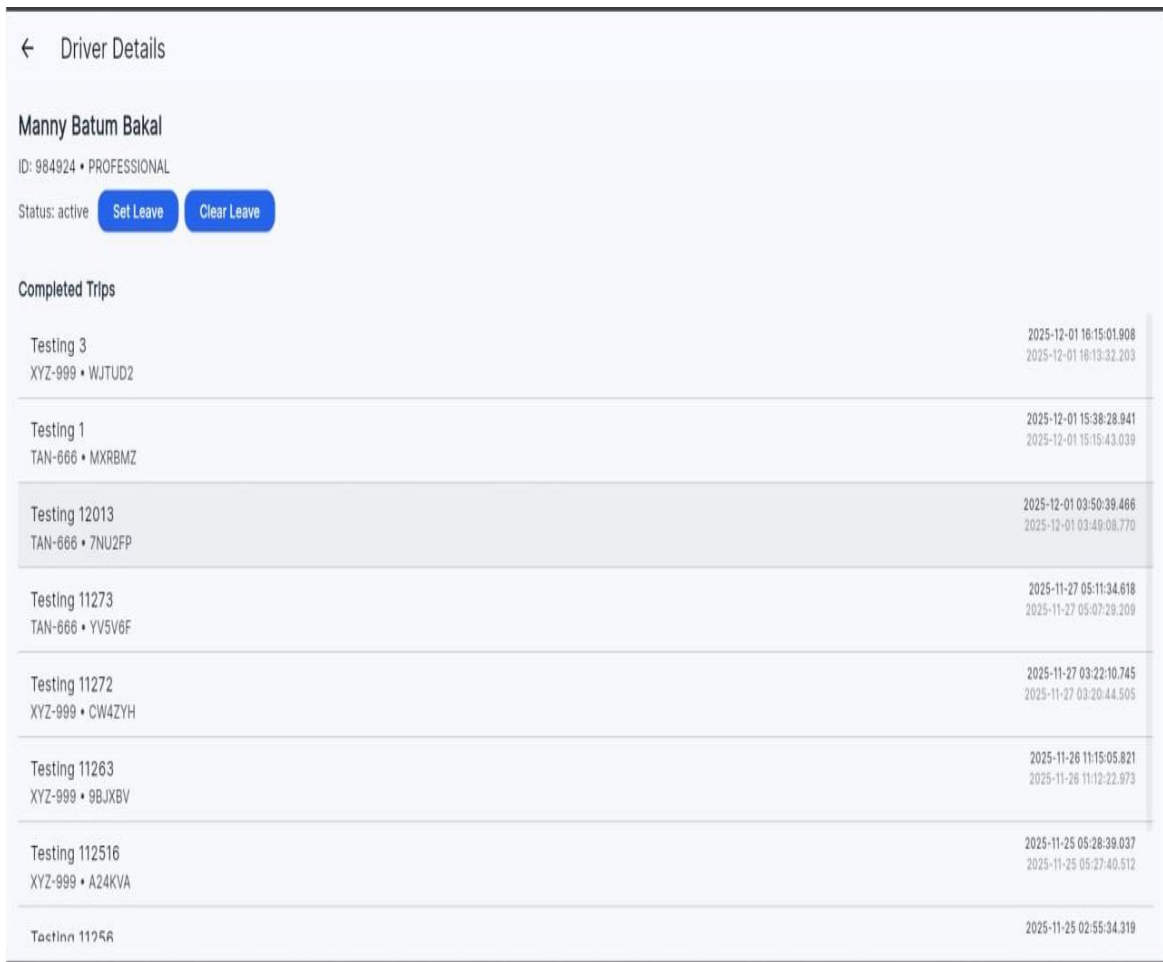
The screenshot displays a mobile application interface for 'Reservation Management'. A modal window titled 'Create New Reservation' is open, featuring a close button (X) in the top right corner. The form is organized into three main sections: 'Requestor Information', 'Official Passengers', and 'Trip Details'. The 'Requestor Information' section contains two text input fields: 'Name *' with the value 'Testing 120201' and 'Office *' with the value 'CCS'. The 'Official Passengers' section includes two text input fields: 'Passengers (comma-separated) *' with the value 'Ronan Gabriel, Raxine Berina, Debie Layron' and 'Passenger emails (comma-separated) - optional' with the value 'ropaat@my.cpsc.edu.ph, radavid@my.cpsc.edu.ph, delayron@my.cpsc.edu.ph'. The 'Trip Details' section has one text input field: 'Travel Order No. (from Department Head) *' with the value 'TO-12021'. At the bottom right of the form are two buttons: 'Cancel' and 'Create Reservation'. Below the form, a status legend shows five colored circles with corresponding labels: 'Rejected' (red), 'Cancelled' (grey), 'Pending' (orange), 'Approved' (green), and 'Completed' (blue).

Figure 21. **Creating New Reservation**

The Create New Reservation interface allowed administrators to manually input and submit reservation requests on behalf of campus personnel. As shown in Figure 13, the form collected

essential trip information, beginning with the requestor's name and office or department. It also included a section for listing official passengers, where the admin entered names and corresponding email addresses in comma-separated format to ensure proper documentation and notification.

Additionally, the form required the Travel Order Number issued by the department head, serving as an authorization reference for the trip. After providing all necessary details, the administrator could proceed by selecting the Create Reservation button, while also having the option to cancel the entry. Overall, this module streamlined the process of generating reservations by offering a structured and user-friendly form for recording complete and accurate trip information.



← Driver Details

Manny Batum Bakal
ID: 984924 • PROFESSIONAL

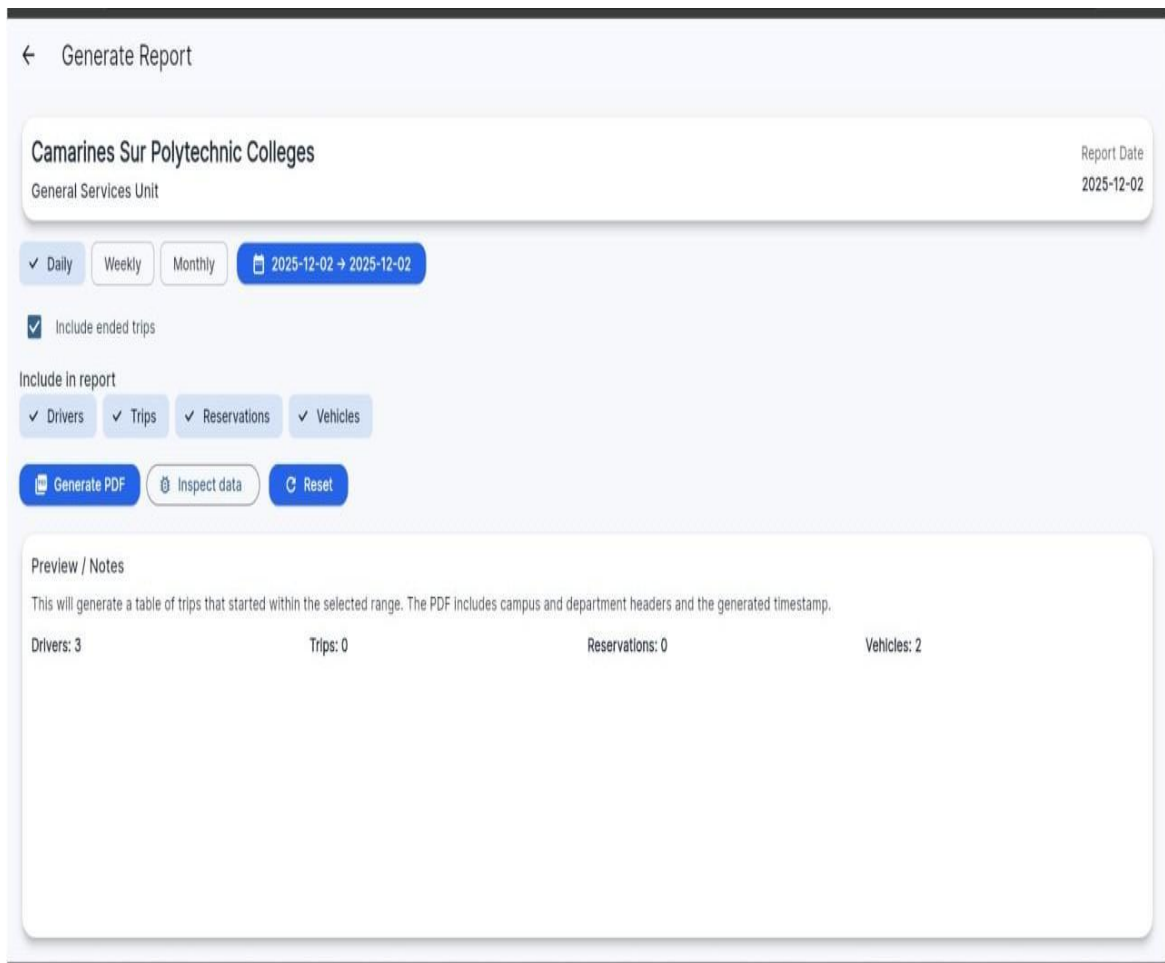
Status: active [Set Leave](#) [Clear Leave](#)

Completed Trips

Testing 3 XYZ-999 • WJTUD2	2025-12-01 16:15:01.908 2025-12-01 16:13:32.203
Testing 1 TAN-666 • MXRBMZ	2025-12-01 15:38:28.941 2025-12-01 15:15:43.039
Testing 12013 TAN-666 • 7NU2FP	2025-12-01 03:50:39.486 2025-12-01 03:49:08.770
Testing 11273 TAN-666 • YV5V6F	2025-11-27 05:11:34.618 2025-11-27 05:07:29.209
Testing 11272 XYZ-999 • CW4ZYH	2025-11-27 03:22:10.745 2025-11-27 03:20:44.505
Testing 11263 XYZ-999 • 9BJXBV	2025-11-26 11:15:05.821 2025-11-26 11:12:22.973
Testing 112516 XYZ-999 • A24KVA	2025-11-25 05:28:39.037 2025-11-25 05:27:40.512
Testing 11256	2025-11-25 02:55:34.319

Figure 22. **Driver Details**

The Driver Details interface displayed comprehensive information about an assigned driver within the TrackCSPC system. As shown in Figure 14 the top section presented the driver's name, identification number, and license classification, along with their current status. Administrators were also provided with two action buttons Set Leave and Clear Leave allowing them to update the driver's availability in real time. Below the driver's basic profile, the interface listed all completed trips associated with the driver. Each trip entry showed the trip name or identifier, the vehicle used, the generated trip code, and the corresponding timestamps, including departure and completion times. These details helped administrators review the driver's performance, verify trip completion, and confirm the chronological order of activities.



← Generate Report

Camarines Sur Polytechnic Colleges
General Services Unit

Report Date
2025-12-02

☒ Daily ☐ Weekly ☐ Monthly

☒ Include ended trips

Include in report

☒ Drivers ☒ Trips ☒ Reservations ☒ Vehicles

Preview / Notes

This will generate a table of trips that started within the selected range. The PDF includes campus and department headers and the generated timestamp.

Drivers: 3 Trips: 0 Reservations: 0 Vehicles: 2

Figure 23. **Generate Report**

Figure 23 shows the Generate Report interface, which enabled administrators to create organized transportation reports for the campus. The module allowed users to set a time range Daily, Weekly, Monthly, or a specific date and choose which data to include, such as Drivers, Trips, Reservations, and Vehicles. They could also opt to include completed trips. The interface provided three actions: Generate PDF to produce a formal report, Inspect Data to preview information, and reset clear selections. A summary preview displayed the number of drivers, trips, reservations, and vehicles based on the chosen filters.

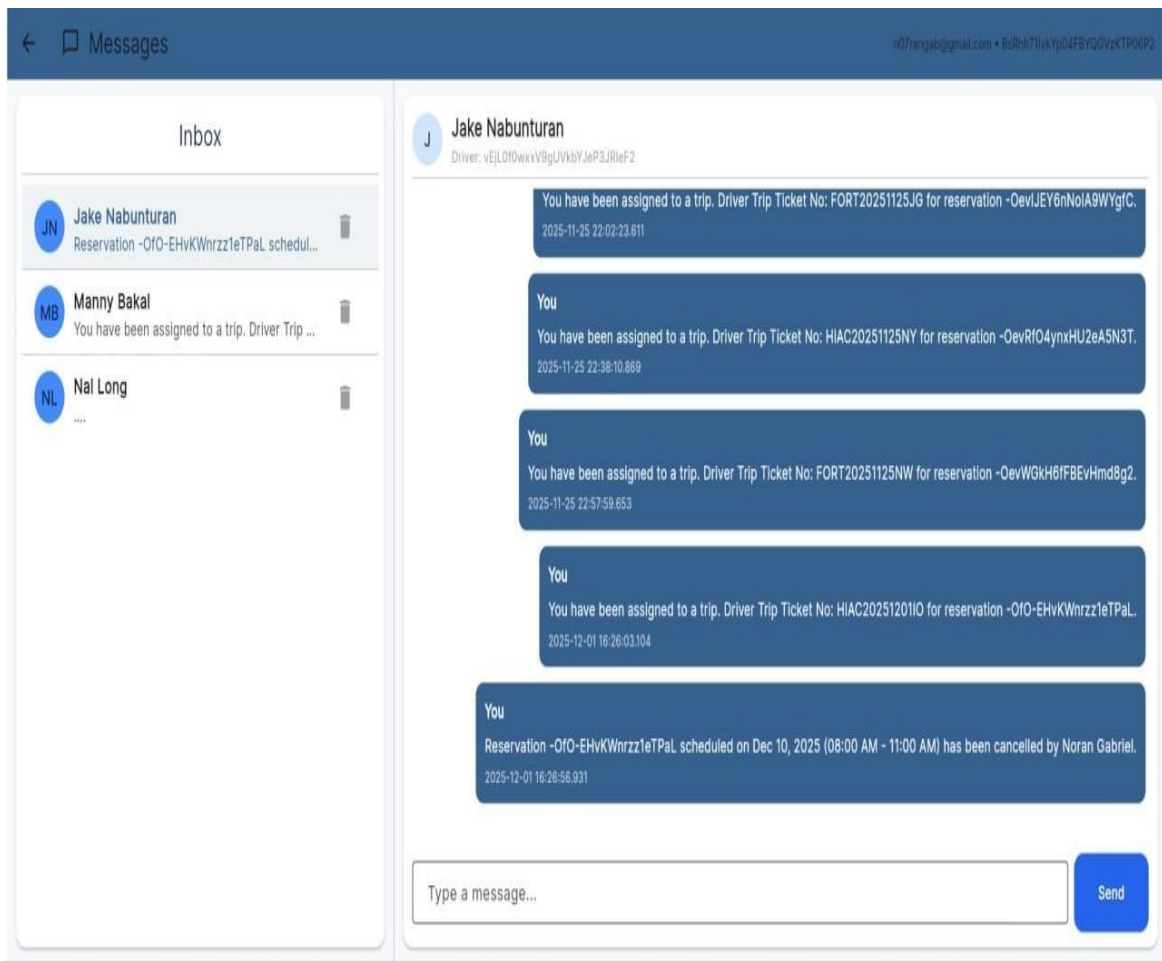


Figure 24. Messages

As shown in Figure 24, the Messaging module provided a dedicated communication interface between the administrator and drivers. The inbox displayed all message threads, allowing

the administrator to send important updates such as trip assignments, reservation notifications, and cancellations. Drivers were able to view these messages in a clear, conversation-based format, ensuring that information was received in a timely and organized manner. The interface featured a message list on the left and a chat window on the right, where users could read the message history and send new replies to support efficient communication for trip and reservation updates. It also helped maintain a record of all communications related to trips and reservations. The streamlined messaging process improved coordination and response time during daily transport operations. Overall, the module contributed to smoother communication and more reliable trip management across the system.

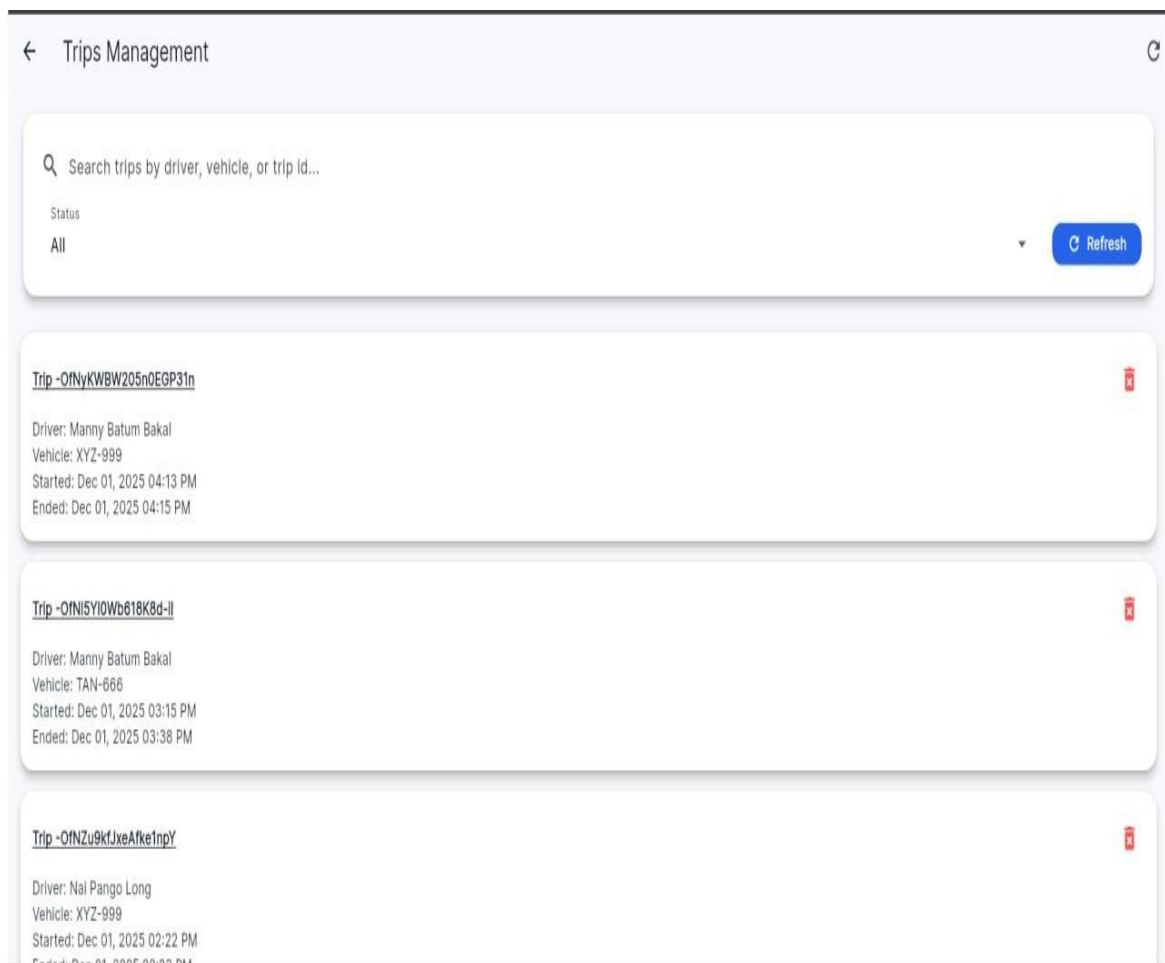


Figure 25. Trips Management

As shown in Figure 25, the Trips Management module provided administrators with a comprehensive interface to oversee all transportation activities recorded in the system. The interface included a search bar and status filter, allowing the admin to quickly locate trips by keywords or narrow the list by categories such as completed, ongoing, or cancelled. Each trip entry displayed key details, including the assigned driver, vehicle, and exact start and end times, while also offering an option to delete records when necessary, which helped maintain accurate and organized trip data usage. Additionally, the module includes an option to delete trip records when necessary, helping maintain accurate and organized trip data.

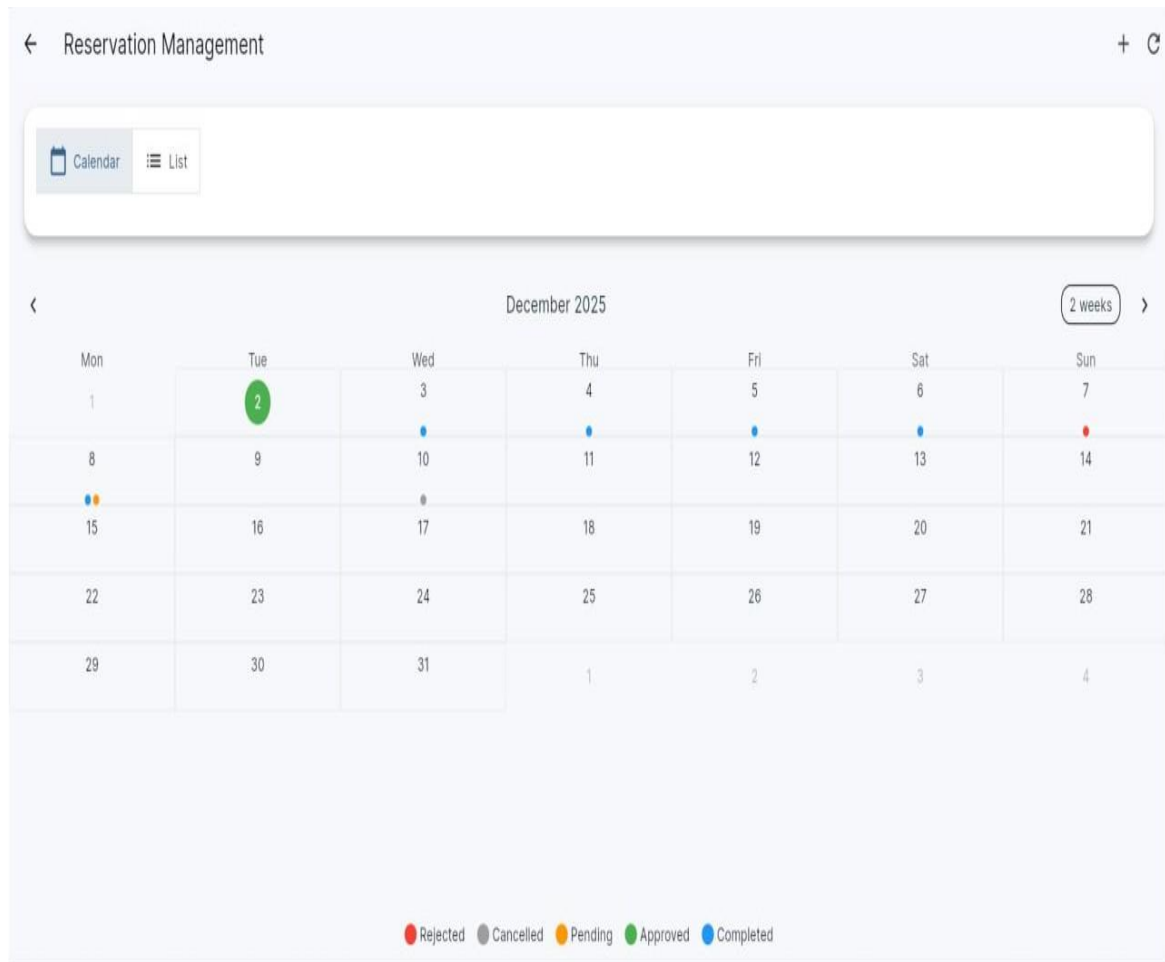


Figure 26. **Reservation Calendar**

As shown in Figure 26, the Reservation Management module provides administrators with a clear and organized interface for viewing and managing all reservation activities within the system. The interface displays a calendar view, allowing users to easily track reservations by date and visually interpret their statuses through color-coded indicators. Each reservation is marked based on its current state—such as rejected, cancelled, pending, approved, or completed—helping staff quickly assess the overall schedule at a glance. By selecting a specific date, users can review reservation details associated with that day, enabling efficient monitoring and planning. The module allows administrators to maintain accurate schedules, prevent overlaps, and ensure that all reservation-related information is kept up to date and well-organized.

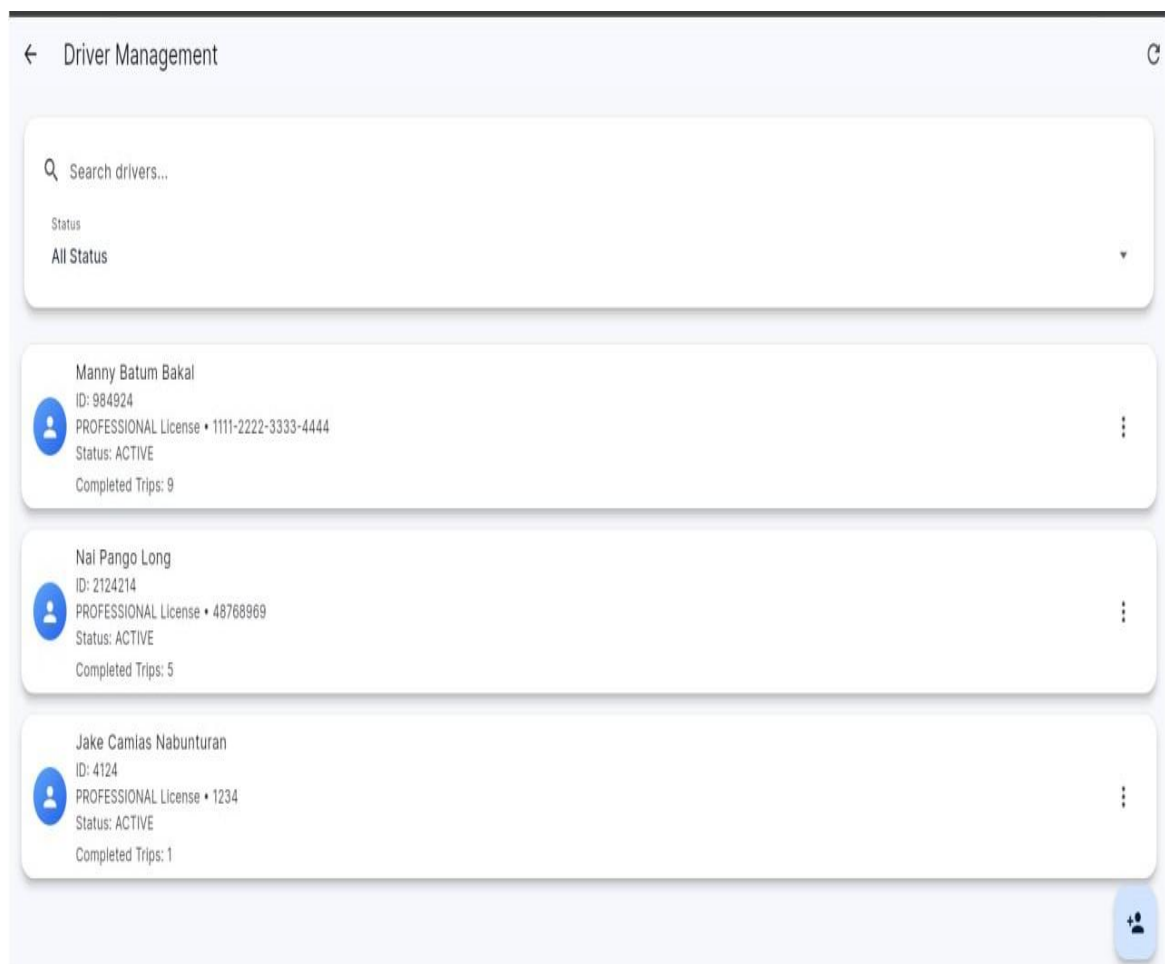


Figure 27. **Driver Management**

As shown in Figure 27, the Driver Management module provides administrators with an organized interface for viewing and managing all registered drivers in the system. A search bar at the top allows the admin to quickly locate specific drivers by typing in names or related keywords, while a status filter enables users to sort drivers based on their current activity level. Each driver entry displays essential information, including the driver's name, ID number, license type, license number, status, and total completed trips. This helps administrators easily verify driver credentials, assess performance, and monitor active participation in trip operations. Additionally, action options are available for each driver, allowing admins to update their information or manage their records when needed, ensuring that the driver database remains accurate and well-maintained.

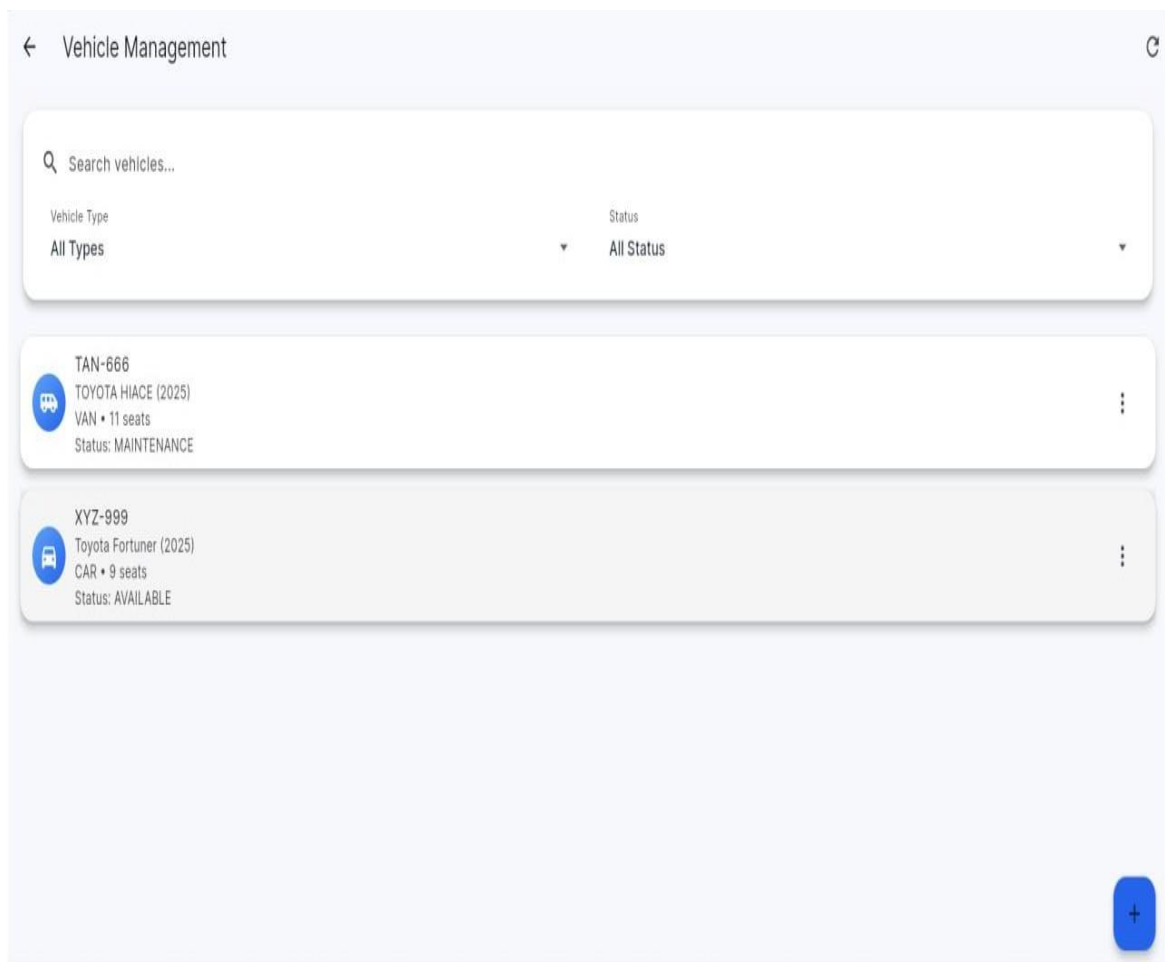


Figure 28. Vehicle Management

As shown in Figure 28, the Vehicle Management module provided administrators with a streamlined interface for monitoring and organizing all vehicles registered in the system. A search bar allowed quick retrieval of specific vehicles, while filters for vehicle type and status helped narrow down the list to match the admin's needs. Each vehicle entry displayed key information, including the plate number, model, seat capacity, and current status—such as available or under maintenance. This allowed the admin to easily assess vehicle readiness, track maintenance activities, and ensure that only operational units were assigned for trips. Additionally, action options were provided for each vehicle, enabling updates or modifications to vehicle details and helping maintain an accurate and up-to-date fleet database. The module ultimately supported efficient fleet management and improved overall operational reliability.

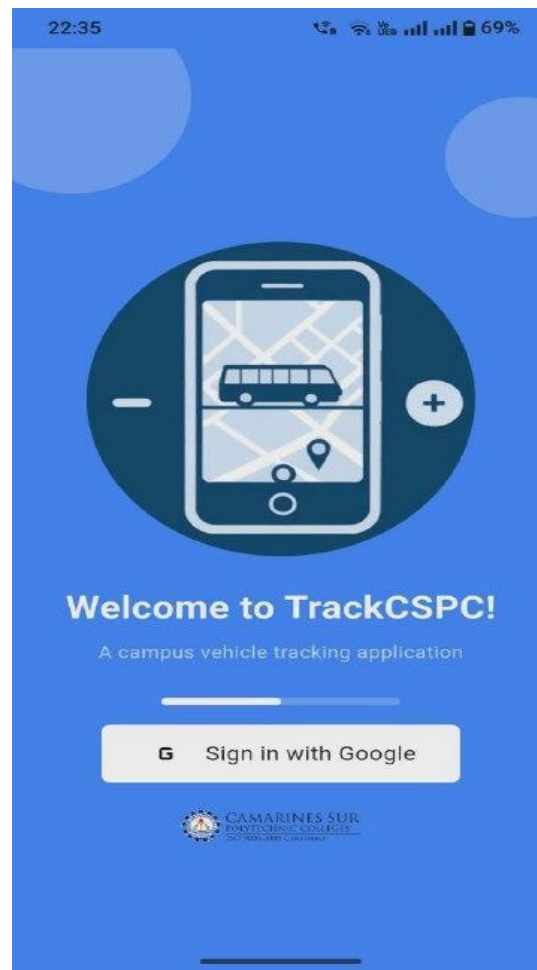


Figure 29. Mobile Login Page

Figure 29 shows the Mobile Sign-In Page of the TrackCSPC application, which serves as the entry point for users accessing the campus vehicle tracking system via mobile devices. Instead of requiring manual input of credentials, the interface utilizes a streamlined “Sign in with Google” button to authenticate users securely and efficiently. This design ensures quick access for students, faculty, drivers, and administrators while maintaining strong security through Google’s verification system. The page features a clean and visually appealing layout with an icon representing mobile vehicle tracking, helping users easily recognize the purpose of the application. By offering a simple, one-tap login method, the mobile interface improves accessibility, reduces login errors, and allows users to immediately access live tracking and reservation updates.

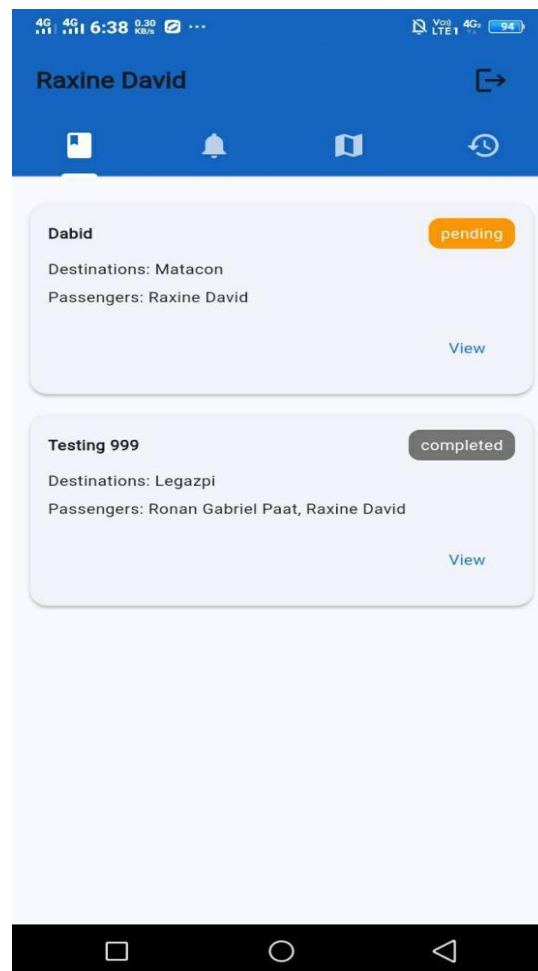


Figure 30. Student App

This interface displays the ongoing and completed vehicle service requests submitted by the user. It serves as the central page where students, faculty, and staff can track the progress of their booking requests and review the details of each scheduled trip. The navigation bar features icons for bookmarks, notifications, maps, and trip history, allowing users to switch seamlessly between different sections of the application. Each request card contains essential trip information, including the destination and the list of passengers assigned to the booking. A colored status label, such as “pending” or “completed,” provides a clear and immediate indicator of the request’s current state. The “View” option at the bottom of each card allows the user to open a more detailed breakdown of the trip, including schedule, assigned driver, and additional notes from the General Services Unit (GSU).

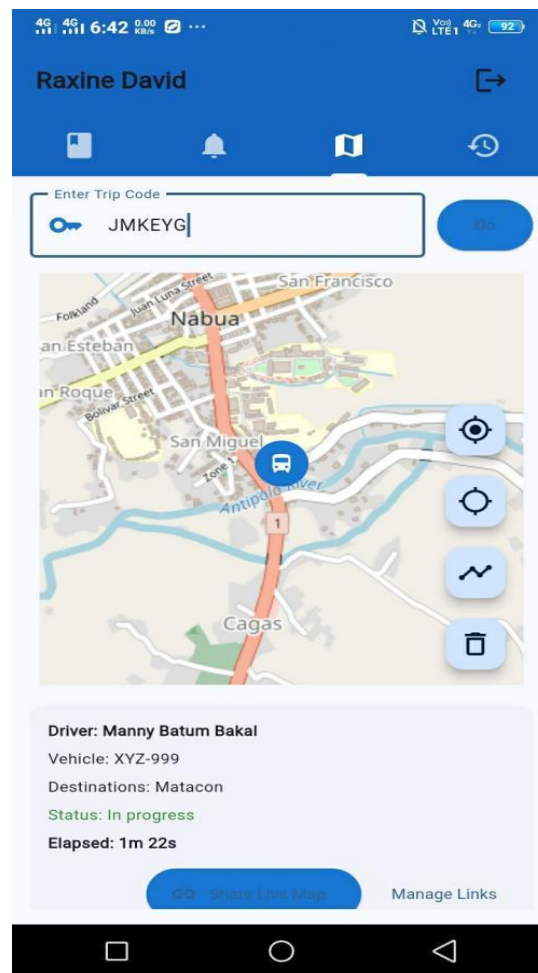


Figure 31. Live Map

This page allows users to track their active trip in real time. A trip code field at the top enables users to enter or confirm their assigned trip, activating the live map display. The interactive map shows the vehicle's current location and route, supported by tools for zooming and recentering.

Below the map, essential trip details are displayed, including the driver, vehicle, destination, status, and an elapsed time counter. Overall, this page provides a clear, streamlined interface for monitoring campus vehicle movement during an ongoing trip.



Figure 32. **Trip tracking History**

This page provides a clear and organized interface for tracking active and recent trips in real time. The user's profile name appears at the top along with quick-access icons for bookmarks, notifications, guides, and trip history. Trip records are displayed in clean, card-style sections, each

showing the trip name, destination, assigned vehicle, and driver, along with the exact start timestamp and total trip duration. Overall, the design delivers a straightforward and efficient way to monitor campus vehicle movement and trip details through a simple, readable list format.

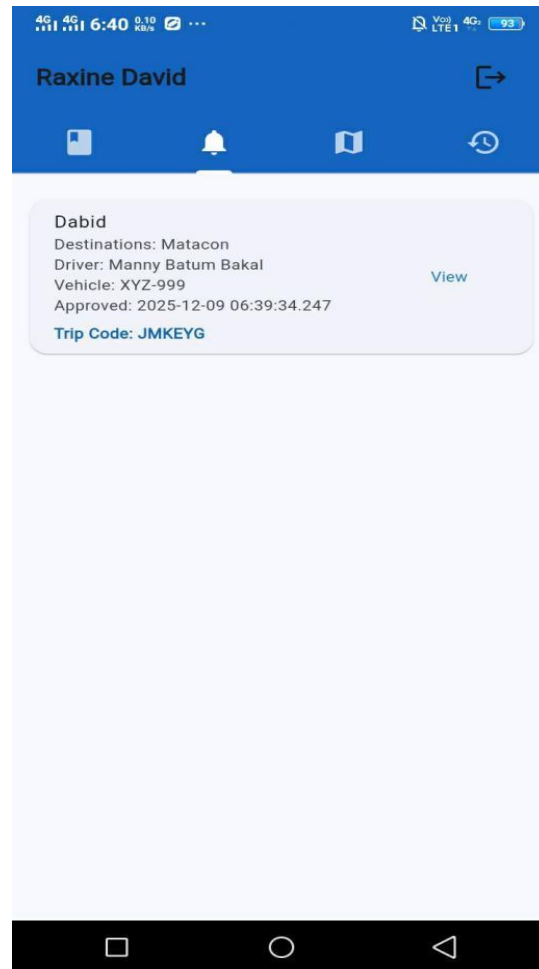


Figure 33. **Approved Trip**

This page provides a clean and focused view of a user's currently approved trip. At the top, the user's name is displayed along with quick-access icons for bookmarks, notifications, guides, and trip history. The main content area presents the active trip in a neatly styled card, showing essential details such as the destination, driver, and assigned vehicle. It also includes the exact approval timestamp and a prominently displayed trip code for easy reference or sharing. A "View" link is provided to allow the user to open more detailed trip information when needed.

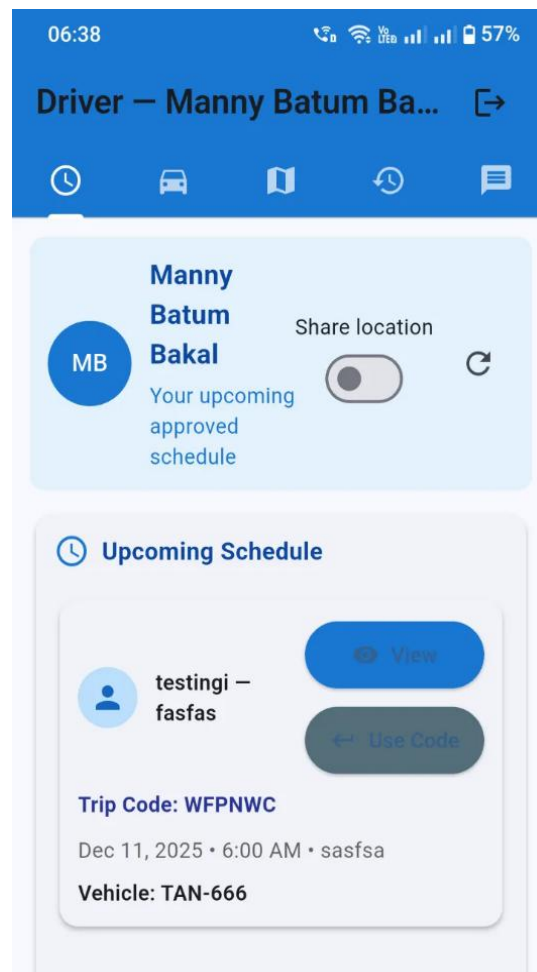
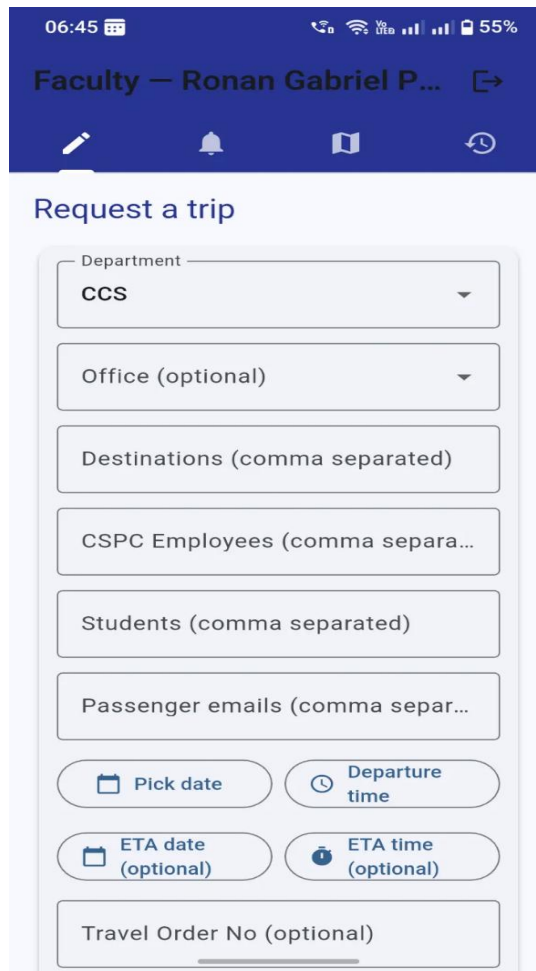


Figure 34. **Driver Dashboard**

This page presents the driver's dashboard in a clear and organized layout, allowing them to easily manage their upcoming trips and location-sharing settings. At the top, the driver's name is displayed along with navigation icons for schedules, vehicles, guides, trip history, and messaging. The main profile card highlights the driver's identity, showing their initials, full name, and a toggle switch to enable or disable location sharing, accompanied by a refresh button for quick updates. Below this, an "Upcoming Schedule" section lists the driver's next approved trip in a structured card format. Each trip card includes the requester's name, a "View" button for accessing full trip details, and a disabled "Use Code" button indicating options based on trip state. Additional trip information—such as the trip code, scheduled date and time, purpose, and assigned vehicle is

clearly displayed. Overall, the interface is designed to help drivers quickly review their duties, manage visibility settings, and stay prepared for their upcoming trips.

The dashboard also integrated real-time updates to keep drivers informed of any changes to their schedules or trip details. The combination of interactive elements, such as buttons for viewing trip details, and visual cues, like status labels, made it easier for drivers to prioritize tasks and respond quickly. By presenting all relevant information in one centralized interface, the dashboard minimized the need for manual tracking or constant checking with administrators. Overall, this design enhanced driver productivity, reduced errors, and promoted a smoother, more organized approach to daily trip management.



06:45 55%

Faculty — Ronan Gabriel P...

Request a trip

Department
CCS

Office (optional)

Destinations (comma separated)

CSPC Employees (comma separa...

Students (comma separated)

Passenger emails (comma separ...

Pick date

Departure time

ETA date (optional)

ETA time (optional)

Travel Order No (optional)

Figure 35. **Faculty Trip Request**

The image shows a trip request form where a faculty member can fill out all necessary travel details. At the top is a blue header displaying the faculty role and name, along with navigation icons for editing, notifications, maps, and history. Below it, the “Request a trip” section contains a clean set of form fields arranged in rounded boxes. These fields allow the user to select a department, optionally choose an office, and enter destinations, employees, students, and passenger emails using comma-separated inputs. Further down are buttons for selecting the pick date, departure time, ETA date, and ETA time, followed by another field for entering an optional travel order number. The layout organized and easy to follow, guiding the user through each step of submitting a trip request.

4.4 Testing

The testing phase focused on verifying that all features and modules functioned according to the defined requirements. It outlined the testing objectives, the resources involved, and the overall strategy used to validate the entire system. The phase also defined the scope of the test cases, specifying which functionalities were included and highlighting any limitations. This section presented the functionality testing conducted on the developed TRACK CSPC: Smart Campus Vehicle Tracking and Reservation Platform.

Table 7. Functionality Testing Results

MODULE	RESULT	INTERPRETATION
User Authentication	1.00	Passed
Real - time GPS Tracking	1.00	Passed
Route and Schedule Display	1.00	Passed
Admin Dashboard	1.00	Passed
Vehicle Reservation	1.00	Passed
Parental Monitoring	1.00	Passed
Arithmetic Mean (Total)	1.00	Passed

Table 7 summarized the functionality testing results for each core module of the system, including User Authentication, Real-time GPS Tracking, Route and Schedule Display, Admin Dashboard, Vehicle Reservation, and Parental Monitoring. All modules received a perfect score of 1.00, which was interpreted as “Passed,” resulting in an overall arithmetic mean of 1.00.

These results demonstrated that all core functional modules of TRACK CSPC operated correctly and consistently. Each module performed as expected, showing that the system was capable of controlling user access, tracking vehicles in real-time, managing reservations, and supporting administrative tasks without errors. The perfect arithmetic mean indicated that no functional failures, broken processes, or incorrect outputs were observed during testing.

The findings suggested that the system was logically sound and reliable. The consistency across all modules confirmed that the platform could effectively support campus transportation processes, ensuring operational efficiency and user satisfaction. This outcome aligned with current software quality standards, particularly ISO/IEC 25010:2023, which emphasizes functional correctness as a key attribute of system quality.

The functionality testing was conducted using Gray Box Testing, which evaluates system functionality from both internal and external perspectives. This approach allowed the study to verify that each module not only produced the correct outputs for given inputs but also followed the expected internal workflows, logic, and data handling procedures.

All modules — User Authentication, Real-time GPS Tracking, Route and Schedule Display, Admin Dashboard, Vehicle Reservation, and Parental Monitoring — scored 1.00, producing an overall arithmetic mean of 1.00 (“Passed”). The alignment between the observed system outputs and the expected functionality confirmed that the software behaved correctly under standard operations. This result is consistent with previous studies on Gray Box Testing, which emphasize that combining external testing with internal knowledge increases the reliability of functional validation (Li et al., 2014).

Furthermore, Gray Box Testing bridged the gap between technical verification and user expectations. Previous research indicated that this method is particularly effective for database-driven and multi-module applications, as it ensures that internal logic aligns with expected results while still evaluating overall system usability (Kumar and Sagar, 2025). The perfect match between test case execution and expected module outputs demonstrated the robustness of TRACK CSPPC, confirming that the system was ready for deployment in a campus environment.

A limitation of this testing was that it did not assess system performance under high load or potential security vulnerabilities, as the primary focus was on functionality and internal workflow correctness. Nevertheless, Gray Box Testing provided a higher level of confidence in system correctness than either Black Box or White Box Testing alone.

In summary, the functionality testing confirmed that TrackCSPPC: Smart Campus Vehicle Tracking and Reservation Platform was fully functional and reliable. The agreement among IT experts, students, drivers, and the executed test cases showed that all modules were logically sound and free of functional errors.

Table 8: Load Testing

USER GROUP	NO. OF THREADS (USERS)	AVERAGE RESPONSE TIME (MS)	THROUGH PUT (REQ/SEC)	ERROR %	SCORE
Students	20	1845	1.25	0.00%	0.90
Drivers	5	2110	0.85	0.00%	0.85
GSU Staff	5	2320	0.72	0.00%	0.80
Faculty	0	N/A	N/A	N/A	N/A
IT Expert	1	1280	0.30	0.00%	1.00
Overall Result	31	1889	0.78	0.00%	0.89

Table 8 summarized the performance of TRACK CSPC under simultaneous user access. The system maintained an overall average response time of 1,889 ms, a throughput of 0.78 requests per second, and an error rate of 0.00%. The faculty group was not included in the test due to the lack of active threads during the testing period.

The results demonstrated that TRACK CSPC could handle multiple concurrent users efficiently without functional errors, maintaining stability across all tested groups. Students, Drivers, and GSU Staff experienced response times within acceptable limits, and IT Experts confirmed that the system maintained expected functionality under the simulated load conditions.

These findings aligned with current performance evaluation studies, which emphasize the importance of systematically measuring response time, throughput, and error rate to assess system readiness and scalability in real-world environments (Neelapu, 2024; Kan et al., 2019). By maintaining functional stability across multiple user threads and achieving a 0.00% error rate, TRACK CSPC demonstrated that its performance met operational requirements for campus-wide deployment.

A limitation of this testing was that extreme load conditions beyond 31 concurrent users were not evaluated. Future testing should include all user groups, particularly Faculty, and simulate higher traffic scenarios to ensure robustness under peak usage conditions. In summary, the load testing confirmed that TRACK CSPC could support multiple simultaneous users efficiently, with minimal errors and acceptable response times. The system demonstrated consistent stability and functional reliability across all tested groups, indicating that it was ready to manage real-time campus vehicle tracking and reservations under normal operational conditions.

This section presented the User Acceptance Testing (UAT) conducted on the developed TRACK CSPC: Smart Campus Vehicle Tracking and Reservation Platform. The primary aim of this testing was to evaluate whether the system met the expectations and satisfaction of its intended end-users, including students, drivers, staff, faculty, and IT experts.

Table 9: User Acceptance Testing (UAT)

USER GROUP	RESULTS	INTERPRETATION
Student	1.00	Acceptable
Driver	1.00	Acceptable
GSU Staff	1.00	Acceptable
Faculty	0.00	Unacceptable
IT Expert	1.00	Acceptable
Arithmetic Mean (Total)	1.00	Acceptable

Table 9 presented the results of the User Acceptance Testing (UAT), showing that most user groups rated the system as acceptable. Students, Drivers, GSU Staff, and IT Experts all gave perfect scores of 1.00, indicating full satisfaction with the platform. The faculty group, however, did not participate effectively during testing, resulting in a score of 0.00 (“Unacceptable”). Despite this, the overall arithmetic mean across all participating groups was 1.00, reflecting broad acceptance of the platform.

These results indicated that TRACK CSPC met the needs and expectations of most user groups, confirming its usability and suitability for managing campus vehicle tracking and reservations. The high scores from Students, Drivers, and IT Experts demonstrated that the platform’s core features—such as real-time GPS tracking, route and schedule display, vehicle reservations, and administrative dashboards—were intuitive, reliable, and aligned with user requirements.

The findings aligned with current user evaluation and performance literature, which emphasizes the importance of systematically assessing system behavior under realistic conditions to ensure readiness and reliability. Studies on User Acceptance Testing highlight that evaluating functional stability, response accuracy, and error handling from the perspective of real users is essential for determining system effectiveness in operational environments (Arif & Soko, 2022;

Mazlan et al., 2025). Combined with the results from load testing, where the system maintained functional stability across multiple user threads and achieved a 0.00% error rate, TRACK CSPC demonstrated that it met operational requirements for campus-wide deployment while satisfying the expectations of its intended users.

The lower score from the faculty group highlighted a limitation in user participation during testing. Future testing should ensure full engagement from all faculty members to validate the system across all user categories and address any specific needs that may not have been captured during this UAT session.

In summary, the User Acceptance Testing confirmed that TRACK CSPC was widely acceptable to its primary users, achieving high satisfaction among students, drivers, staff, and IT experts. The platform was deemed ready for deployment in the campus environment, though further engagement with faculty users is recommended to ensure complete adoption and satisfaction across all user groups.

This section presented a consolidated overview of the testing outcomes for the TRACK CSPC: Smart Campus Vehicle Tracking and Reservation Platform. The purpose of this summary was to provide a clear picture of the system's performance, functionality, and acceptance across all evaluated dimensions, including Functional Testing, Load Testing, and User Acceptance Testing (UAT).

Table 10: Summary of Results

TESTING CATEGORY / RESPONDENTS	RESULTS	INTERPRETATION
Functional Testing	1.00	Completely Functional
Load Testing	1.00	Completely Functional
User Acceptance Testing (UAT)	1.00	Completely Functional
Arithmetic Mean (Total)	1.00	Acceptable / Completely Functional

Table 10 presented the overall results of each testing phase. Functional Testing scored 1.00, indicating that all system modules—including User Authentication, Real-time GPS Tracking, Route and Schedule Display, Admin Dashboard, Vehicle Reservation, and Parental Monitoring performed correctly without errors or failures. The system was confirmed to be fully functional.

Load Testing also achieved a perfect score of 1.00, demonstrating that the platform could handle multiple concurrent users efficiently, maintaining acceptable response times, throughput, and a 0.00% error rate, and showing operational stability under simulated real-world conditions.

User Acceptance Testing (UAT) received a score of 1.00 as well. Students, Drivers, GSU Staff, and IT Experts rated the system as fully acceptable. Although Faculty participation was limited, the overall results still reflected widespread user satisfaction.

The arithmetic mean (total) of 1.00 confirmed that across all testing categories, the system met its intended requirements and performance expectations, earning the interpretation of “Acceptable / Completely Functional.”

4.4.1 Deployment

In the Software Development Life Cycle (SDLC) implementing through the Iterative Waterfall Model, the deployment phase is a stage where the developed system is released from testing or local environment into a live environment for actual use [22]. This stage is a major step as it offers the defined functionalities of the system to the users in an effective way. Deployment that is done correctly allows for easy access to new features, which in turn can increase user satisfaction, thus the use of the system can be taken to a higher level.

4.4.2 Deployment Diagram

A deployment diagram is a type of illustration that helps in understanding which hardware, such as a computer, server, or cloud, the components of a system will be placed. It indicates the various hardware that are used and their connection in order to give a better idea of the system operation [10]. After the testing phase, the system undergoes on a deployment stage, where the

system is released in a target environment and made available for practical use by the users for whom it was designed. To plan this, the researchers designed a deployment diagram that indicates the distribution of the application's software components along with the specified nodes and their interaction.

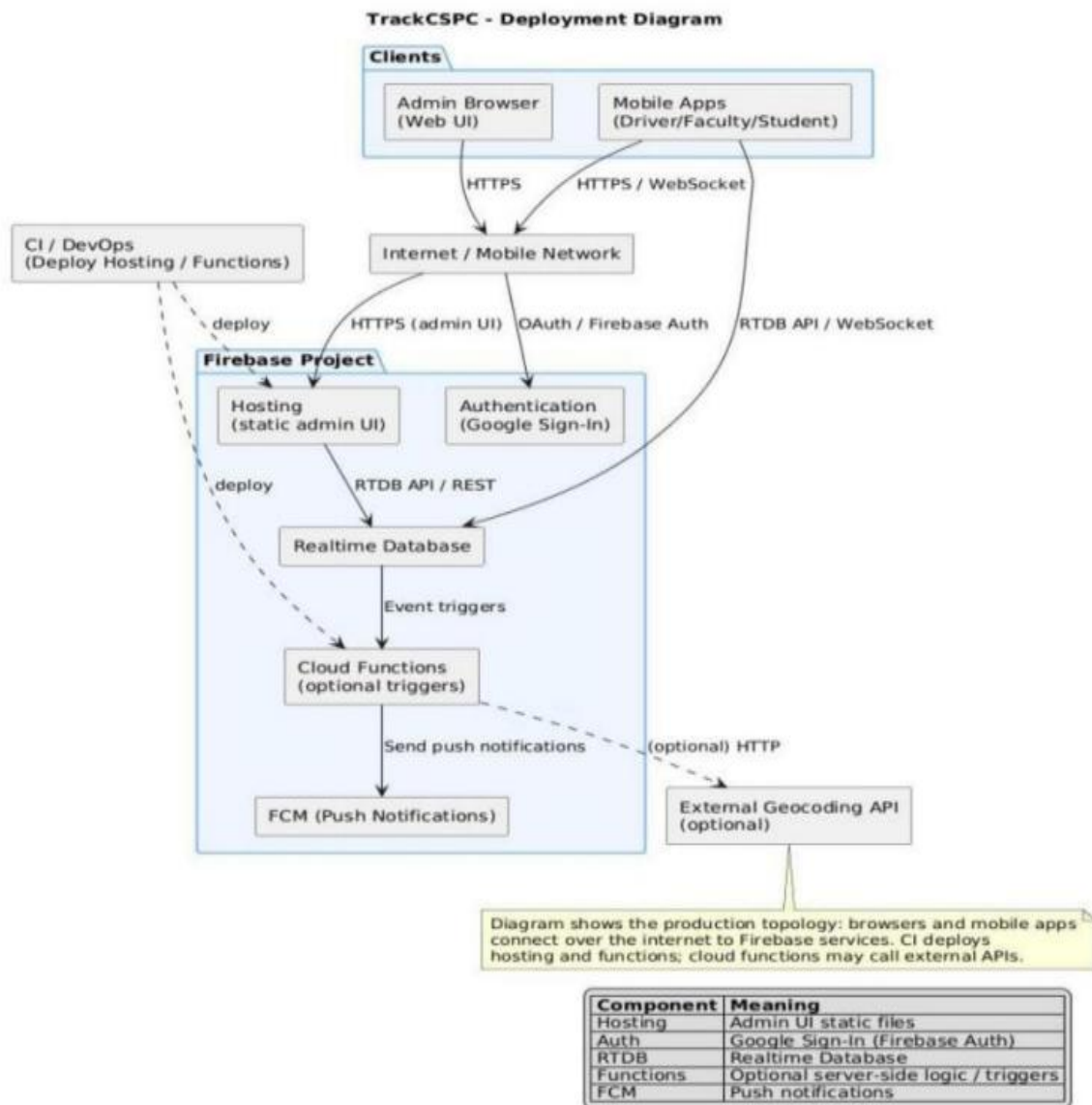


Figure 36. **Deployment Diagram**

The deployment diagram illustrates how the TrackCSPC system operates across its main components: the client devices, the internet, and the Firebase backend. Users—or mobile apps using secure HTTPS and WebSocket connections.

These requests pass through the internet and connect to the *Firebase Project*, where the system's core services are hosted. Firebase Hosting provides the admin web interface, while Firebase Authentication manages secure logins and role-based access. The Realtime Database handles all reservation and vehicle-tracking data, ensuring instant synchronization across all connected devices.

Optional Cloud Functions automate backend tasks such as sending push notifications through Firebase Cloud Messaging. The system may also interact with an external geocoding API for additional location-based features.

Notes

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CHAPTER 5

SUMMARY, FINDINGS, CONCLUSIONS AND RECOMMENDATIONS

This chapter presents the summary of the study, findings, conclusions, and recommendations regarding the design, development, and evaluation of TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform for Camarines Sur Polytechnic Colleges – General Services Unit (GSU). All information in this chapter was gathered from the processes of system development, deployment, testing, and user evaluation.

5.1 Summary

This study presented the development and evaluation of TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform, which was designed to modernize and streamline the transportation operations of the Camarines Sur Polytechnic Colleges – General Services Unit (GSU). The project addressed several long-standing issues in the traditional manual process, such as slow request handling, lack of centralized documentation, difficulty in monitoring ongoing trips, and limited transparency for both requestors and administrators.

TrackCSPC was developed as an integrated solution composed of a mobile application for submitting reservation requests and tracking status updates, and a web-based administrative dashboard equipped with tools for approving reservations, assigning drivers, managing vehicle availability, reviewing trip histories, and monitoring real-time vehicle locations using GPS technology. The development followed a systematic software engineering approach, ensuring that system requirements, interface design, implementation, and testing adhered to institutional needs and user expectations.

Through pilot deployment and user evaluation, the system demonstrated significant improvements in operational efficiency, data accuracy, and communication between requestors, drivers, and the GSU. Overall, the study successfully provided a functional and user-friendly platform that enhanced campus transportation management. Tables included in this chapter were

aligned to the left and presented with clear headers to ensure readability and proper interpretation of results.

5.2 Findings

The primary goal of the researchers was to design and develop TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform that would streamline reservation processing and enhance the overall transportation operations of the Camarines Sur Polytechnic Colleges – General Services Unit (GSU). Based on the system development process, pilot implementation, and user evaluation, the following findings were established:

1. The analysis of the existing manual vehicle reservation and tracking process used by the General Services Unit (GSU), conducted through interviews and questionnaires, revealed several operational issues. The results of interviews and questionnaires validated the need for a digital and centralized solution to address these operational challenges.
2. Based on the identified challenges, the TrackCSPC mobile application and web-based platform were successfully designed and implemented. The integration of automated reservation processing and real-time GPS-based vehicle monitoring provided a centralized solution that effectively addressed the limitations of the previous manual process and improved transportation management and coordination within the campus.
3. The usability and performance of TrackCSPC were evaluated through functional testing, load testing, and User Acceptance Testing (UAT). The results confirmed that the system is functional, reliable, and user-friendly across both mobile and web platforms.
 - 3.1 The system obtained a mean score of 1.00 from IT experts, interpreted as Completely Functional, indicating that all major modules operated correctly and complied with the required specifications.
 - 3.2 Students, GSU staff, drivers, and IT experts rated the system with a mean score of 1.00, interpreted as Completely Functional, confirming that the system handled

multiple users and simultaneous operations effectively. However, faculty respondents recorded a mean score of 0.00, interpreted as Not Completely Functional, indicating the need for minor refinements to better address specific faculty-related usage requirements.

3.3 The UAT results showed high levels of user satisfaction, ease of navigation, and successful task completion across both mobile and web platforms. These findings confirmed that the system interface and functionality were acceptable and suitable for daily campus transportation operations.

4. The successful pilot implementation of TrackCSPC within Camarines Sur Polytechnic Colleges demonstrated that the system is operationally ready for institutional use. Actual user interaction during deployment confirmed the system's applicability to real-world transportation operations and its capability to support the GSU's daily vehicle reservation and monitoring activities.

The findings indicate that the TrackCSPC system successfully met its objectives, addressed the limitations of the previous manual process, and significantly improved the efficiency, accuracy, and transparency of campus transportation management.

5.3 Conclusions

Based on the findings of the study, the researchers arrived at the following conclusions regarding the design, development, and evaluation of TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform for Camarines Sur Polytechnic Colleges – General Services Unit (GSU):

1. In relation to the analysis of the existing vehicle reservation and tracking system, this study concludes that the manual process currently used by the General Services Unit (GSU) at Camarines Sur Polytechnic Colleges is inefficient and prone to delays, inconsistent documentation, and limited monitoring capability.

The results of interviews and questionnaires clearly established the necessity of a digital and centralized solution to address these operational challenges.

2. In relation to system development, the TrackCSPC mobile application and web-based platform were successfully designed and implemented based on the identified limitations of the manual process. The integration of automated reservation processing and real-time GPS-based vehicle monitoring provided a centralized system that effectively improved coordination, monitoring, and overall transportation management within the campus.
3. In relation to system evaluation, the results of functional testing, load testing, and User Acceptance Testing (UAT) confirmed that TrackCSPC is functional, reliable, and user-friendly across both mobile and web platforms. The system met its intended requirements and enabled students, faculty, drivers, and GSU staff to perform their respective tasks efficiently, validating the system's usability and performance.
4. In relation to system deployment, the successful pilot implementation of TrackCSPC within Camarines Sur Polytechnic Colleges demonstrated that the system is operationally ready for institutional use. Actual user interaction during deployment confirmed its applicability to real-world transportation operations and its capability to support the GSU's daily vehicle reservation and monitoring activities.

In conclusion, TrackCSPC proved to be an efficient, secure, and innovative platform that enhances operational performance, automates reservation workflows, strengthens vehicle monitoring, and improves user experience within CSPC. The study confirms that the system is ready for deployment and has the potential to significantly transform the General Services Unit's transportation management processes.

5.4 Recommendations

In line with the conclusions of this study, the researchers recommend the following actions to further enhance and strengthen the TrackCSPC Smart Campus Vehicle Tracking and Reservation System.

1. Provide regular training and orientation programs for students, faculty, drivers, GSU staff, and administrators to ensure proper use of the system, familiarize users with new features, and maximize the platform's benefits for campus transportation.
2. Implement continuous system maintenance and scheduled updates to ensure security, reliability, and optimal performance. This includes resolving bugs, improving system responsiveness, and upgrading database structure and security protocols.
3. Enhance mobile compatibility across major device ecosystems, ensuring smooth operation on Huawei, iPhone (iOS), and Xiaomi (Android) devices. This will allow seamless access regardless of brand or operating system.
4. Improve GPS mapping capabilities, particularly in accurately reading and processing latitude and longitude coordinates, to refine real-time vehicle location updates, route visualization, and tracking precision.
5. Explore future improvements such as adding online payment options, offline mobile access, AI-based trip scheduling, and better reporting tools to make the system more efficient and easier to use.
6. Regularly check the system and gather user feedback to fix issues, improve the experience, and make sure it continues to meet the needs of the campus.

By following these recommendations, Camarines Sur Polytechnic Colleges can make full use of the TrackCSPC platform, ensure safe and efficient campus transportation, and continue improving the system and user experience for everyone who uses it.

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APPENDICES

APPENDIX A

EVALUATION TOOL

System Testing Questionnaire

SYSTEM TESTING QUESTIONNAIRE

Name: _____

Email Address: _____

Role: _____

College/Unit: _____

Instructions:

Please perform the following tasks using the system.

After completing each task, mark ☒ under Pass (1) if the module works correctly, or

Fail (0) if it does not.

Functionality Testing

Module	Task Description	Score (1 + Pass, 0 = Fail)	Remarks
User Authentication	Logged in successfully using student credentials		
	Logged out successfully using student credentials		
Real - time GPS Tracking	Viewed accurate vehicle location		

	Viewed updated vehicle location		
Route and Schedule Display	Viewed correct route information		
	Viewed correct schedule information		
Notification and Alert Module	Notification are received when request status changes		
Admin Dashboard	Viewed system information correctly (<i>student-accessible view</i>)		
Vehicle Reservation	Successfully made a vehicle reservation		
	Successfully viewed vehicle reservation		
Parental Monitoring	Parent-related tracking or viewing function worked correctly		

User Acceptance Testing (UAT)

UAT ID	Acceptance Statement	Score (1 + Pass, 0 = Fail)	Remarks
ID 1	The system is easy to navigate and user - friendly		

ID 2	The system supports campus vehicle request procedure effectively		
ID 3	The system reduces time and processes compared to manual procedures		
ID 4	Date entered into the system is saved correctly		
ID 5	The system improves communication effectiveness between administrator and end - users		
ID 6	The system reduces confusion when multiple requests are submitted simultaneously		
ID 7	The system meets the transportation needs for the campus		

Load Testing

Item	Load Test Statement	Score (1 + Pass, 0 = Fail)	Remarks
ID 1	The system remains stable without crashing during peak load conditions		
ID 2	Mobile app access under slow network conditions		



ID 3	Web Dash board receives simultaneous GPS updates from all active vehicles		
ID 4	GSU Staff uploads attachments (e.g, travel order documents) during request submission		
ID 5	Driver update trip start and end times while GPS is being updated		
ID 6	Multiple user log in repeatedly at the same time		
	Multiple user log out repeatedly at the same time		
ID 7	System performs real - time notifications to multiple users at once		

Manuscript Guide Questionnaire

Team Members:

- Ronan Gabriel V. Paat – *Project Head*
- Raxine D. Beriña
- Debie S. Layron

MANUSCRIPT GUIDE QUESTIONNAIRE

current process

(starting from **reserving a vehicle** until to the **process of returning the vehicle**, what happens next?)

further explanation of the Vehicle Request Form

(detailed explanation of the form for example what does PR No. mean?)

the management of the requests

(how do you manage the list of requests? manual listing? or _____??)

driver allocation

(how do you choose the person to be the driver in that certain trip? driver's availability? if yes, is there a list of driver's schedules or are they required to be on standby 24/7? or you'll just contact them and tell "may byahé ka sa gantong araw" and ask for their availability?)

driver per trip

(how many driver/s are required to be in a certain trip?)



passenger checking

(how do you verify that the passengers boarding the vehicle match the names on the passenger list?)

expenses tracking

(how do you calculate the expenses to spend on a certain trip based on the Vehicle Request Form?)

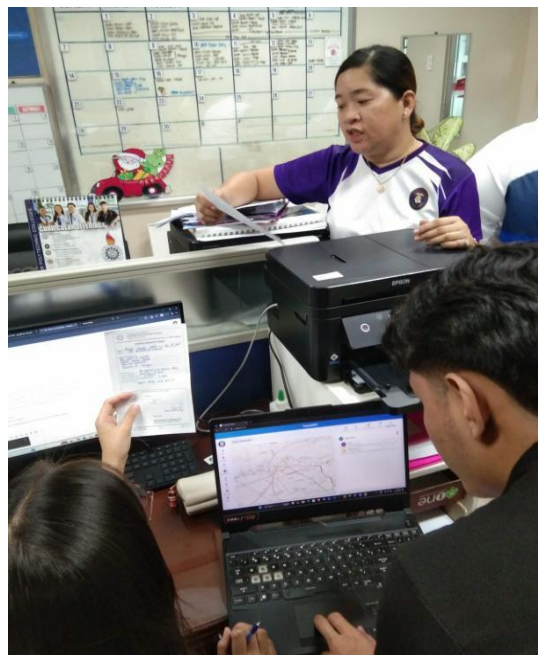
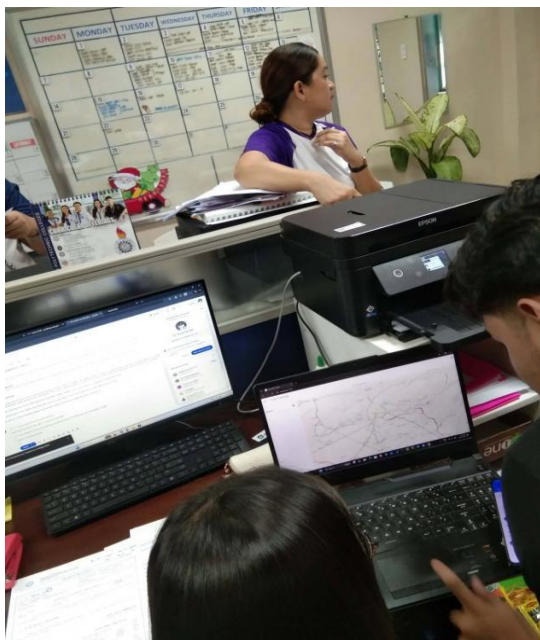
trip emergencies

(how does the driver communicate in case of a vehicle issue or any emergency situation during the trip?)

APPENDIX B

DOCUMENTATION

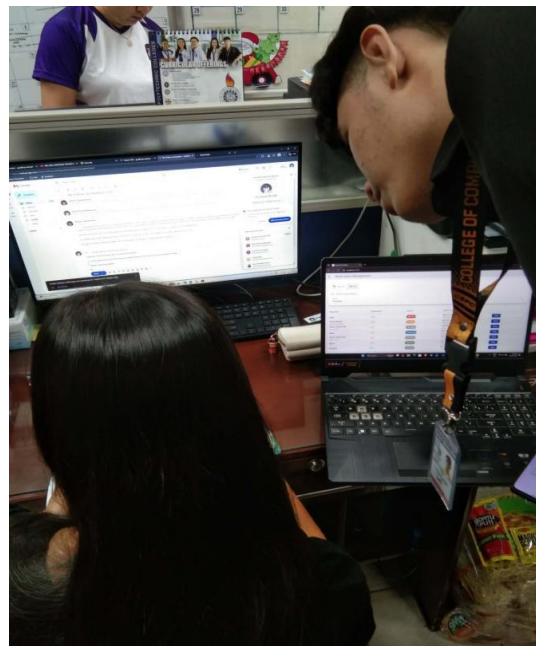
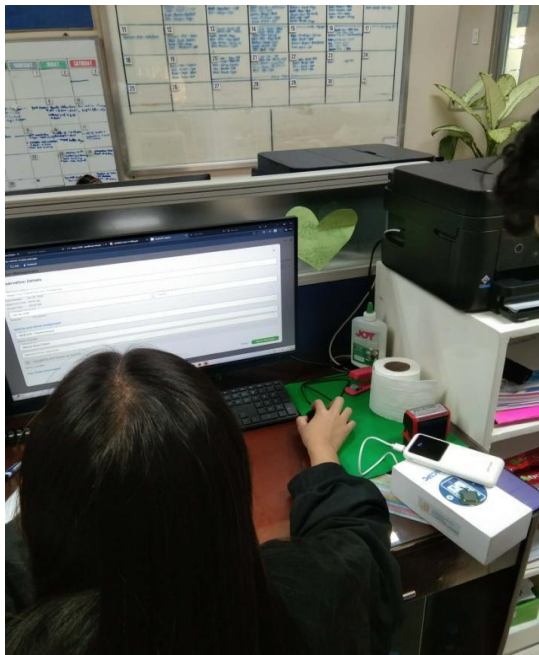
Researchers Conducting suggestions with the client



Researcher Conducting survey questionnaire in GSU employee



Researcher Turn over TRACKCSPC to the GSU Office



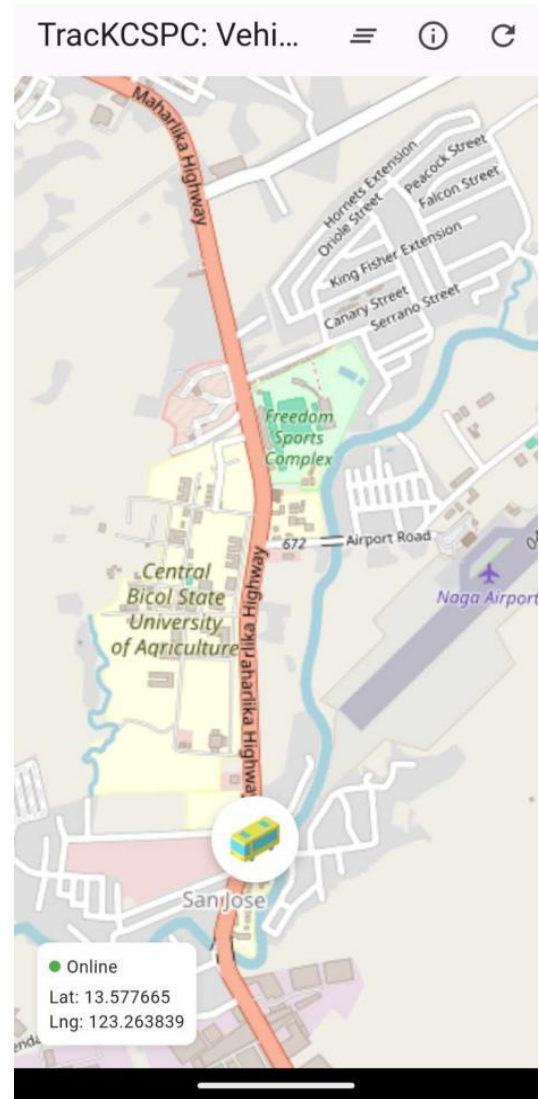
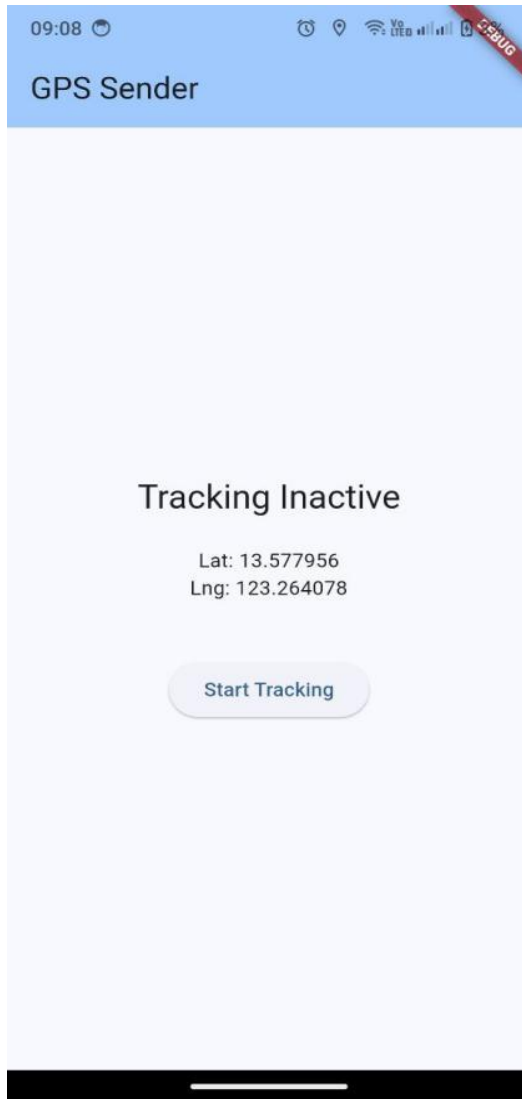
The researcher successfully passed the Title and Final Defenses



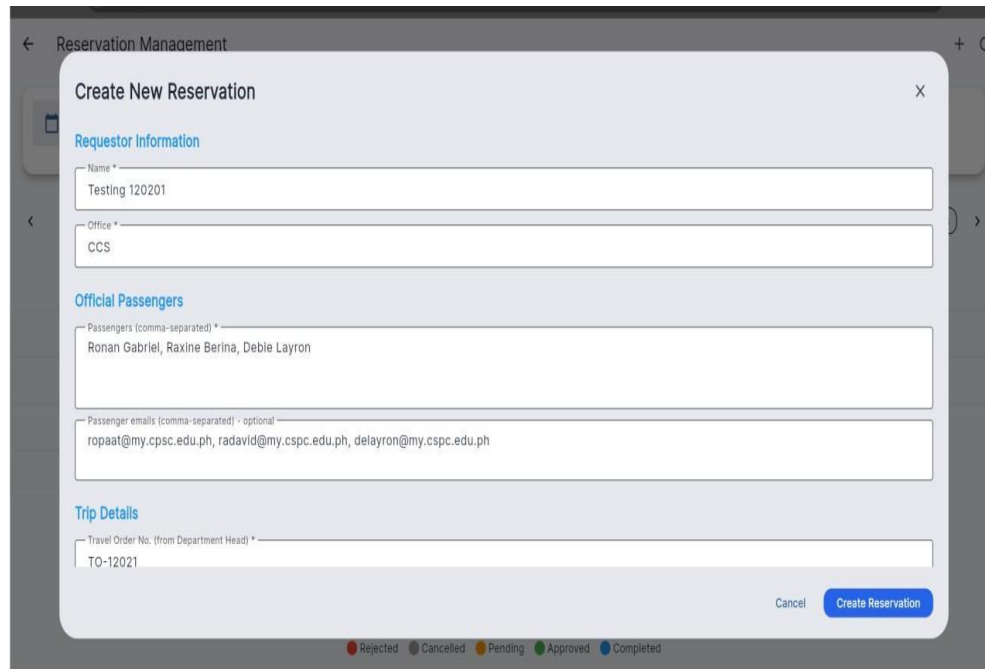
APPENDIX C

SAMPLE INPUT/OUTPUT/REPORTS

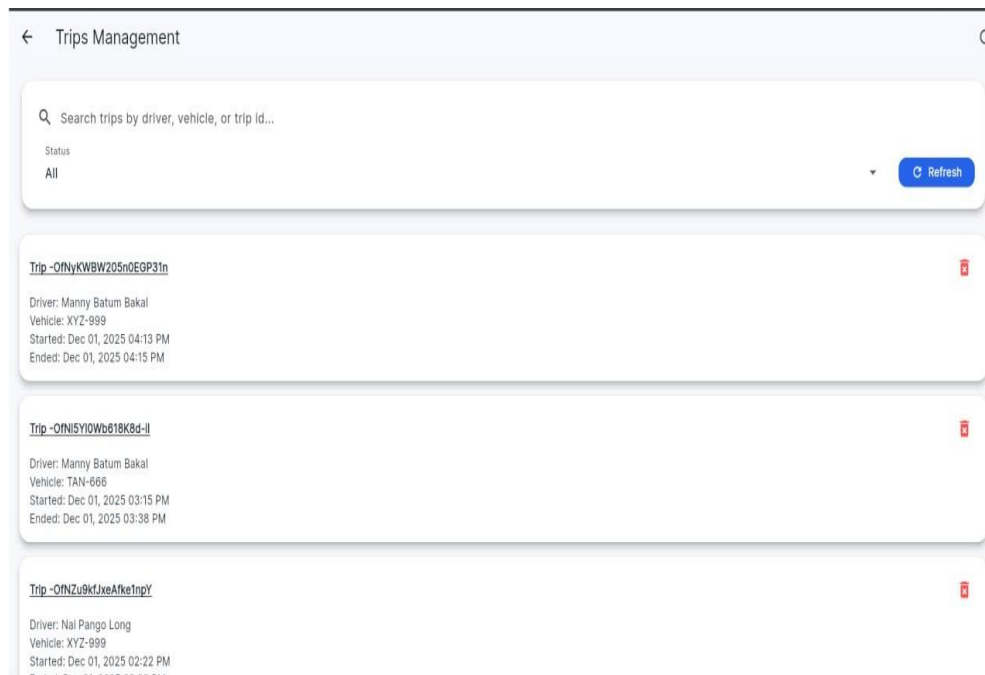
(A) Input



(B) Output

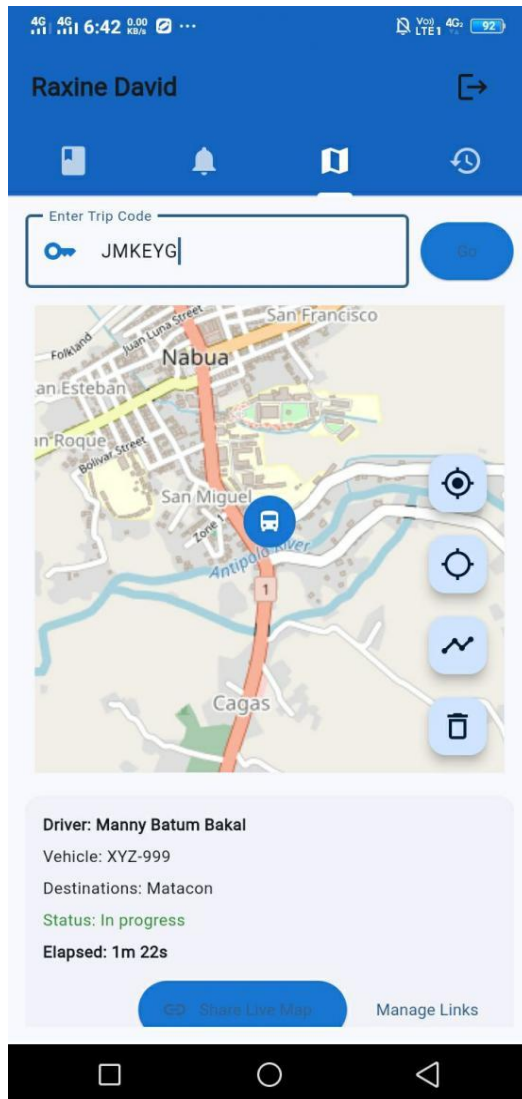


The screenshot shows a mobile application interface for 'Reservation Management'. A modal titled 'Create New Reservation' is open. It contains three sections: 'Requestor Information' with fields for 'Name' (Testing 120201) and 'Office' (CCS); 'Official Passengers' with fields for 'Passengers (comma-separated)' (Ronan Gabriel, Raxine Berina, Debie Layron) and 'Passenger emails (comma-separated) - optional' (ropaat@my.cpsc.edu.ph, radavid@my.cpsc.edu.ph, delayron@my.cpsc.edu.ph); and 'Trip Details' with a field for 'Travel Order No. (from Department Head)' (TO-12021). At the bottom right are 'Cancel' and 'Create Reservation' buttons. A legend at the very bottom shows status icons: Rejected (red), Cancelled (grey), Pending (yellow), Approved (green), and Completed (blue).



The screenshot shows the 'Trips Management' screen of the app. It features a search bar at the top, a status filter dropdown set to 'All', and a 'Refresh' button. Below is a list of three trips, each with a unique ID, driver name, vehicle, and start/end times.

Trip ID	Driver	Vehicle	Started	Ended
-OFNyKWBW205n0EGP3In	Manny Batum Bakal	XYZ-999	Dec 01, 2025 04:13 PM	Dec 01, 2025 04:15 PM
-OFNl5YlOWb618K8d-II	Manny Batum Bakal	TAN-666	Dec 01, 2025 03:15 PM	Dec 01, 2025 03:38 PM
-OFNZu9KfJxeAfkeTnpY	Nai Pango Long	XYZ-999	Dec 01, 2025 02:22 PM	Dec 01, 2025 02:33 PM



(C) Reports

Generate Report

Camarines Sur Polytechnic Colleges

General Services Unit

Report Date
2025-12-02

✓ Daily

Weekly

Monthly

2025-12-02 → 2025-12-02

✓ Include ended trips

Include in report

✓ Drivers

✓ Trips

✓ Reservations

✓ Vehicles

Generate PDF

Inspect data

Reset

Preview / Notes

This will generate a table of trips that started within the selected range. The PDF includes campus and department headers and the generated timestamp.

Drivers: 3 Trips: 0 Reservations: 0 Vehicles: 2

Trips Management

Search trips by driver, vehicle, or trip id...

Status

All

Refresh

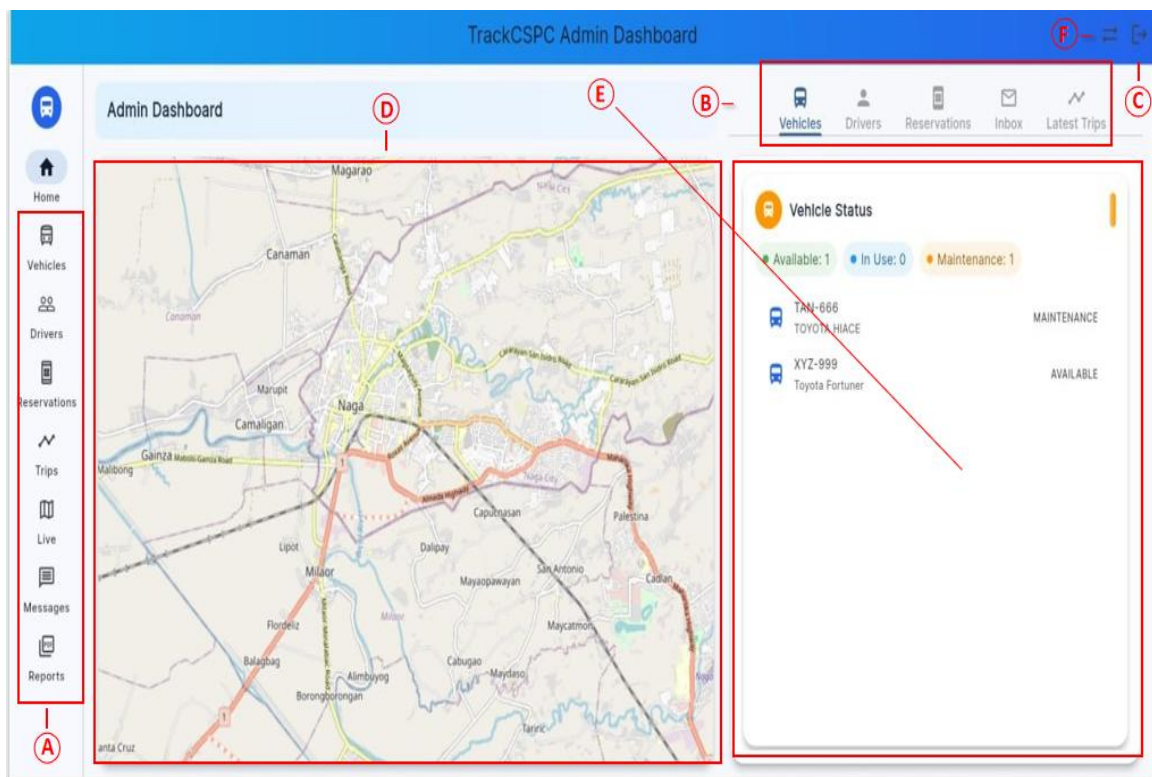
Trip ID	Requestor	Driver	Vehicle	Destination	Status
-OJUmvcifRkJ85mPQXMv	Ronan Gabriel Paat	Manny Batum Bakal	KOL 146	Unep	Completed View
-OJUILxdc-ZvVjKTKo8	Ronan Gabriel Paat	Manny Batum Bakal	SEA 1192	Secret	Completed View
-OJUOuTq8PqVFXU9W0IM	DSADDSDA	Manny Batum Bakal	SEA 1192	Pafcpic - Fort Bonifacio, Bayani Road, Taguig, Metro Manila, Philippines	Completed View
-OJTnWB4CYDBuZt8uMlw	AJEJEJE	Manny Batum Bakal	SHG 318	Villa Caceres Hotel, Magsaysay Avenue, Naga City, Camarines Sur, Philippines	Completed View
-OJTfSHcPWC3QmslQWd	AKO	Manny Batum Bakal	SHA 1063	Villa Caceres Hotel, Magsaysay Avenue, Naga City, Camarines Sur, Philippines	Completed View
-OJTcseRNF8xdQ-dNefK	Ronanult	Manny Batum Bakal	SEA 1192	Villa Caceres Hotel, Magsaysay Avenue, Naga City, Camarines Sur, Philippines	Completed View
-OJTaM2x4y6tTF8Una76	RONAN	Manny Batum Bakal	K1 J404	Villa Caceres Hotel, Magsaysay Avenue, Naga City, Camarines Sur, Philippines	Completed View

APPENDIX D

USER'S GUIDE

Dashboard

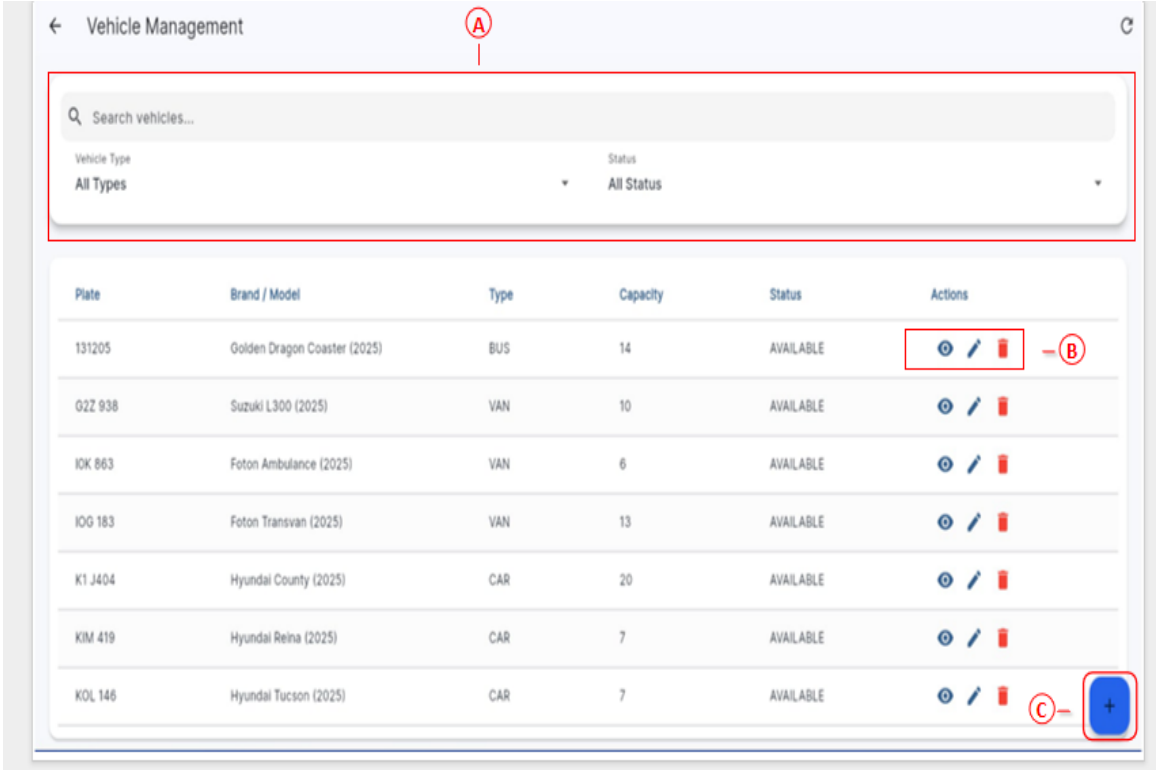
The Dashboard serves as the main interface of the system, providing users with an overview of vehicle status, navigation options, and real-time information. It allows users to easily monitor vehicles and access system features efficiently.



- A. **Side Panel** - Contains the main menu for navigating different system features.
- B. **Tabbed Panel** - Displays organized tabs for quick access to related information.
- C. **Sign Out** - Allows the user to safely log out of the system.
- D. **Map** - Shows the real-time location and movement of vehicles.
- E. **Tabbed Panel' Card** - Displays the current information on each tabbed panel
- F. **Reconcile Button** - Updates the dashboard to show the latest data.

Vehicle Management

This section allows the user to manage, organize, and maintain vehicle information within the system. It enables the user to efficiently monitor vehicle records, view detailed vehicle information, and update necessary details as needed. This section also helps ensure that all vehicle data remains accurate, organized, and up to date for proper system operation and decision-making.



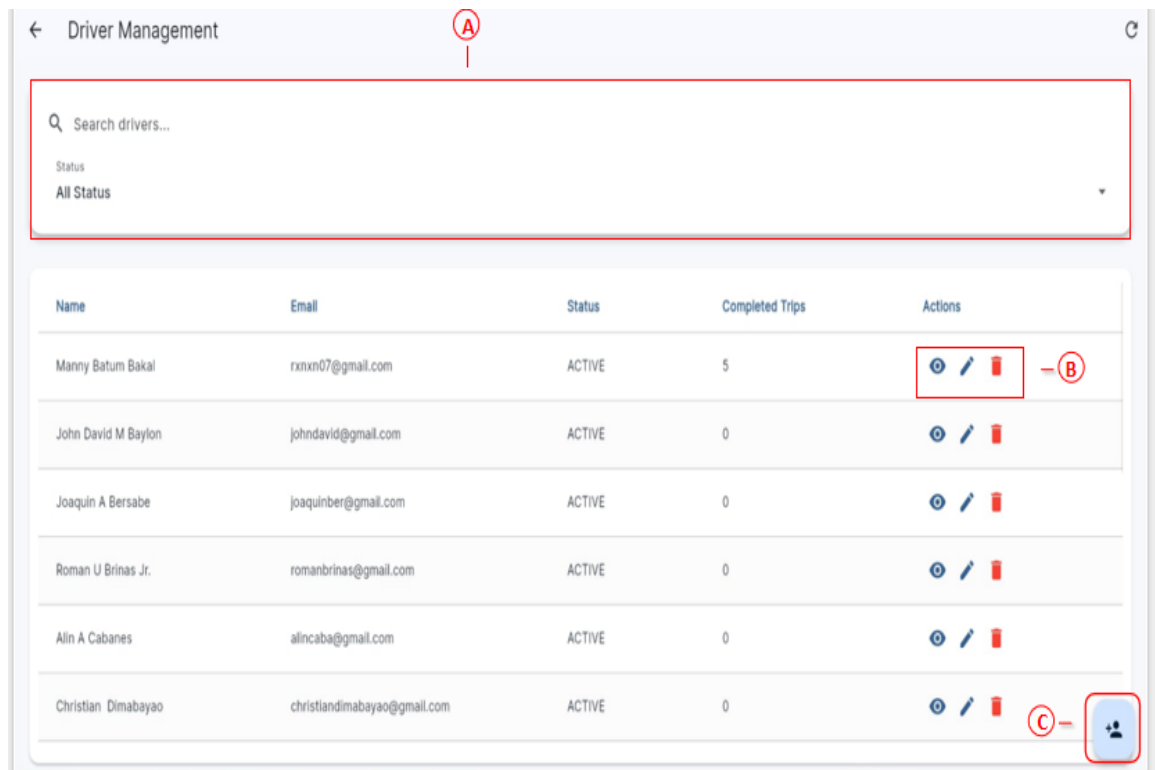
The screenshot displays the 'Vehicle Management' interface. At the top, there is a search bar labeled 'Search vehicles...' and two dropdown filters: 'Vehicle Type' (set to 'All Types') and 'Status' (set to 'All Status'). Below these filters is a table with the following columns: Plate, Brand / Model, Type, Capacity, Status, and Actions. The table lists seven vehicles, all with a status of 'AVAILABLE'. The 'Actions' column for each vehicle contains three icons: a magnifying glass (View), a pencil (Edit), and a trash can (Delete). A red box labeled 'A' highlights the search bar. A red box labeled 'B' highlights the action icons for the first vehicle (131205). A red box labeled 'C' highlights a blue button with a plus sign at the bottom right of the table, used for adding new vehicles.

Plate	Brand / Model	Type	Capacity	Status	Actions
131205	Golden Dragon Coaster (2025)	BUS	14	AVAILABLE	   
G2Z 938	Suzuki L300 (2025)	VAN	10	AVAILABLE	  
IKK 863	Foton Ambulance (2025)	VAN	6	AVAILABLE	  
IOG 183	Foton Transvan (2025)	VAN	13	AVAILABLE	  
K1 J404	Hyundai County (2025)	CAR	20	AVAILABLE	  
KIM 419	Hyundai Reina (2025)	CAR	7	AVAILABLE	  
KOL 146	Hyundai Tucson (2025)	CAR	7	AVAILABLE	  

- A. **Sort all type** - Allows the user to filter and search drivers based on status or name.
- B. **View, Edit, Delete** - Provides options to view driver details, update information, and remove a driver.
- C. **Add Vehicle** - Enables the user to add a new driver to the system.

Driver Management

The Vehicle Management section allows the user to manage, organize, and maintain all vehicle-related information within the system. This section enables the user to add new vehicles, view detailed vehicle records, update existing information, and remove vehicles when necessary. It also provides sorting and filtering options to easily monitor vehicle status, availability, and other relevant details, helping ensure accurate records and efficient vehicle monitoring.

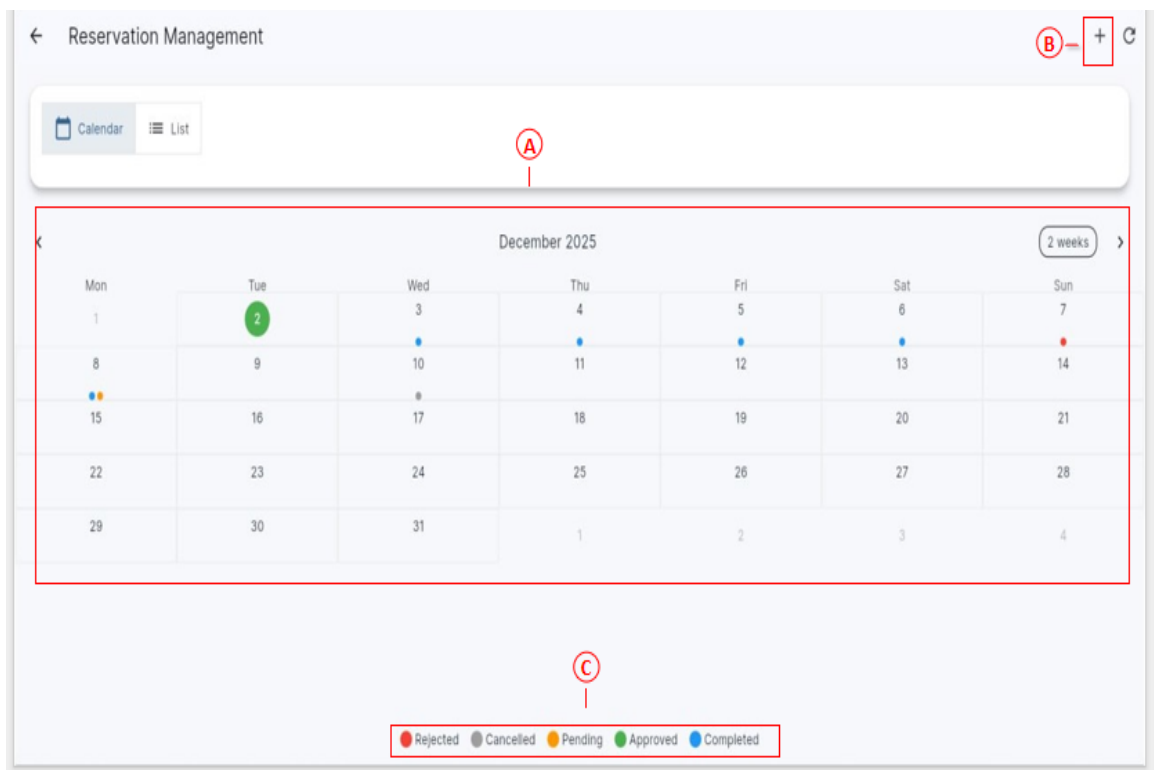


- A. **Sort** – Allows sorting and filtering of drivers.
- B. **View, Edit, Delete** – Provides actions to view details, edit information, or delete a driver.
- C. **Add Driver** – Used to add a new driver to the system.

Reservation Management

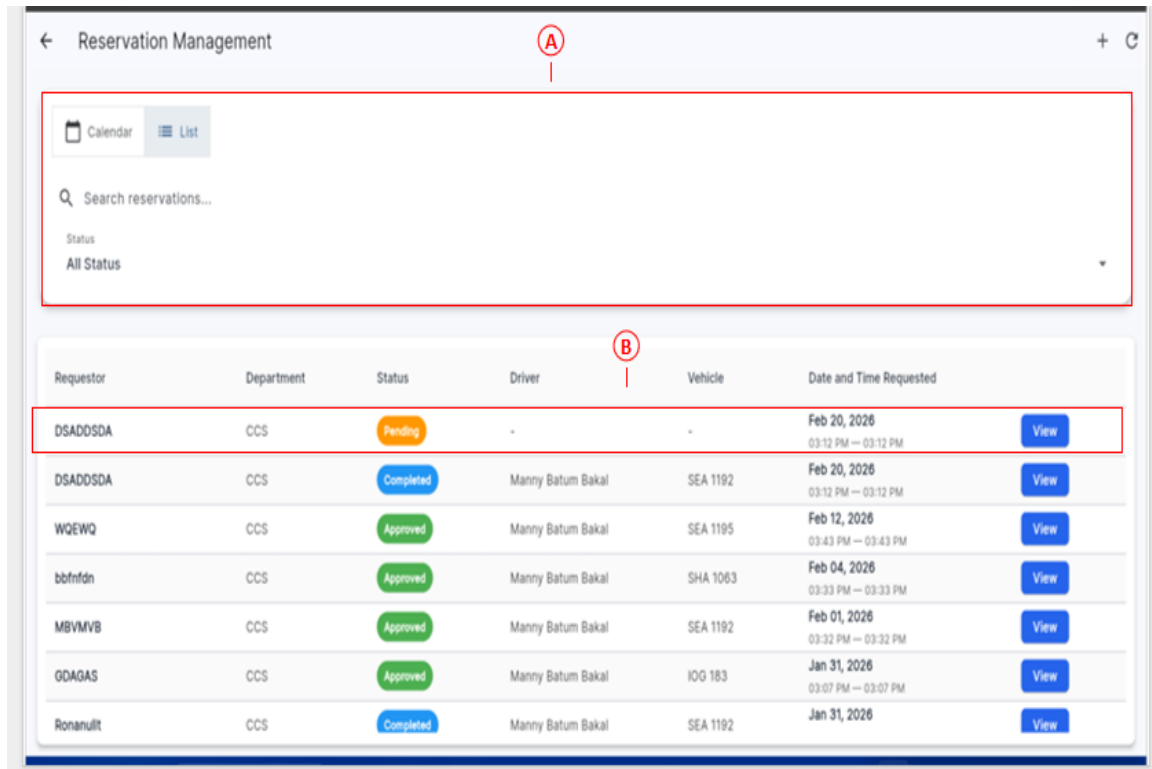
The **Reservation Management** section allows the user to create, view, monitor, and manage vehicle reservation requests within the system. This section helps ensure proper scheduling, tracking, and organization of reservations, making it easier to monitor reservation status and vehicle availability.

Calendar View



- A. **Calendar** - Displays reservation schedules by date for easy monitoring.
- B. **Legend** - Shows the color indicators and their corresponding reservation status.
- C. **Add Reservation** - Allows the user to create a new reservation request.

List View



Reservation Management

Calendar List

Search reservations...

Status
All Status

Requestor	Department	Status	Driver	Vehicle	Date and Time Requested	
DSADSDA	CCS	Pending	-	-	Feb 20, 2026 03:12 PM -- 03:12 PM	View
DSADSDA	CCS	Completed	Manny Batum Bakal	SEA 1192	Feb 20, 2026 03:12 PM -- 03:12 PM	View
WQEWQ	CCS	Approved	Manny Batum Bakal	SEA 1195	Feb 12, 2026 03:43 PM -- 03:43 PM	View
bbfnfdn	CCS	Approved	Manny Batum Bakal	SHA 1063	Feb 04, 2026 03:33 PM -- 03:33 PM	View
MBVMVB	CCS	Approved	Manny Batum Bakal	SEA 1192	Feb 01, 2026 03:32 PM -- 03:32 PM	View
GDAGAS	CCS	Approved	Manny Batum Bakal	IOG 183	Jan 31, 2026 03:07 PM -- 03:07 PM	View
Ronanult	CCS	Completed	Manny Batum Bakal	SEA 1192	Jan 31, 2026	View

- A. **Sort All Type** - Use this section to search reservations and filter by status (All, Pending, Completed, Approved).
- B. **View Reservation Status** - This table displays all reservations with their status, requester, driver, vehicle, and date/time.

Trips Management

The Trips Management section allows the user to monitor and manage all trips made using the system. It provides tools for searching, filtering, and viewing detailed trip information, helping users keep track of trip progress and completed activities. This section ensures that trip records are organized, easily accessible, and accurately maintained for reporting and operational purposes.

Trips Management

Search trips by driver, vehicle, or trip id...

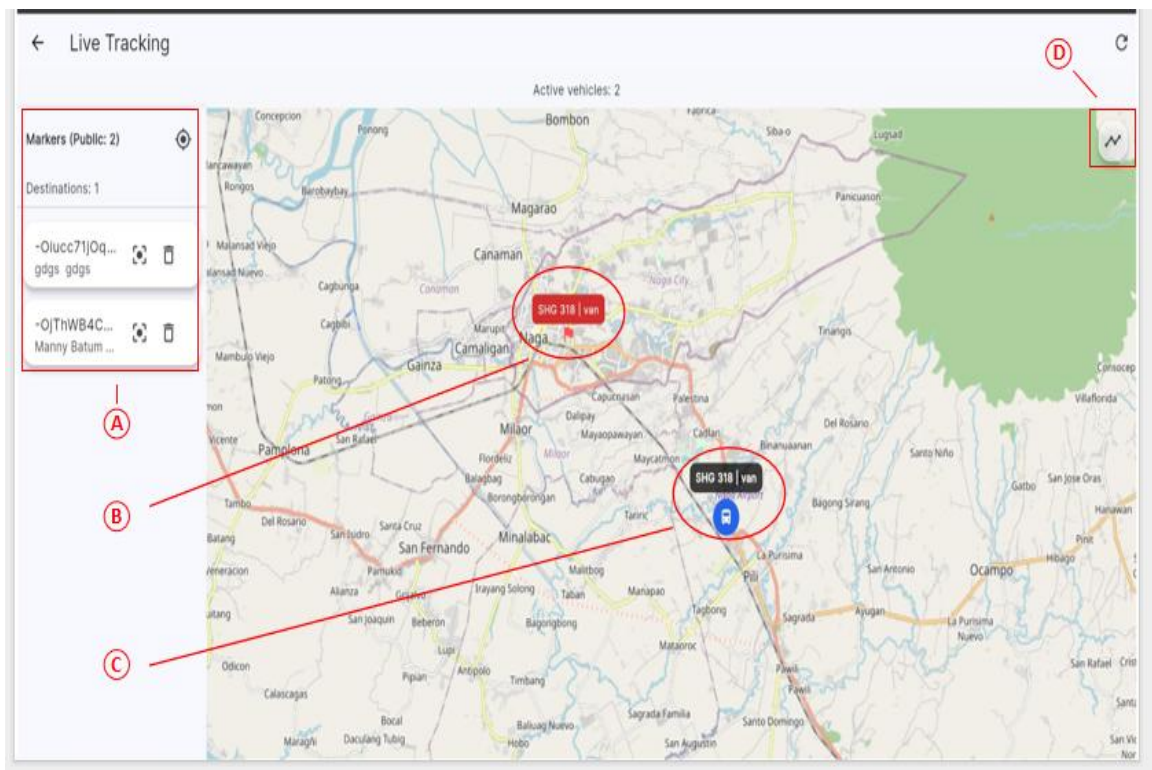
Status: All Refresh

Trip ID	Requestor	Driver	Vehicle	Destination	Status
-OJUmvcHfRkJB5mPQXMv	Ronan Gabriel Paat	Manny Batum Bakal	KOL 146	Unep	Completed View
-OJUILxdc-ZvVjKTKo8	Ronan Gabriel Paat	Manny Batum Bakal	SEA 1192	Secret	Completed View
-OJUOuTq8PqVFXU9W0IM	DSADSDA	Manny Batum Bakal	SEA 1192	Patcpic - Fort Bonifacio, Bayani Road, Taguig, Metro Manila, Philippines	Completed View
-OJThWB4CYDBuZt8uMlw	AJEJEJE	Manny Batum Bakal	SHG 318	Villa Caceres Hotel, Magsaysay Avenue, Naga City, Camarines Sur, Philippines	Completed View
-OJTfShHCpWC3QmsIQWd	AKO	Manny Batum Bakal	SHA 1063	Villa Caceres Hotel, Magsaysay Avenue, Naga City, Camarines Sur, Philippines	Completed View
-OJTcseRNF8xdQ-dNefK	Ronanullit	Manny Batum Bakal	SEA 1192	Villa Caceres Hotel, Magsaysay Avenue, Naga City, Camarines Sur, Philippines	Completed View
-OJTaM2+4y8ITF8Una76	RONAN	Manny Batum Bakal	K1 J404	Villa Caceres Hotel, Magsaysay Avenue, Naga City, Camarines Sur, Philippines	Completed View

- A. **Search and Filter Trips** - Use this section to search trips by driver, vehicle, or trip ID and filter by status.
- B. **View Completed Trips** - This table shows the list of trips with details such as requester, driver, vehicle, destination, and current status (Completed).

Live Tracking Map

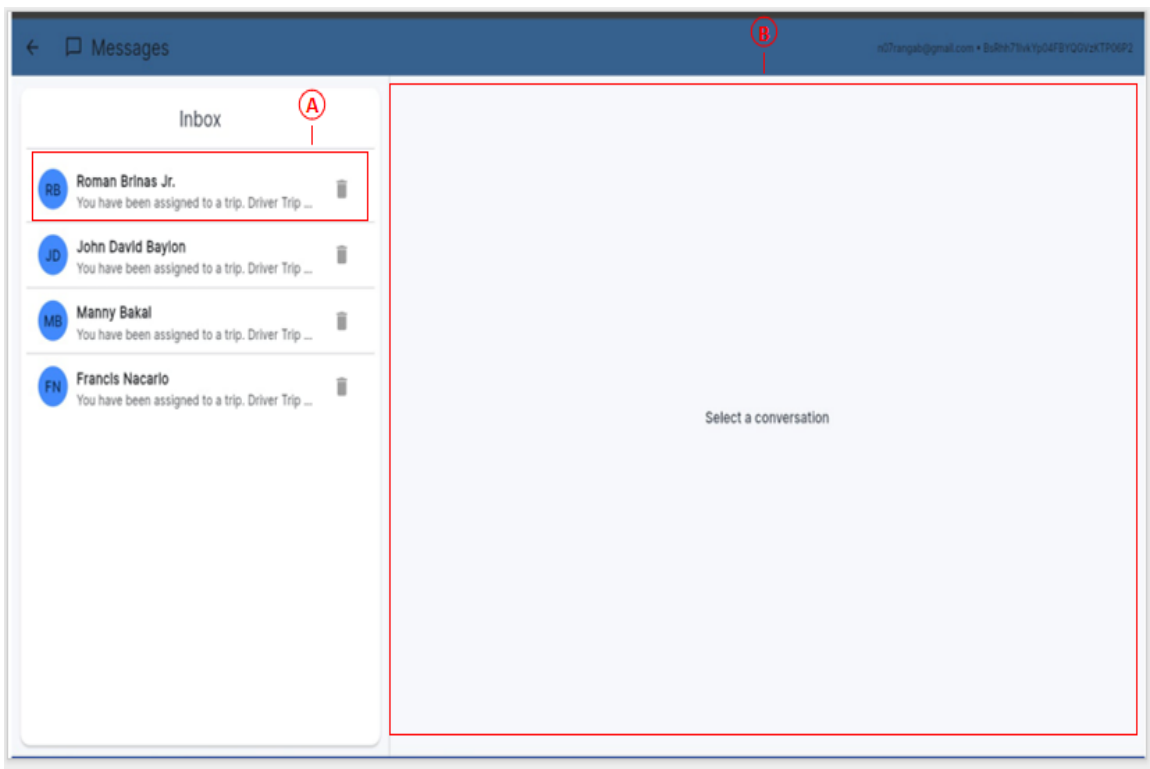
The Live Tracking section allows the user to monitor the real-time location and movement of active vehicles within the system. This feature provides visual representation of vehicle positions, routes, and destinations, helping users ensure timely monitoring, route management, and operational efficiency. Live Tracking improves oversight of trips and enhances coordination between drivers and system users.



- A. **Marker** - Shows the current location of active vehicles on the map.
- B. **Destination** - Indicates the target or assigned destination for the selected trip.
- C. **Start Destination** - Displays the starting point of the trip on the map.
- D. **Toggle Lines** - Turn route lines on or off to show or hide the travel path.

Message Inbox

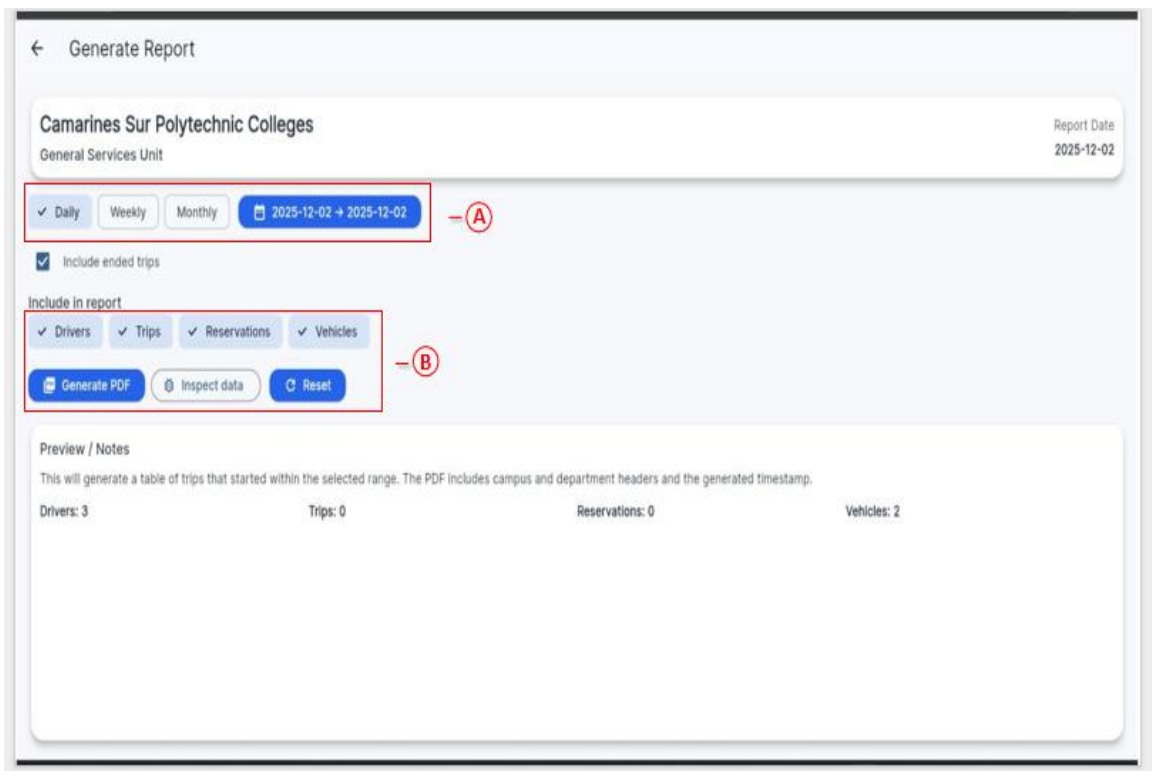
The Message Inbox section allows the user to communicate directly with drivers through the system. It provides a platform to send, receive, and manage messages in real time, ensuring smooth coordination and quick response between users and drivers. This feature helps maintain effective communication for trip updates, notifications, and other important information.



- A. **Message button** - Displays all your conversations; click a message to open it.
- B. **Conversation panel** - Shows the selected conversation where you can read and reply to messages.

Reports

The **Reports** section allows the user to generate detailed summaries of system activities and data for analysis and record-keeping. Users can customize reports by selecting the desired time period and the specific information to include, making it easier to monitor operations, track performance, and make informed decisions. This section ensures that all relevant system data can be compiled into organized, printable formats for future reference.



← Generate Report

Camarines Sur Polytechnic Colleges
General Services Unit

Report Date
2025-12-02

☒ Daily ☐ Weekly ☐ Monthly — **A**

☒ Include ended trips

Include in report

☒ Drivers ☒ Trips ☒ Reservations ☒ Vehicles

— **B**

Preview / Notes

This will generate a table of trips that started within the selected range. The PDF includes campus and department headers and the generated timestamp.

Drivers: 3 Trips: 0 Reservations: 0 Vehicles: 2

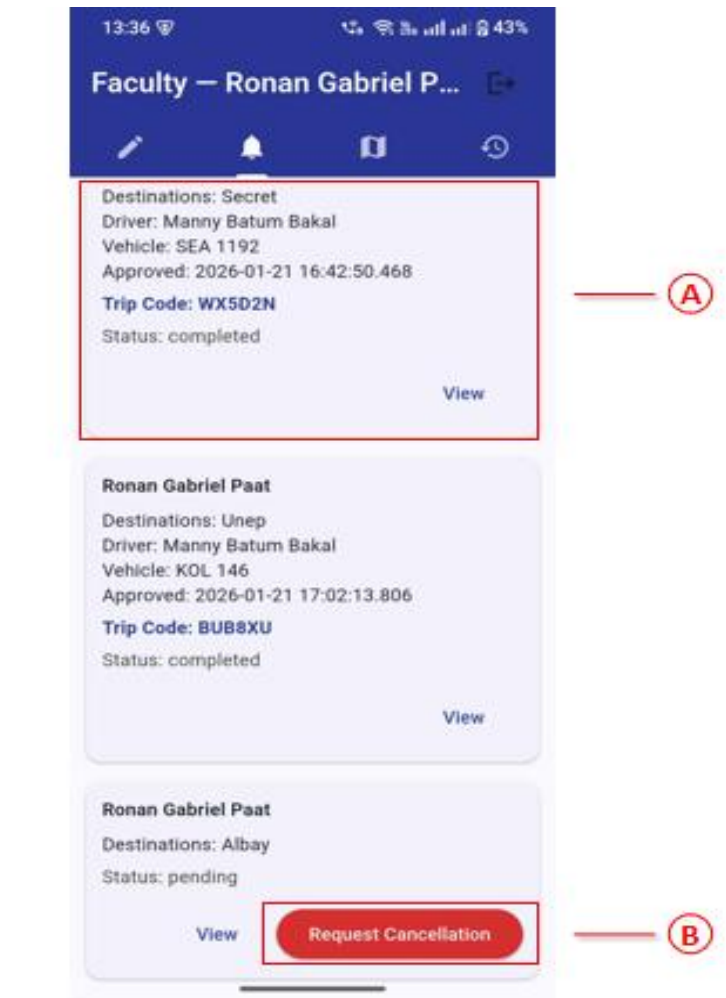
- A. **Date Report: Daily, Weekly, Monthly** - Choose the report period: Daily, Weekly, or Monthly, and select the date range.
- B. **Include Report Options** – Select the data to include in the report (Drivers, Trips, Reservations, Vehicles) and click **Generate PDF** to create the report.

Faculty Side

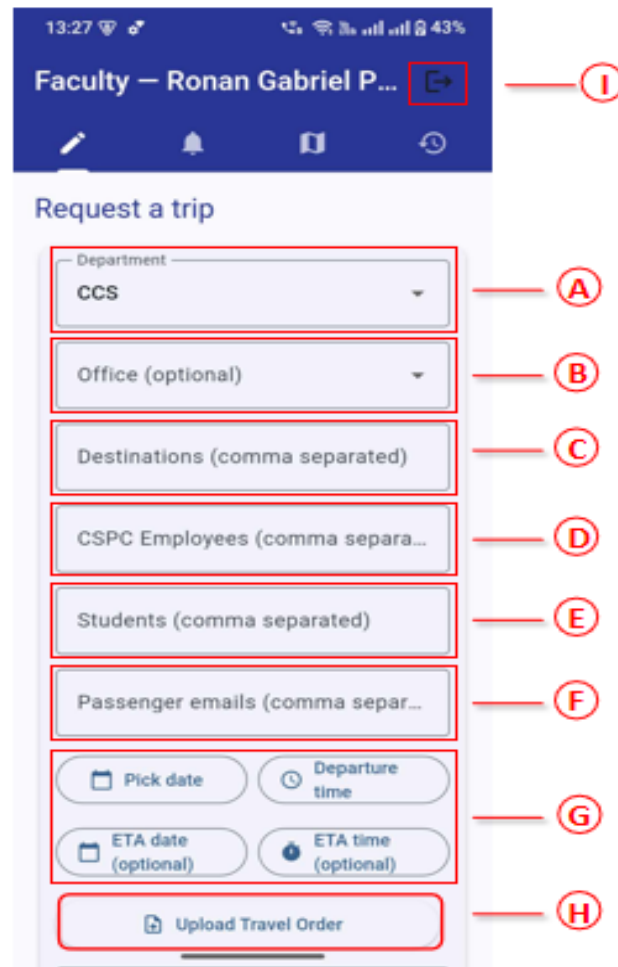
The **Faculty Side** section allows faculty members to manage and track their vehicle and travel requests within the system. This interface provides tools for signing in, submitting destination requests, entering detailed trip information, monitoring assigned trips, and reviewing travel history. It ensures that all travel-related activities are organized, accurately recorded, and easily accessible for both faculty and the system administrators.



- A. **CSPC Email Login** – Faculty, Student, Driver sign in using their official CSPC email by selecting **Sign in with Google**.

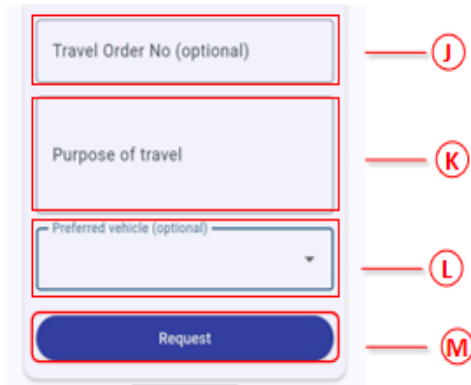


- A. **Destination Request** – Shows trip details such as destination, date, time, and vehicle information.
- B. **Request Cancellation** – Allows faculty to cancel a submitted request if needed.



The screenshot shows a mobile application interface for requesting a trip. At the top, a dark blue header bar displays the user's name 'Faculty — Ronan Gabriel P...' and a red box with a right-pointing arrow, labeled 'I'. Below the header is a white section titled 'Request a trip'. This section contains several input fields and buttons, each labeled with a red letter in a circle: 'A' for the 'Department' dropdown (currently showing 'CCS'), 'B' for the 'Office (optional)' dropdown, 'C' for the 'Destinations (comma separated)' text field, 'D' for the 'CSPC Employees (comma separa...)' text field, 'E' for the 'Students (comma separated)' text field, 'F' for the 'Passenger emails (comma separ...)' text field, 'G' for a group of four buttons: 'Pick date', 'Departure time', 'ETA date (optional)', and 'ETA time (optional)', and 'H' for the 'Upload Travel Order' button at the bottom. The status bar at the very top shows the time 13:27 and 43% battery.

- A. Department Dropdown** – Select your department.
- B. Offices Dropdown** – Choose the office responsible for the trip.
- C. Destination** – Enter the travel destination(s), separated by commas if more than one.
- D. Name of CSPC Employee** – Input the name(s) of the CSPC employee(s) involved.
- E. Name of Passenger (e.g., Student)** – Enter passenger name(s), if applicable.
- F. Passenger Email** – Provide the passenger's email address.
- G. Set Pickup & Travel Details** – Select pickup date, departure time, ETA date, ETA time
- H. Upload Travel Order** – Attach the travel order document (if available).
- I. Travel Order No.** – Enter the travel order number, if applicable.



Travel Order No (optional) — **J**

Purpose of travel — **K**

Preferred vehicle (optional) — **L**

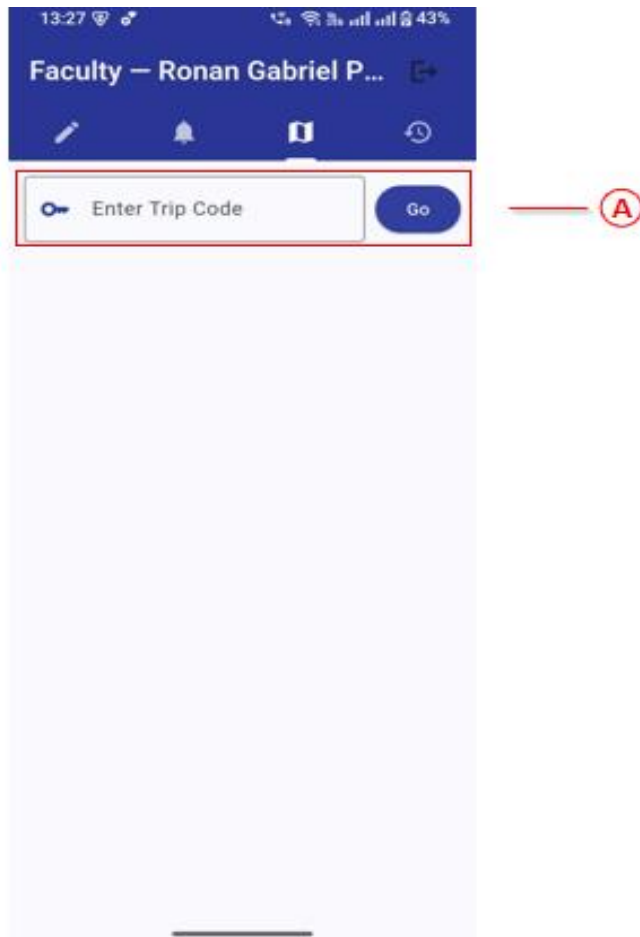
Request — **M**

J. Purpose of the Travel – Specify the reason for the trip.

K. Request Button – Click **Request** to submit the travel request.

L. Preferred Vehicle – Choose a preferred vehicle for the trip

M. Request Button – Finally save the reservation

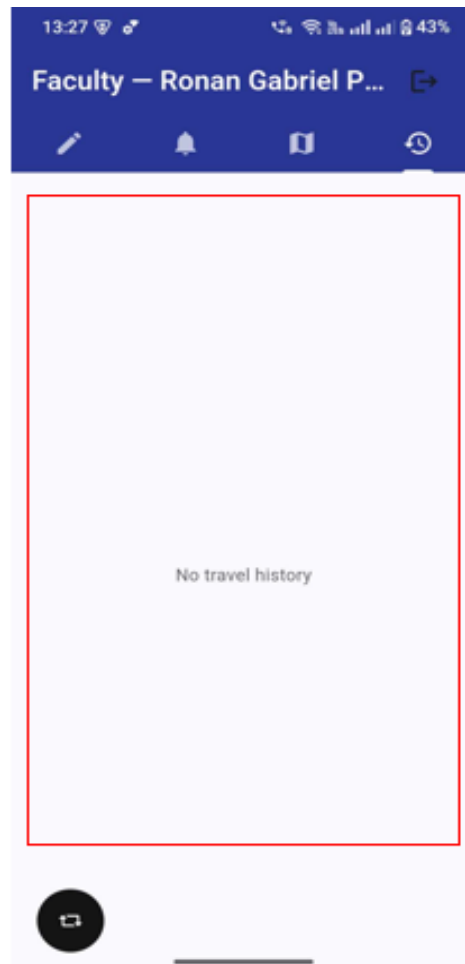


13:27 43%

Faculty — Ronan Gabriel P...

Enter Trip Code — **A** **Go**

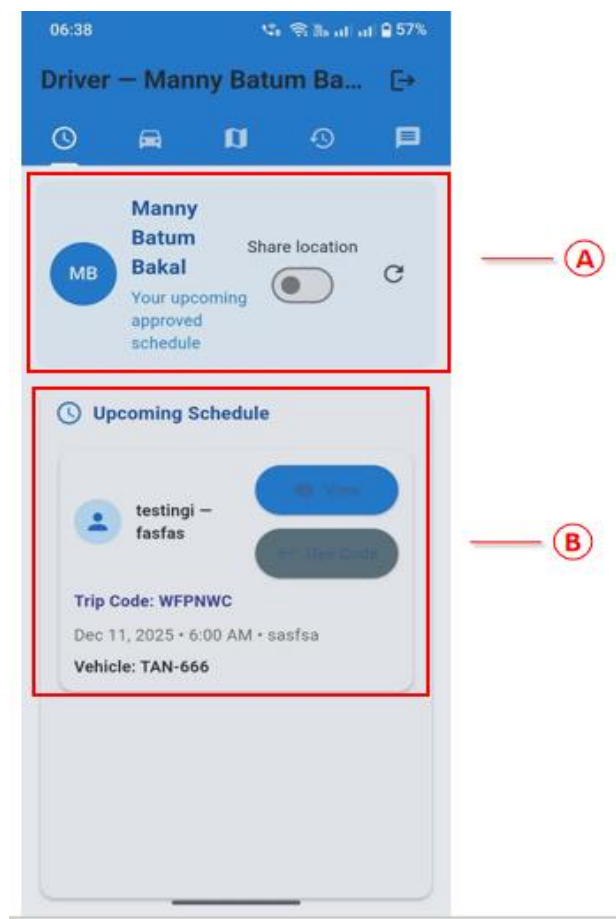
A. Enter Trip Code – Input the provided trip code to track the assigned trip.



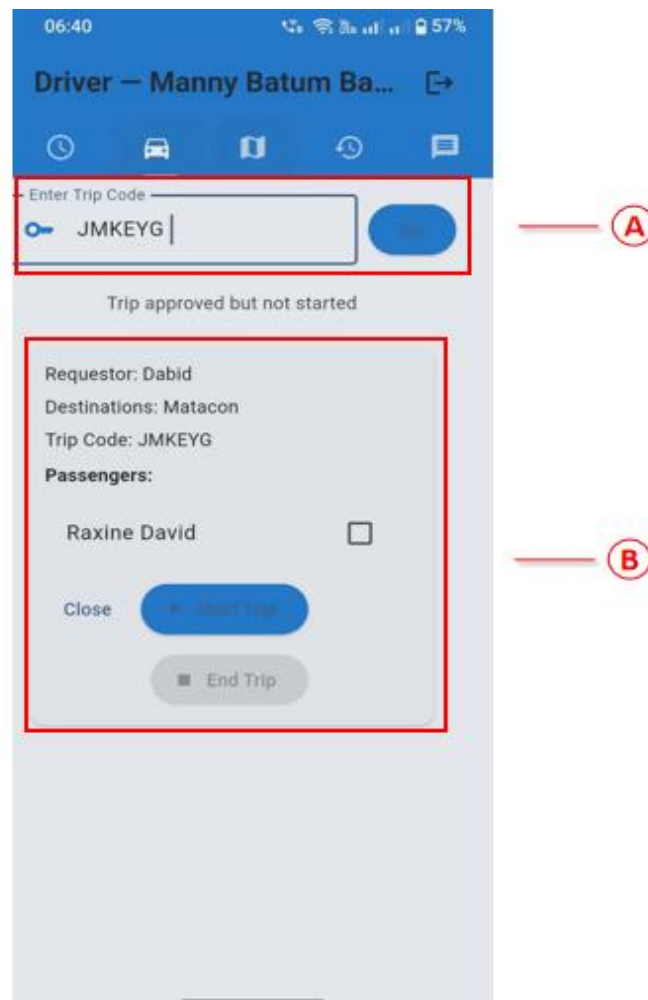
- A. History of Trips** – Displays previous and current trip records. If no trips are available, the screen will show *No travel history*.

Driver Side

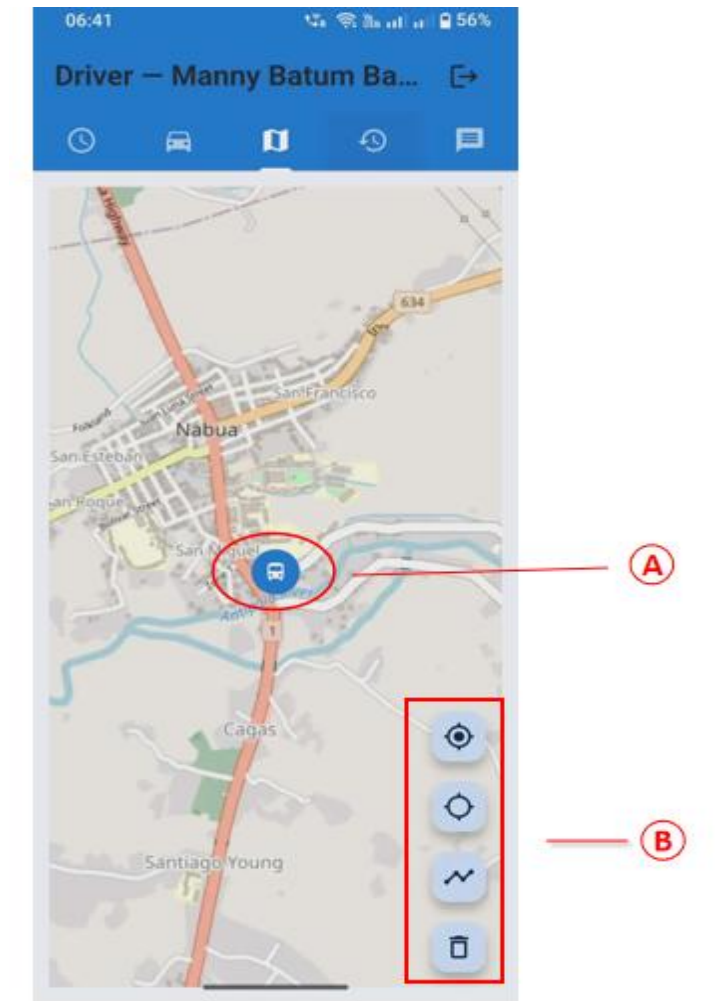
The **Driver Side** section allows drivers to manage and monitor their assigned trips, share their location, access navigation tools, and communicate with requestors through the system. This interface ensures that drivers can efficiently carry out travel requests, track their schedules, and maintain real-time updates for better coordination and trip management.



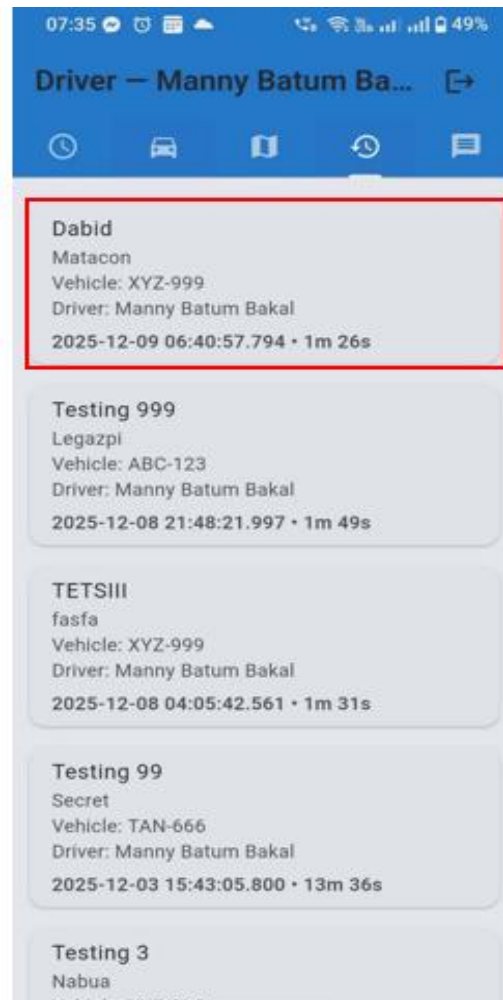
- A. **Share Location & Refresh Button** – Allows the driver to share real-time location and refresh trip details.
- B. **Upcoming Schedule** – Shows the driver's upcoming trips, including trip code, date, time, and vehicle information.



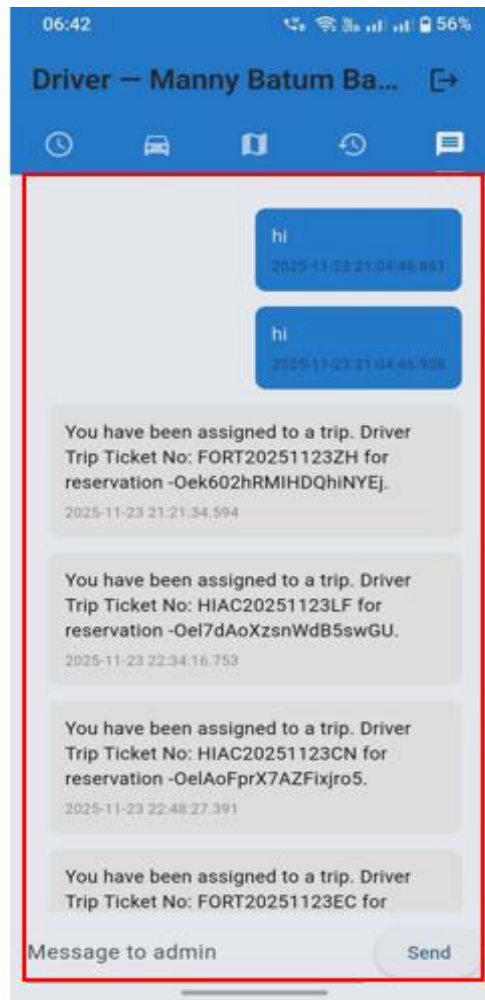
- A. **Trip Code** – Displays the assigned trip code used to identify the travel request.
- B. **Start Trip / End Trip** – Tap **Start Trip** to begin the journey and **End Trip** once the trip is completed.



- A. **Vehicle Location** – Displays the real-time location of the vehicle on the map.
- B. **Navigation Controls** – Provides map tools to assist the driver in route guidance.



- A. **Trip History Tab**– Shows the selected trip’s requestor information, including name, assigned vehicle, and schedule details.

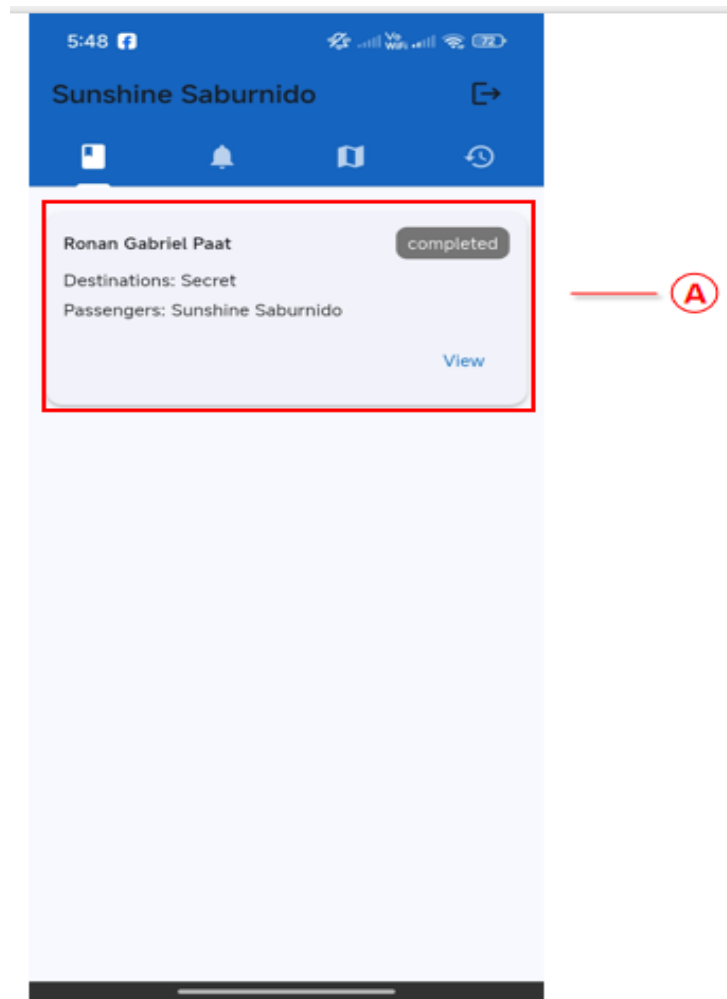


A

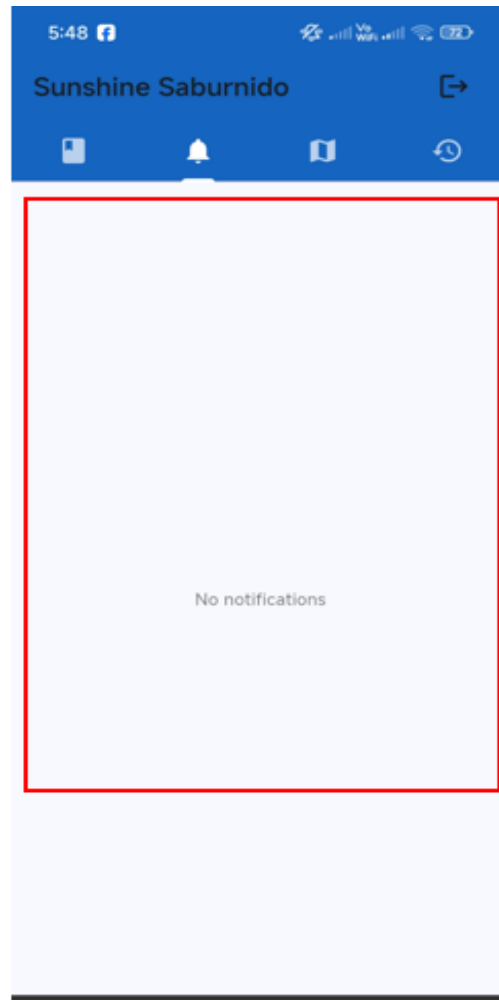
- A. **Conversation Tab** – Displays system and trip-related messages sent to the driver for updates and notifications.

Student Side

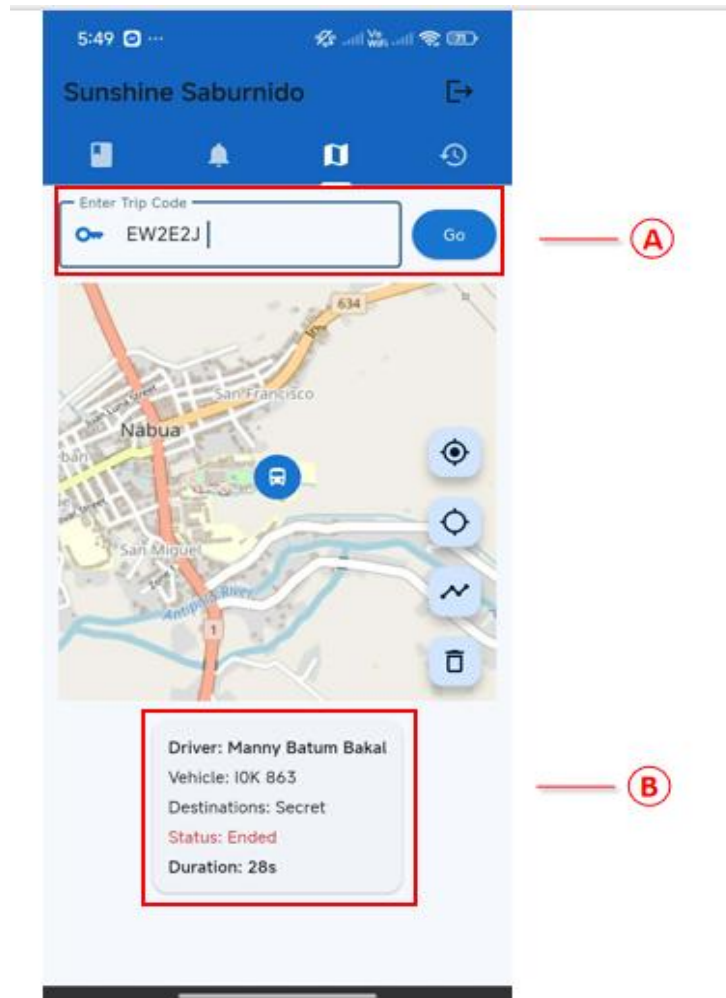
The Student Side section allows students to view, track, and manage their trips within the system. This interface provides tools to monitor completed trips, receive notifications, access trip details, and review travel history. It ensures students can stay informed about their travel requests and easily access important trip information.



- A. Reservation Tab**– Shows the list of trips, including destination details, passenger name, and trip status.

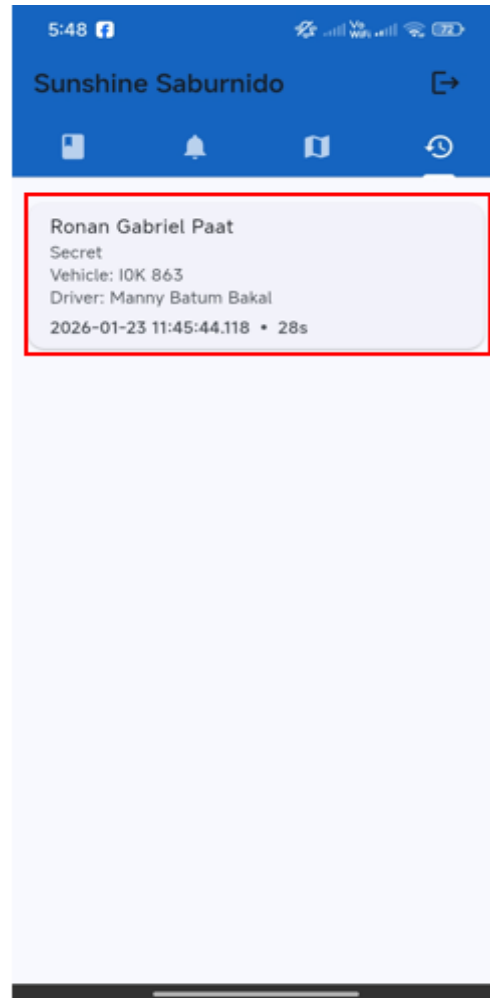


A. Notification Tab – Displays notifications related to trip updates and status



A. Trip Code – Shows the unique code assigned to the trip for identification.

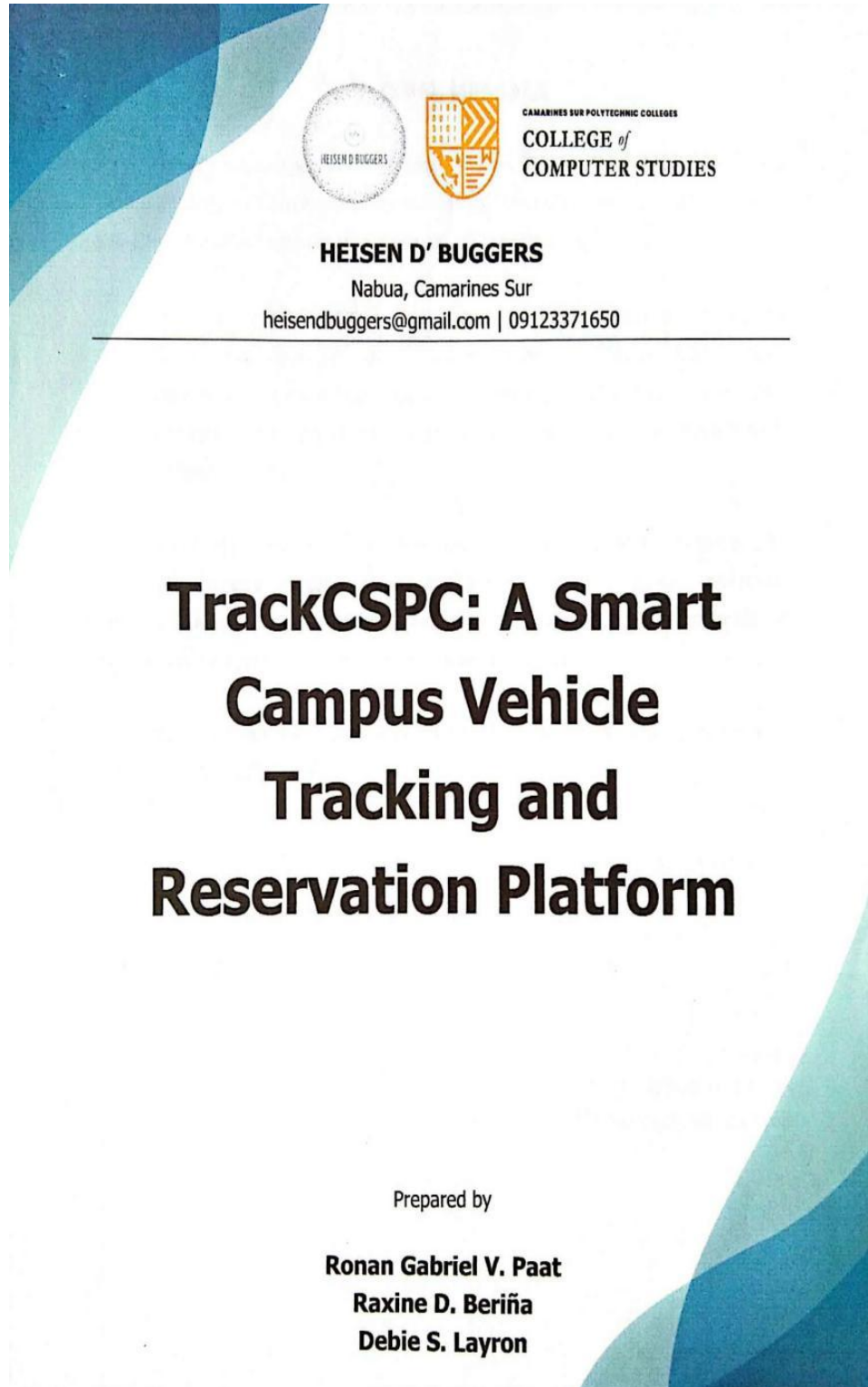
B. Trip Status – Indicates the trip status and displays the trip details.



- A. **Trip History Tab** – Displays the list of submitted trips with basic details such as name, purpose, location, and date.

APPENDIX E

COMMUNICATION LETTERS



  CAMARINES SUR POLYTECHNIC COLLEGES
COLLEGE of
COMPUTER STUDIES

HEISEN D' BUGGERS
Nabua, Camarines Sur
heisendbuggers@gmail.com | 09123371650

TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform

Prepared by

Ronan Gabriel V. Paat
Raxine D. Beriña
Debie S. Layron

WELCOME MESSAGE

We are pleased to present our proposal for TrackCSPC: A Smart System for Campus Vehicle Tracking and Reservations. This innovative solution is designed to enhance efficiency, safety, and overall management of campus transportation services.

By integrating real-time GPS tracking, a streamlined reservation system, and parental monitoring features, TrackCSPC aims to address common challenges such as vehicle scheduling conflicts, long wait times, and communication gaps. This system will empower your department with a comprehensive platform that simplifies vehicle coordination while improving service reliability.

We firmly believe that TrackCSPC will significantly benefit the Camarines Sur Polytechnic Colleges community by providing a safer and more organized transportation experience. We are committed to collaborating with your department to ensure the system meets your operational needs and enhances your existing processes.

We look forward to discussing this opportunity further and working together to bring this innovative solution to life.

Sincerely,
HEISEN D' BUGGERS
heisendbuggers@gmail.com

I. Executive Summary

This proposal presents a comprehensive plan aimed at establishing our project as an innovative solution in campus transportation management. Our primary objectives include enhancing vehicle tracking efficiency, improving user convenience, and ensuring student safety through real-time monitoring. By integrating GPS technology, mobile applications, and web platforms, our system offers a smart solution for tracking campus vehicles, streamlining reservations, and improving communication between students, staff, and parents.

Through effective development practices, rigorous testing, and continuous improvement, this project is designed to enhance campus mobility, reduce waiting times, and promote a safer and more organized transportation system for the Camarines Sur Polytechnic Colleges community.

II. Introduction

The increasing demands for efficient campus transportation systems present both challenges and opportunities, which this project aims to address. Our proposed solution, TrackCSPC, is designed to revolutionize campus mobility by providing real-time vehicle tracking and reservation platform. By incorporating GPS technology, web and mobile applications, and intuitive user interfaces, this system aims to streamline campus transportation while ensuring student safety and improving overall convenience.

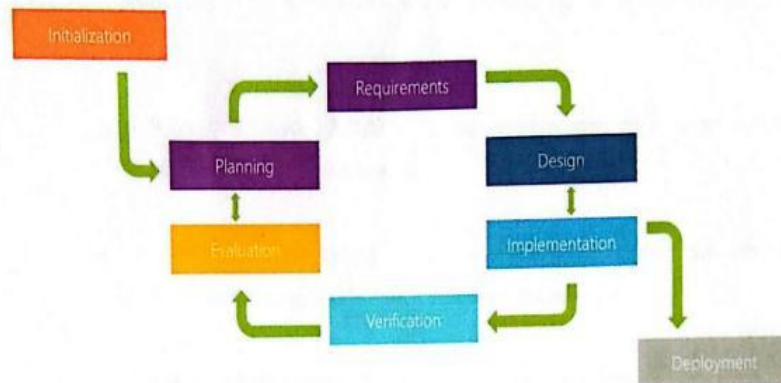
In an environment where technology continues to reshape everyday experiences, TrackCSPC aligns with modern innovations to offer a smarter and more organized solution for managing campus vehicle operations. Our commitment to leveraging emerging technologies drives this initiative, ensuring that students, staff, and parents benefit from a more efficient and secure transportation experience.

III. Objectives of the Study

Our project objectives for TrackCSPC focus on creating an efficient and reliable campus transportation system:

VI. Software Methodology

Iterative Model



- **Planning & Requirements Gathering** – Identifying user needs and system specifications.
- **Design & Prototyping** – Creating wireframes and initial system architecture.
- **Development & Implementation** – Coding the mobile and web application using appropriate technologies.
- **Testing & Evaluation** – Conducting functional and usability tests to ensure system reliability.
- **Deployment & Maintenance** – Releasing the system for use and providing updates as needed.

VII. Program Specification

1. **User Authentication Module** – Secure login and registration for students, staff, and parents.
2. **Real-time GPS Tracking Module** – Displays the live location of campus vehicles on a map using either a GPS tracker installed in the vehicle or the driver's mobile device.

- Provide accurate real-time vehicle tracking to improve reliability and accessibility.
- Develop a seamless reservation system for authorized users to manage campus vehicle bookings effectively.
- Promote user engagement by encouraging students, staff, and parents to utilize the platform's features.
- Enhance student safety by offering parents a convenient way to monitor their child's location while onboard campus vehicles.

IV. Scope of the Study

This study focuses on developing a mobile and web-based application that provides real-time tracking of campus vehicles at Camarines Sur Polytechnic Colleges. The system will integrate GPS tracking and an interactive map to allow students and staff to monitor vehicle locations, check estimated arrival times, and plan their trips efficiently.

Additionally, the system will include a reservation feature in the admin web-based application. This will allow authorized users to reserve campus vehicles for academic trips, such as nursing students traveling to hospitals for duty, field trips, and other official purposes. Parents or guardians will also be able to track their child's location while they are on a campus vehicle, ensuring their safety.

V. Limitations of the Study

- The system will only track designated campus vehicles within the premises of Camarines Sur Polytechnic Colleges.
- GPS tracking is dependent on network connectivity and device compatibility.
- The reservation system is limited to authorized users such as faculty, department heads, and student leaders.
- The application will be accessible only via registered student, staff, and parent accounts.

3. **Route and Schedule Display Module** – Shows designated vehicle routes and estimated arrival times.
4. **Notification and Alert Module** – Sends real-time updates on vehicle locations and delays.
5. **Admin Dashboard Module** – Allows administrators to manage vehicles, schedules, user access, and reservations.
6. **Vehicle Reservation Module** – Enables faculty, department heads, and authorized users to reserve campus vehicles for school-related activities.
7. **Parental Monitoring Module** – Allows parents to track their child's location while onboard a campus vehicle.

We strongly believe that TrackCSPC has the potential to transform campus transportation at Camarines Sur Polytechnic Colleges. By adopting this system, your department can improve efficiency, enhance safety, and provide a more organized travel experience for students and staff. This solution is designed to support your team in managing campus vehicles with greater ease and reliability.

We are excited about the positive impact this project can have, and we hope for your support in making it a reality.

☒ Accepted

☐ Rejected


DAN RANDOLF P. FRANCIA, LPT
Administrative Officer V (GSU)

Noted by:


ROSEL O. ONESA, MIT

Capstone Adviser


ROSEL O. ONESA, MIT

CCS Dean - OIC

APPENDIX F

NON-DISCLOSURE AGREEMENT FORM

Non-Disclosure Agreement Form

EFFECTIVE DATE: August 19, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and Philip Alger M. Serrano, MIT hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:
PHILIP ALGER M. SERRANO, MIT – Panel Member 1

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:
ROSEL O. ONESA, MIT – Dean

2. The confidential information disclosed under this Agreement is described as:
Contents of the TCP by:

RONAN GABRIEL V. PAAT, RAXINE D. BERINA, DEBIE S. LAYRON

which is entitled:

TRACKCSPC: A SMART CAMPUS VEHICLE TRACKING AND RESERVATION PLATFORM

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.

4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.

5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:

- marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
- if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.

6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.

7. The obligations imposed upon an Expert hereunder do not apply to information:

- which is or becomes publicly available without breach of this Agreement;
- which is already known to the Recipient prior to its disclosure hereunder;
- which is independently developed by the Expert.

8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed.



8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special, or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.
9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
 - a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

CAMARINES SUR
POLYTECHNIC COLLEGES


PHILIP ALGER M. SERRANO, MIT

Expert


ROSEL O. ONESA, MIT

CSPC Representative

DATE:

DATE: AUG 22 2025

Non-Disclosure Agreement Form

EFFECTIVE DATE: March 10, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and Vincent B. Cortez hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:
VINCENT B. CORTEZ, MIT

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:
ROSEL O. ONESA, DIT - Dean

2. The confidential information disclosed under this Agreement is described as:
Contents of the TCP by:

Ronan Gabriel V. Paat,
Raxine D. Beriña,
Debie S. Layron

which is entitled:

TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
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 - a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
 - b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
 - a. which is or becomes publicly available without breach of this Agreement;
 - b. which is already known to the Recipient prior to its disclosure hereunder;
 - c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special, or consequential damages in connection with or arising out of the performance or use of any portion of the confidential



- information.
9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
 10. Neither Party:
 - a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
 11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
 12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
 13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
 14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

**CAMARINES SUR
POLYTECHNIC COLLEGES**


VINCENT B. COBTEZ, MIT
Expert


ROSEL O. ONESA, MIT
CSPC Representative

DATE: March 10, 2025

DATE: March 10, 2025

Non-Disclosure Agreement Form

EFFECTIVE DATE: September 16, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies- Camarines Sur Polytechnic Colleges and Ella Mae S. Alamo hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:
ELLAMAE S. ALAMO - Grammarian

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:
ROSEL O. ONESA, MIT - Dean

2. The confidential information disclosed under this Agreement is described as: **Contents of the TCP by:**

RONAN GABRIEL V. PAAT, RAXINE D. BERIÑA, DEBIE S. LAYRON

which is entitled:

TRACKCSPC: A Smart Campus Vehicle Tracking and Reservation Platform

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
- a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
 - b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
- a. which is or becomes publicly available without breach of this Agreement;
 - b. which is already known to the Recipient prior to its disclosure hereunder;
 - c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special, or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.



9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
 - a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:


ELLANRE S. ALAMO
Expert

DATE: September 16, 2025

**CAMARINES SUR
POLYTECHNIC COLLEGES**


ROSEL O. ONESA, MIT

CSPC
Representative

DATE: September 16, 2025

Non-Disclosure Agreement Form

EFFECTIVE DATE: August 19, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies- Camarines Sur Polytechnic Colleges and Dr. Jocelle B. Monreal hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:
DR. IAN P. BENITEZ – Panel Chair

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:
ROSEL O. ONESA, MIT – Dean

2. The confidential information disclosed under this Agreement is described as:
Contents of the TCP by:

RONAN GABRIEL V. PAAT, RAXINE D. BERIÑA, DEBIE S. LAYRON

which is entitled:

TRACKCSPC: A SMART CAMPUS VEHICLE TRACKING AND RESERVATION PLATFORM

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
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 - a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
 - b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
 - a. which is or becomes publicly available without breach of this Agreement;
 - b. which is already known to the Recipient prior to its disclosure hereunder;
 - c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied



warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular, the Expert shall not be liable for any direct, indirect, special, or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.

9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
 - a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

CAMARINES SUR
POLYTECHNIC COLLEGES

DR. IAN P. BENITEZ

Expert

ROSEL O. ONESA, MIT

CSPC Representative

DATE: AUG 19, 2025

DATE:

Non-Disclosure Agreement Form

EFFECTIVE DATE: August 19, 2025

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and Sheila Dona S. Manlapaz, MIT hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:
MARICRIS L. RAMIZARES, MIT – Panel Member 2

On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:
ROSEL O. ONESA, MIT – Dean

2. The confidential information disclosed under this Agreement is described as:
Contents of the TCP by:

RONAN GABRIEL V. PAAT, RAXINE D. BERIÑA, DEBIE S. LAYRON

which is entitled:

TRACKSPC: A SMART CAMPUS VEHICLE TRACKING AND RESERVATION PLATFORM

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
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 - b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
 - a. which is or becomes publicly available without breach of this Agreement;
 - b. which is already known to the Recipient prior to its disclosure hereunder;
 - c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed.

In particular, the Expert shall not be liable for any direct, indirect, special, or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.

9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
 - a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
 - b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
 - c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

**CAMARINES SUR
POLYTECHNIC COLLEGES**


MARICRIS L. RAMIZARES, MIT
Expert


ROSEL O. ONESA, MIT
CSPC Representative

DATE: AUG 19, 2025

DATE:

APPENDIX G

JOINT AFFIDAVIT OF UNDERTAKING

We, RAXINE D. BERIÑA of legal age, single and a resident of ZONE 5, PUROK OLIVA, BONBON, LIBON, ALBAY, DEBIE S. LAYRON, of legal age, single and a resident of ZONE 1, VELASCO STREET, LIBON ALBAY, RONAN GABRIEL V. PAAT, of legal age, single and a resident of ZONE 3, SAN JOSE, PILL, CAMARINES SUR, after having been sworn to in accordance with law, do hereby take oath and state:

- (i) That, we are officially enrolled for the thesis/capstone project on the topic titled TRACKCSPC: A SMART CAMPUS VEHICLE TRACKING AND RESERVATION PLATFORM in the COLLEGE OF COMPUTER STUDIES of CAMARINES SUR POLYTECHNIC COLLEGES.
- (ii) That, the contents of our thesis/ capstone project submitted to the Camarines Sur Polytechnic Colleges, for award of BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY Degree are original and our own work, and is not plagiarized.
- (iii) That, if, after completing the thesis/ capstone project, are found copied or come under plagiarism, we will be solely responsible for it and College shall have sole right to cancel my research work ab-initiation.
- (iv) That, this work has not been submitted for the award of any other Degree/Diploma in any other University/ Institute.
- (v) That, we shall be responsible for any legal dispute/case(s) for violation of any provisions of the Copyright Act relating to our thesis/ capstone project.

IN WITNESS WHEREOF, I have hereunto set my name this 26 JAN 2026 day of 2026 in

_____, Philippines.


RAXINE D. BERIÑA

Affiant
221007012

DEBIE S. LAYRON

Affiant
221007763


RONAN GABRIEL V. PAAT
Affiant
221006919

26 JAN 2026
SUBSCRIBED AND SWORN TO before me this _____ day of _____ at _____
Philippines, affiants exhibiting to me their competent proofs of identity above stated.

Doc. No. 168 :
Page No. 39 :
Book No. XCVIII :
Series of 202 26

ATTY. JOSELITO F. FIGURACION
Notary Public
Until December 31, 2025
Roll of Attys. No. 52026
PTR No. 37615160 1/2/2025 Nabua Cam. Sur
IBP O.A. No. 540225 02/18/2025
MCLE Compliance No. VII-0013316 valid 04/14/28
San Francisco, Nabua, Camarines Sur

APPENDIX H

PROJECT TEAM ASSIGNMENT FORM

PROJECT TEAM ASSIGNMENTS FORM		
Team Alias	HEISEN D' BUGGERS	
Course Code	IT 3211	
Subject adviser/ CaPSA	JOCELLE B. MONREAL, DIT	

Name and Signature	Project Role	Email address and mobile#(s)
 RONAN GABRIEL V. PAAT	PROJECT HEAD/ LEAD PROGRAMMER	ropaat@my.cspc.edu.ph 09123371650
 RASHED D. BERINA	DATABASE DESIGNER/ DOCUMENTATION WRITER	radavid@my.cspc.edu.ph
 DEBIE S. LAYRON	UI DESIGNER/ NETWORK DESIGNER	delayron@my.cspc.edu.ph

APPENDIX I
FINAL PROJECT TITLE FORM

FINAL PROJECT TITLE FORM				
Proponents/Researchers:				
<table border="1"><tr><td>1. Raxine D. Beriña</td></tr><tr><td>2. Debie S. Layron</td></tr><tr><td>3. Ronan Gabriel V. Paat</td></tr></table>		1. Raxine D. Beriña	2. Debie S. Layron	3. Ronan Gabriel V. Paat
1. Raxine D. Beriña				
2. Debie S. Layron				
3. Ronan Gabriel V. Paat				
Proposed Thesis/ Capstone Project Title:				
TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform				
Submitted by:	Noted by:			
 RONAN GABRIEL V. PAAT (Signature of Project Head over printed name)	 JOCELLE B. MONREAL, DIT (Signature of Subject adviser over printed name)			
Date: <u>March 18, 2025</u>	Date: <u>March 18, 2025</u>			
Recommending Approval:	Approved:			
 ROSEL O. ONESA, MIT (Signature of Adviser over printed name)	 ROSEL O. ONESA, MIT OIC DEAN, CCS			
Date: <u>March 18, 2025</u>	Date: <u>March 18, 2025</u>			

APPENDIX J
CAPSTONE HEARING FORM (TD, POD, FOD)

CAPSTONE PROJECT HEARING FORM

Date Filed: May 9, 2025 [☒] Title Proposal [☐] Pre-Oral [☐] Final Oral

Date of Hearing: May 29, 2025 Time: 1:00am – 3:00am Venue: CS Lab

Department: COLLEGE OF COMPUTER STUDIES (CCS)

Research Title:

**TRACKSPC: A SMART CAMPUS VEHICLE TRACKING AND RESERVATION
PLATFORM**

Proponent/s:

RAXINE D. BERIÑA

RONAN GABRIEL V. PAAT

DEBIE S. LAYRON

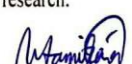
Recommended by:


ROSEL O. ONESA, MIT
Capstone Project Adviser

CERTIFICATION

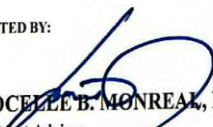
The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research.


PHILIP ALGER M. SERRANO, MIT
PANEL MEMBER 1


MARICRIS L. RAMIZARES, MIT
PANEL MEMBER 2


DR. IAN P. BENITEZ
PANEL CHAIR

NOTED BY:


JOCELLE B. MONREAL, DIT
Subject Adviser

APPROVED:


ROSEL O. ONESA, MIT
OIC Dean, CCS



CAPSTONE PROJECT HEARING FORM

Date Filed: December 3, 2025 ☐ Title Proposal ☒ Pre-Oral ☐ Final Oral

Date of Hearing: December 9, 2025 Time: 1:00pm-4:00pm Venue: _____

Department: COLLEGE OF COMPUTER STUDIES (CCS)

Research Title:

TRACKCSPC: A SMART CAMPUS VEHICLE TRACKING AND RESERVATION
PLATFORM

Proponent/s:

RAXINE D. BERIÑA

RONAN GABRIEL V. PAAT

DEBIE S. LAYRON

Recommended by:

ROSEL O. ONESA, MIT
Capstone Project Adviser

--CERTIFICATION--

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research.

PHILIP ALGER M. SERRANO, MIT
PANEL MEMBER 1

MARICRIS L. RAMIZARES, MIT
PANEL MEMBER 2

DR. IAN P. BENITEZ
PANEL CHAIR

NOTED BY:

JOCELLE B. MONREAL, DIT
Subject Adviser

APPROVED:

ROSEL O. ONESA, MIT
OIC Dean, CCS



CAPSTONE PROJECT HEARING FORM

Date Filed: Dec 22, 2025 ☐ Title Proposal ☐ Pre-Oral ☒ Final Oral

Date of Hearing: Dec 22, 2025 Time: 4:00pm – 5:00pm Venue: Mac Lab

Department: COLLEGE OF COMPUTER STUDIES (CCS)

Research Title:

TRACKCSPC: A SMART CAMPUS VEHICLE TRACKING AND RESERVATION
PLATFORM

Proponent/s:

RAXINE D. BERIÑA

RONAN GABRIEL V. PAAT

DEBIE S. LAYRON

Recommended by:

ROSEL O. ONESA, MIT
Capstone Project Adviser

-- CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research.

PHILIP ALGER M. SERRANO, MIT
PANEL MEMBER 1

MARICRIS L. RAMIZARES, MIT
PANEL MEMBER 2

DR. IAN P. BENITEZ
PANEL CHAIR

NOTED BY:

JOCELLE B. MONREAL, DIT
Subject Adviser

APPROVED:

ROSEL O. ONESA, MIT
OIC Dean, CCS

APPENDIX K

PANEL RSC (TD, POD, FOD)

Title : TRACKCSPC: A SMART CAMPUS VEHICLE TRACKING AND RESERVATION PLATFORM
 Alias : TEAM HEISEN D' BUGGERS
 Date : May 29, 2025 ☒ TD ☐ POD ☐ FOD
 Secretary : JACQUES NICO L. BELMONTE

MANUSCRIPT			
CHAPTER	PAGE NO.	RECOMMENDATIONS, SUGGESTIONS AND COMMENT (RSC)	ACTION TAKEN
ALL	1-49	- Fix the page number formatting so that the first page is numbered 1, and the rest follow accordingly. - Check all the number citations. The [n] (e.g. [1]) must be placed before the period.	Page numbering has been renumbered starting from 1 through page 49. All in-text numeric citations [n] are now correctly placed before the period.
1	2-4	Project Context - Add references/ citations in project context second paragraph. - Make it specific to the GPS tracking of what you are tracking.	Additional citations were added in the second paragraph. We clearly identify exactly what is being tracked via GPS. See Page 2-4
	4-6	Purpose and Description of the Project Ensure that the year you cited is placed within square brackets [].	All years in the "Purpose and Description" section are properly enclosed in square brackets []. See Page 4-6

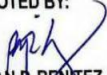
	6-7	Objectives of the Project Add the 4th objective for the deployment.	The 4th deployment objective has been added. See Page 6-7
	9-10	Scope and Limitations of the Project Ensure that each specific objective is thoroughly discussed within the scope section.	Each Specific objectives are now discussed in detail. See Page 9-10
	10-13	Project Dictionary - In "TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform," separate the terms and provide clear definitions for each component. - Must define the terms including Functional Testing, User Acceptance Testing, Feedback Analysis.	"TrackCSPC," "Vehicle Tracking," and "Reservation Platform" are now individually defined. Definitions for testing terms such as Functional Testing, User Acceptance Testing, and Feedback Analysis have been added. See Page 10-13
	13-16	Notes Use ACM citation.	The notes have been updated to adhere to ACM citation style. See Page 13-16
2	17-25	Related Literature and Systems - Ensure that an introduction section is included in the document. - The flow of the Review of Related Literature (RRL) should follow the sequence of your specific objectives. Each specific objective should be supported by related literature organized by topic. - Change from Bus Tracking and Reservation to Vehicle Tracking and Reservation.	A brief introduction is presented before the literature review. The RRL now follows the sequence of the specific objectives, with each objective supported by topical literature.



			Wording updated from "Bus Tracking and Reservation" to "Vehicle Tracking and Reservation." See Page 17-25
3	33-35	Table 1: Hardware Specifications Add an introduction for Table 1 before the actual table.	The Table 1 introduction has been added. See Page 33-35
	35-37	Table 2: Software Specifications - Add an introduction for Table 2 before the actual table. - Must be "software", not "hardware" in the table.	Introductory added and label corrected from hardware to software. See Page 35-37
	37-40	Program Specifications - Add an introduction for Figure 1 before the actual table. - The specific FR should be discussed in table format.	The Figure 1 Introductory explanation added. Each functional requirement (FR) is now clearly outlined in table format. See Page 37-40
	40-42	Programming Environment Identify and discuss each component.	All components have been clearly identified and discussed. See Page 40-42
	42-44	Testing Plan Revise Figure 2.	Figure 2 has been revised to align with the updated plan. See Page 42-44

	45	Testing Tools and Framework Remove the redundant testing tools and frameworks.	Removed redundant tool listings; no more references to JUnit, Selenium, or Android Studio Testing. See Page 45
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NOTED BY:


IAN P. BENITEZ, DIT
PANEL CHAIRMAN


PHILIP ALGER M. SERRANO, MIT
PANEL, MEMBER 1


MARICRIS L. RAMIZARES, MIT
PANEL, MEMBER 2

Title : TRACKCSPC: A Smart Campus Vehicle Tracking and Reservation Platform
Alias : HEISEN D' BUGGERS
Date : December 09, 2025 ☐ TD ☒ POD ☐ FOD
Secretary : JOHN PETER G. ANDALIS

MANUSCRIPT			
CHAPTER	PAGE NO.	RECOMMENDATIONS, SUGGESTIONS AND COMMENT (RSC)	ACTION TAKEN
ALL	ALL	Fix the full document's format	The entire document was reformatted to ensure consistency and compliance with the required format.
ALL	ALL	Fix all spacing/extra spaces	All extra spaces and spacing inconsistencies were corrected throughout the document.
ALL	Tables	Fix format. Not to exceed the document's margin	All tables were adjusted to fit within the document margins.
		The actual UI's should be in the Development section	All actual system UI screenshots were transferred to the Development section.
	3.3.1 Types of testing	Add an intro paragraph	An introductory paragraph was added to the Types of Testing section.
	Instrument	Content /discussion of questionnaire should be in paragraph format	The instrument and questionnaire discussion were rewritten in paragraph form.
	Testing	Testing Interpretations change "Pass" to "Passed"	All instances of "Pass" were revised to "Passed."
		PIIILS discussion should be applied in the discussion of the results	The PIIILS framework was incorporated into the discussion of the results.
	Table 9	Summarize full result for every testing, no need to separate them by respondents	Table 9 was revised to present a summarized overall testing result.
	Finding 3	Include the numerical result of testing	Numerical testing results were added to Finding 3.

SOFTWARE PROGRAM		
MODULE NO. (DFD)	RECOMMENDATIONS, SUGGESTIONS, AND COMMENTS (RSC)	ACTION TAKEN
	In the Reservation Management list, show in list format/tabular	The reservation management page is now in a tabular format and has a much organize view
	The approval and rejection inside a specific reservation	Approval and rejection button was moved inside a reservation when clicking the view button.
	In the reservation, will input the other data once it is approving the request	The system already implemented the saving of all the information regarding the request once approved
	In the reservation, automatic will have 2 drivers if the travel is in Manila, and if in only in the Bicol region, 1 driver only	The destination has now a search bar that can autocomplete the desired destination and can verify if the



		destination is outside of the Bicol region or not
	The admin should see the approved TO document in PDF file format	When the faculty request using the faculty's mobile app, he/she is able to upload either a pdf file or picture of the travel order document.
	It should have no deletion in the system	Deletion buttons inside the reservation management page and trip management page was removed
	In the Reservation Management, it will be a view of the request only; there is no button for approve or reject yet	Approval and rejection button was moved inside a reservation when clicking the view button
	In cancellation or rejection of the request, it should have a reason indicated	When cancelling, a dialog with a text field will pop-up to type the reason for cancellation both for the admin and the requestor
	Add a feature for changing of driver and vehicle	When viewing the approved reservation, a "Replace" button was added to change the driver or the vehicle in certain circumstances. It also messages both old and new drivers during replacement
	In requesting, it will appear the available vehicle preferred	A feature was added on the faculty's mobile app where they can choose their preferred available vehicle
	The user has a feature to cancel the request	The requestor can now cancel a request whatever the status of his/her reservation
	Both the user and the admin will receive a notification if the user cancels the request. The admin will receive a notification that the user canceled the request. If the admin cancels or rejects the request, the user will receive a notification that the admin canceled or rejected the request	Both the admin and the requestor can see the reason of cancellation of the reservation
	Actual data and actual transaction	An actual data and transaction were inserted to the system from the GSU itself
	All the notifications to the driver should be push notifications	Push notification feature wasn't implemented due to the incompatibility of the versions of the packages used in the system
	Put the plate number and type of vehicle in the icon vehicle	The marker of the trip on the live map has the plate number and vehicle type on its label
	Accurately the red line where the vehicle goes and make it toggle	Added a feature where the marker has tracking polylines and made it toggle
	Add a feature that will show where the vehicle will go or where the destination is	Added a feature where the destination will be flagged on the live map
	All notifications make it push notifications	Push notification feature wasn't implemented due to the incompatibility of the versions of the packages used in the system
	Add a notification to the requestor if the vehicle suddenly goes under maintenance or breaks down, if will change the vehicle, and if the driver is absent	Instead of notifying the requestor when a vehicle goes under maintenance or the driver is absent, a replacement feature was added for changing the driver or the vehicle in certain circumstances



	In selecting the date of leave selections must be multiple selecting	Maintenance and Leave can be added more than once
	The passenger must have contact with the driver that the message of the passenger will be push notification to the driver	Push notification feature wasn't implemented due to the incompatibility of the versions of the packages used in the system
	Provide a GPS for 2 vehicles	Two GPS devices were created with a portable power source
	Actual List of vehicles and drivers	Given by the information provided by the GSU, the system has now its actual list of vehicles and drivers including their email.
	In viewing, the reservation must be sorted by driver and by vehicle	Added a feature for sorting by vehicle and driver when clicking the title of the column of the table, Applied on both reservation management page and trip management page
	Should have an actual reservation	An actual data and transaction were inserted to the system from the GSU itself
	Expect ALL possible scenarios	The system is ready for an actual processing and implementation. Most realistic scenarios and logics were implemented.

NOTED-BY;


DR. IAN P. BENITEZ
PANEL CHAIRMAN


PHILIP ALGER M. SERRANO, MIT
PANEL MEMBER 1


MARICRIS L. RAMIZARES, MIT
PANEL MEMBER 2

Title : **TRACKCSPC: A Smart Campus Vehicle Tracking and Reservation Platform**
Alias : **HEISEN D' BUGGERS**
Date : December 09, 2025 ☐ TD ☐ POD ☒ FOD
Secretary : **JOHN PETER G. ANDALIS**

MANUSCRIPT			
CHAPTER	PAGE NO.	RECOMMENDATIONS, SUGGESTIONS AND COMMENT (RSC)	ACTION TAKEN
	Abstract	1 paragraph only, include the numerical result of testing	The abstract was revised into a single coherent paragraph and updated to include the numerical testing result (overall mean = 1.00, interpreted as Completely Functional).
ALL	Table of Contents	Fix Table of Contents	The Table of Contents was corrected and updated to match the final page numbering and headings.
		Schematic diagram of the GPS circuit to be installed in the vehicle – Put it in Methodology	The schematic diagram of the GPS circuit was transferred and properly placed in the Methodology chapter.
4	86	Fix the hanging content	The hanging content was corrected and properly formatted.
	103 - 106	PIIILS discussion should be applied in the discussion of the results.	The PIIILS framework was integrated into the discussion of the results.
5	114	Findings Include the sub-items	Sub-items were added.

SOFTWARE PROGRAM		
MODULE NO. (DFD)	RECOMMENDATIONS, SUGGESTIONS, AND COMMENTS (RSC)	ACTION TAKEN
3	In the reservation, automatic will have 2 drivers if the travel is in Manila, and if in only in the Bicol region, 1 driver only	The destination has now a search bar that can autocomplete the desired destination and can verify if the destination is outside of the Bicol region or not
3	Add a feature for changing of driver and vehicle	When viewing the approved reservation, a "Replace" button was added to change the driver or the vehicle in certain circumstances. It also messages both old and new drivers during replacement
3	In requesting, it will appear the available vehicle preferred	A feature was added on the faculty's mobile app where they can choose their preferred available vehicle
	All the notifications to the driver should be push notifications	Push notification feature wasn't implemented due to the incompatibility of the versions of the packages used in the system
6	Put the plate number and type of vehicle in the icon vehicle	The marker of the trip on the live map has the plate number and vehicle type on its label
6	Accurately the red line where the vehicle goes and make it toggle	Added a feature where the marker has tracking polylines and made it toggle



6	• Add a feature that will show where the vehicle will go or where the destination is	Added a feature where the destination will be flagged on the live map
	• All notifications make it push notifications	Push notification feature wasn't implemented due to the incompatibility of the versions of the packages used in the system
3	• Add a notification to the requestor if the vehicle suddenly goes under maintenance or breaks down, it will change the vehicle, and if the driver is absent	Instead of notifying the requestor when a vehicle goes under maintenance or the driver is absent, a replacement feature was added for changing the driver or the vehicle in certain circumstances
	• The passenger must have contact with the driver so that the message to the passenger will be a push notification to the driver	Push notification feature wasn't implemented due to the incompatibility of the versions of the packages used in the system
6	• Provide a GPS for 2 vehicles	Two GPS devices were created with a portable power source
3 and 5	• In viewing, the reservation must be sorted by driver and by vehicle	Added a feature for sorting by vehicle and driver when clicking the title of the column of the table, Applied on both reservation management page and trip management page

NOTED BY:


DR. IAN P. BENITEZ
PANEL, CHAIRMAN


PHILIP ALGER M. SERRANO, MIT
PANEL, MEMBER 1


MARICRIS L. RAMIZARES, MIT
PANEL, MEMBER 2

APPENDIX L

CONSULTATION LOGS FORM

CONSULTATION LOGS FORM							
Capstone Project Title:	TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform.						
Proponents:	1. Ronan Gabriel V. Paat 2. Raxine D. Berina 3. Debie S. Layron						
Alias:	Heisen 'D Bugger						
Total # of Modules:				as approved by the CAPSA /TSA:			
PROTOTYPE							
	Date of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Manager's Signature	CAPA/CAPSA/ Consultant/ Grammarian Signature
System	01/23/20						
	Remarks:						
	- recommended for upgrade						
Deadline:							
	Remarks:						
Deadline:							



CONSULTATION LOGS FORM

Capstone Project Title:	TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform.						
Proponents:	1. Ronan Gabriel V. Paat 2. Raxine D. Berina 3. Debie S. Layron						
Allas:	Heisen 'D Bugger						
Total # of Modules:				as approved by the CAPSA /TSA:			
PROTOTYPE	Date of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Manager's Signature	CAPA/CAPSA/ Consultant/ Grammarian Signature
chapter 1-5	01/23/24					<i>Paat</i>	<i>Debie</i>
	Remarks:						
	Perin Findings based on objectives, and conclusion based on findings.						
Deadline:							
	Remarks:						
Deadline:							



CONSULTATION LOGS FORM

Capstone Project Title:	TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform.						
Proponents:	1. Ronan Gabriel V. Paat 2. Raxine D. Berina 3. Debie S. Layron						
Alias:	Heisen 'D Bugger						
Total # of Modules:				as approved by the CAPSA /TSA:			
PROTOTYPE	Date of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Manager's Signature	CAPA/CAPSA/ Consultant/ Grammarian Signature
MANUSCRIPT	01/23/24					<i>[Signature]</i>	<i>[Signature]</i>
	Remarks: Recommended for blue book						
Deadline:							
	Remarks:						
Deadline:							



CONSULTATION LOGS FORM

Capstone Project Title:	TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform.						
Proponents:	1. Ronan Gabriel V. Paat 2. Raxine D. Berina 3. Debie S. Layron						
Alias:	Heisen 'D Bugger						
Total # of Modules:				as approved by the CAPSA /TSA:			
PROTOTYPE							
	Date of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Manager's Signature	CAPA/CAPSA/ Consultant/ Grammarian Signature
Chapter 1-5	01/12/2026						
	Remarks:						
	- Revise Notes/Bibliography, follow the correct format						
	- manuscript is ok, reviewed according to RSC						
Deadline:							
	Remarks:						
Deadline:							



CONSULTATION LOGS FORM

Capstone Project Title:	TRACKCSPC: A SMART CAMPUS VEHICLE TRACKING SYSTEM AND RESERVATION PLATFORM						
Proponents:	Raxine D. Beriña, Debie S. Layron, Ronan Gabriel V. Paat						
Alias:	Heisen D' Buggers						
Total # of Modules:				as approved by the CAPSA /TSA:			
PROTOTYPE	Date of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Manager's Signature	CAPA/CAPSA/ Consultant/ Grammarian Signature
Chapters 1-3							
	Remarks: Please check my comments / suggest a full manuscript. Good to go over revised.						
Deadline:							
	Remarks:						
Deadline:							



CONSULTATION LOGS FORM

Capstone Project Title:	TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform.						
Proponents:	1. Ronan Gabriel V. Paat 2. Raxine D. Berina 3. Debie S. Layron						
Alias:	Heisen 'D Bugger						
Total # of Modules:				as approved by the CAPSA /TSA:			
PROTOTYPE	Date of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Manager's Signature	CAPA/CAPSA/ Consultant/ Grammarian Signature
Menu 1-5	01-27-26					<i>[Signature]</i>	<i>[Signature]</i>
	Remarks:						
	OK for blue book						
Deadline:							
	Remarks:						
Deadline:							

APPENDIX M
SECRETARY CERTIFICATE

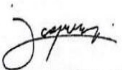
SECRETARY'S CERTIFICATION

This is to certify that the undersigned has provided true and correct recommendations, suggestions, and comments unanimously agreed and approved by the panel of examiners during the oral examination of the capstone project entitled "TRACKCSPC: A SMART CAMPUS VEHICLE TRACKING AND RESERVATION PLATFORM" prepared and submitted by Raxine D. Beriña, Debie S. Layron, Ronan Gabriel V. Paat, and that the same have not been amended, modified or obliterated.

Signed:


MR. JOHN PETER G. ANDALIS
Secretary

Date signed: 01-20-26


MR. JACQUES NICO L. BELMONTE
Secretary

Date signed: 01-23-26

APPENDIX N
GRAMMARIAN CERTIFICATE

GRAMMARIAN'S CERTIFICATION

This is to certify that the undersigned has reviewed and went through all the pages of the capstone project entitled “TRACKCSPC: A SMART CAMPUS VEHICLE TRACKING AND RESERVATION PLATFORM” as against the set of structural rules that govern research writing in accord with the composition of *sentences*, *phrases*, and *words* in the English language.

Signed:



MS. ELLA MAE S. ALAMO

Grammarian

Date signed: 1 - 23 - 26

APPENDIX O
CERTIFICATE OF TRANSFER



The certificate is a formal document with a white background and a decorative teal and blue curved border on the left and bottom right. At the top center, it features the Camarines Sur Polytechnic Colleges logo (a gear with a torch) and the text "CAMARINES SUR POLYTECHNIC COLLEGES ISO 9001:2015 Certified". Below this, the title "CERTIFICATE OF TRANSFER" is written in a large, blue, serif font. The main body of the certificate contains two paragraphs of text. The first paragraph certifies the transfer of a thesis project titled "TRACKCSPC: A SMART CAMPUS VEHICLE TRACKING AND RESERVATION PLATFORM" developed by four students to the CSS Dean's Office for further research. The second paragraph states that the certification is issued on a specific day in January 2026 at the CCS Dean's Office for whatever legitimate purpose it may serve. At the bottom left, there is a signature line with a handwritten signature in blue ink, followed by the printed name "Rosel O. Onesa, MIT" and the title "OIC Dean, CCS".

  **CAMARINES SUR**
POLYTECHNIC COLLEGES
ISO 9001:2015 Certified

**CERTIFICATE
OF TRANSFER**

This is to certify that **“TRACKCSPC: A SMART CAMPUS
VEHICLE TRACKING AND RESERVATION PLATFORM”**
developed by Raxine D. Beriña, Debie S. Layron, Ronan
Gabriel V. Paat, Camarines Sur Polytechnic Colleges (CSPC) -
College of Computer Studies (CCS) students, as their
undergraduate thesis project was transferred to the CSS
Dean’s Office , Camarines Sur Polytechnic Colleges for
further research

This certification is issued this _ day of January 2026 at CCS
Dean’s Office for whatever legitimate purpose it may serve.



Rosel O. Onesa, MIT
OIC Dean, CCS

APPENDIX P
CERTIFICATE OF UTILIZATION



The certificate is a formal document with a decorative border on the left and bottom. It features the logos of Camarines Sur Polytechnic Colleges and the College of Computer Studies at the top. The main title 'CERTIFICATE OF UTILIZATION' is prominently displayed in the center. The body of the certificate contains two paragraphs: the first certifies the adoption and utilization of a specific software platform developed by students, and the second states the date of issuance and the purpose of the certification. The bottom of the certificate includes a signature line for an official from the Administrative Office V (GSU).

  **CAMARINES SUR**
POLYTECHNIC COLLEGES
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**CERTIFICATE
OF UTILIZATION**

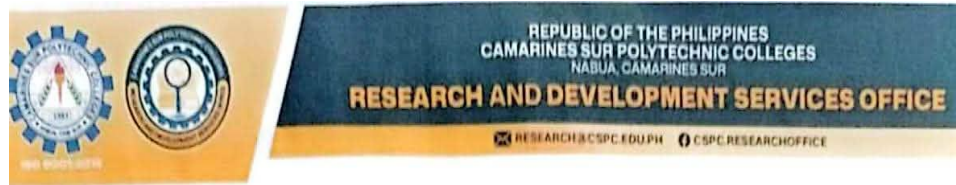
This is to certify that **“TRACKCSPC: A SMART CAMPUS
VEHICLE TRACKING AND RESERVATION PLATFORM”**
developed by Raxine D. Beriña, Debie S. Layron, Ronan
Gabriel V. Paat, Camarines Sur Polytechnic Colleges (CSPC) -
College of Computer Studies (CCS) students, as their
undergraduate thesis project was adopted and utilized by
General Services Unit (GSU) from December 2025 to present
date.

This certification is issued this 17th December, 2025 at
General Services Unit (GSU) for whatever legitimate purpose
it may serve.


Dan Randolph P. Francia, LPT
Administrative Office V (GSU)

APPENDIX Q

CERTIFICATE OF PLAGIARISM CHECK



CERTIFICATION

Date of Release: February 5, 2026
Submission ID: 23367:127560376
Output Title: TRACK CSPC: SMART CAMPUS VEHICLE TRACKING AND RESERVATION PLATFORM.
Author(s): Raxine D. Beriña
Debie S. Layron
Ronan Gabriel V. Paat
Program: Bachelor of Science in Information Technology

Authenticity Report:

Similarity Index Report: 9%

Internet Sources: 2%
Publications: 1%
Student Papers: 8%

- Interpretation: The similarity index report means that 9% of the output is similar to the sources in Turnitin's (authenticity software) online repository and databases.

AI-Generated Report: -

AI-generated only: -

AI-generated text that was AI-paraphrased: -

- Interpretation: The AI generated report means that the entire output is less than 20% and includes the possibility of false positives, although some text is likely AI-generated.


HAROLD JAN R. TERANO, Ph.D.
Director, Research & Development Services Office

APPENDIX R
ACM FORMAT
TRACKCSPC A SMART CAMPUS VEHICLE
TRACKING AND RESERVATION
PLATFORM

Raxine D. Beriña

College of Computer Studies, Camarines Sur Polytechnic Colleges, radavid@my.cspc.edu.ph

Debie S. Layron

College of Computer Studies, Camarines Sur Polytechnic Colleges, delayron@my.cspc.edu.ph

Ronan Gabriel V. Paat

College of Computer Studies, Camarines Sur Polytechnic Colleges, ropaat@my.cspc.edu.ph

ABSTRACT

This study presents TrackCSPC, a smart vehicle tracking and reservation platform developed for Camarines Sur Polytechnic Colleges (CSPC) to address inefficiencies in manual vehicle management, including scheduling delays, inconsistent documentation, and limited monitoring. The system integrates a mobile app for reservations and real-time tracking with a web-based dashboard for administrative tasks. Evaluation through functional testing, load testing, and User Acceptance Testing (UAT) yielded an overall mean of 1.00, indicating the system is Completely Functional. TrackCSPC improves workflow, communication, and centralized vehicle monitoring, with minor refinements suggested for some users. Future enhancements may include AI-based scheduling and online payment integration.

CCS CONCEPTS

Information system → Location-Tracking System • Web Applications → Embedded System

KEYWORDS

GPS Tracking, Vehicle Reservation System, Vehicle Tracking, Live Map, Esp32 GPS Tracker

1. INTRODUCTION

The rapid growth of digital technologies has improved transportation management in educational institutions through GPS tracking and automated reservation systems, yet many Philippine universities still rely on manual, paper-based processes that cause delays, conflicts, and limited monitoring. To address these issues, this study developed TrackCSPC, a smart campus vehicle tracking and reservation platform for Camarines Sur Polytechnic Colleges (CSPC). Transportation management is essential for large educational campuses that depend on service vehicles for academic and administrative operations. While studies show that GPS tracking and digital reservation platforms improve vehicle monitoring, scheduling accuracy, and user satisfaction [8], [9], many Philippine universities—including Camarines Sur Polytechnic Colleges (CSPC)—still rely on manual, paper-based systems that lead to delays, conflicts, and limited transparency [14]. Previous research confirms that integrating GPS and automated reservations enhances efficiency

and real-time fleet visibility [13]. In response, TrackCSPC was developed as a smart campus vehicle tracking and reservation platform that combines real-time GPS monitoring with mobile and web-based access to improve coordination, reliability, and overall transportation management at CSPC.

2. METHODOLOGY.

2.1 Software Development Methodology

TrackCSPC was developed using the Iterative Waterfall Model. This allowed repeated cycles of planning, design, development, testing, and deployment. Feedback after each iteration ensured early issue detection, continuous improvement, and alignment with stakeholder requirements.

2.1.1 Planning & Requirements Gathering

This phase of the Iterative Waterfall Model established the project objectives, scope, and essential requirements to address issues such as manual scheduling, inefficient monitoring, and lack of real-time tracking.

2.1.2 Design & Prototyping

The design phase translated the system requirements into detailed system models and user interface designs for TrackCSPC. Data Flow Diagrams, Use Case Diagrams, and role-based interface mockups were developed to clearly define system processes, user interactions, and workflows for administrators, drivers, and requesters. This phase emphasized usability, accessibility, and efficiency, providing a clear structural foundation for system development.

2.1.3 Development and Implementation

Development and implementation were carried out iteratively, with coding based on the approved system specifications and continuous collaboration among developers. Each iteration concluded with a review to ensure functionality and performance standards were met.

2.1.4 Testing & Evaluation

The TrackCSPC system underwent functional testing, usability testing, and performance evaluation to identify errors, assess system reliability, and ensure user satisfaction. Feedback from users and experts was used to refine system features and improve overall quality before proceeding to the next iteration.

2.2 Material and Statistical Tools

2.2.1 Research Design

The researchers used mixed-method research design where both quantitative and qualitative approaches were mixed together to critically assess the system developed in relation to the goals of the study. A quantitative research design involves the systematic gathering and analysis of numerical data for the purpose of determining trends and assessing performance [4].

2.2.2 Research Method

This section explains the methods that were used in the study, including the research instruments, data-gathering procedures, sampling technique, respondents, and statistical tools applied. These components were essential in collecting meaningful data that evaluated the functionality, usability, and overall performance of the system.

2.2.3 Instrument

The researchers collected data using interviews and surveys with GSU staff, drivers, faculty, and

students to identify issues in CSPC's manual vehicle reservation system and gather insights for designing TrackCSPC. This data informed evaluations of system performance, user satisfaction, and potential improvements.

2.2.4 Sampling Technique

The study employed purposive sampling, selecting GSU staff, drivers, faculty, IT experts, and students directly involved in CSPC's vehicle reservation process to gather reliable feedback on TrackCSPC's usability and performance, reflecting real campus operations.

2.2.5 Statistical Test

The study used the arithmetic mean to assess system functionality and acceptability on a binary scale (1 = pass, 0 = fail). Frequency and percentage were applied to describe respondent demographics, calculating percentages by dividing each group's frequency by the total sample and multiplying by 100.

3. RESULTS AND DISCUSSION

This section presents the outcomes of the analysis, design, and development of the TrackCSPC system, a smart campus vehicle tracking and reservation platform. Through continuous coordination between the developers and stakeholders, system requirements, development tools, and implementation strategies were clearly defined, guiding the project from conceptualization to deployment. Key issues in the manual vehicle reservation and monitoring process were addressed through features such as real-time GPS tracking, vehicle reservation and assignment, driver management, role-based access control, and request status updates. The results indicate that TrackCSPC enhances the efficiency, accuracy, and usability of campus transportation management. Consistent communication among stakeholders ensured alignment of expectations and early identification of potential issues, resulting in a structured development workflow and a smooth transition to system testing and evaluation.

3.1 Planning

Developers and stakeholders collaboratively defined TrackCSPC's requirements and solutions, addressing issues in the manual reservation system and establishing a clear project scope for efficient development.

3.1.1 Identified Problems

CSPC's manual vehicle reservation system was inefficient, causing scheduling conflicts, communication delays, and limited visibility, which led to confusion for users and administrators.

3.1.2 Proposed Solution

TrackCSPC is a web- and mobile-based platform that automates vehicle reservations, provides real-time GPS tracking, and centralizes data, improving efficiency and convenience for CSPC users.

3.2 Designing

The TrackCSPC system flow and user interfaces were designed to streamline processes and improve usability. Intuitive, user-friendly interfaces and visual diagrams ensured efficient navigation for administrators, faculty, students, and drivers.

3.2.1 Output and User-Interface Design

The User Interface provides administrators with a secure, user-friendly entry point to manage vehicles, drivers, and reservations, using Google Sign-In for convenient and safe access.

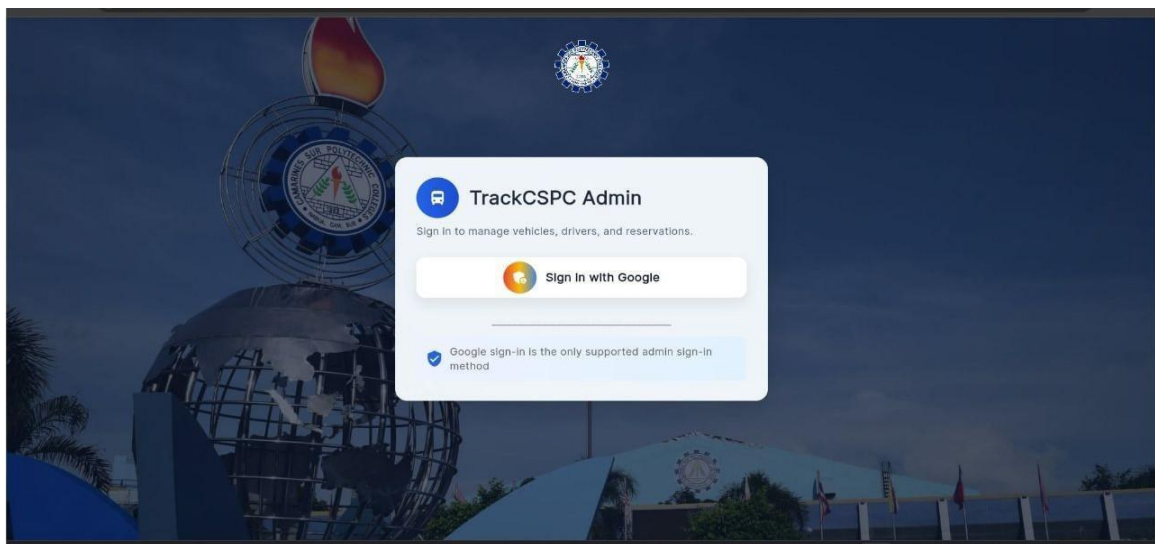


Figure 1. User Interface

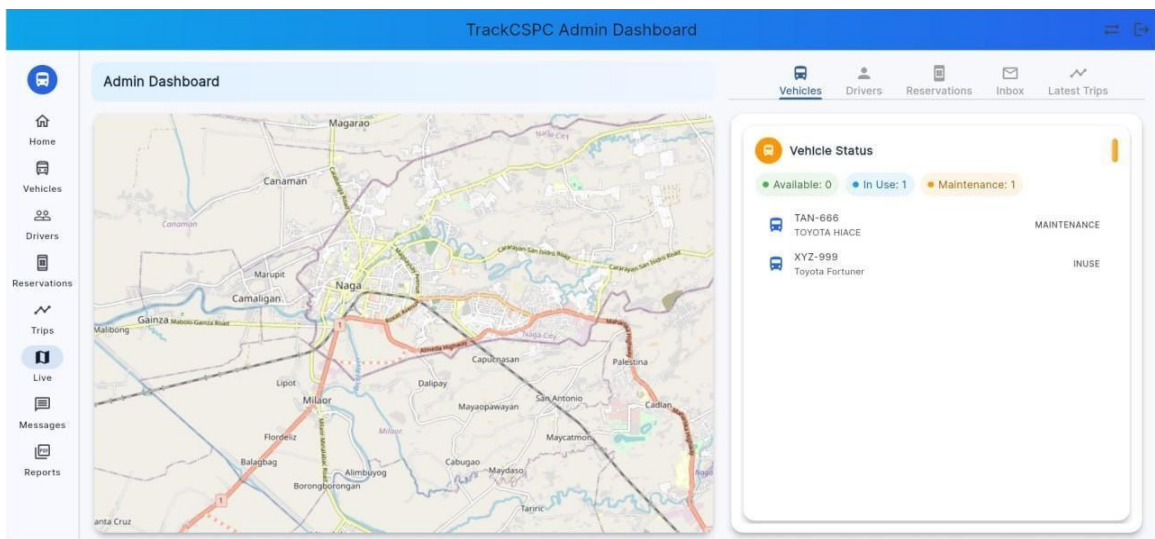


Figure 2. Admin Dashboard

The Admin Dashboard provided administrators with a central control panel to monitor vehicle locations, fleet status, and routes in real time, using an interactive map for efficient management.

3.3 Developing

The development of TrackCSPC used an iterative approach to ensure alignment with approved specifications and project objectives. Developers worked collaboratively, reviewing each cycle to verify functionality and performance. Core features, including real-time GPS tracking, reservation processing, trip authentication, notifications, and integration of the web admin dashboard with the mobile app—were implemented, producing a reliable and user-friendly system.

3.3.1. Hardware Specification

The hardware for TrackCSPC includes a laptop or PC for development, an ESP32 DevKit V1 to manage GPS and GSM modules, and Android devices for the mobile app. All components were

selected for compatibility, efficiency, and reliability, supporting development, testing, and real-world use.

3.3.2 Software Specification

TrackCSPC uses Arduino IDE, Flutter with Geolocator, Firebase, and HTML/CSS/JS for its mobile and web platforms, chosen for compatibility, real-time tracking, and maintainability. These technologies were selected to ensure the system remains easy to develop, update, and adapt as campus needs evolve over time.

3.3.3 Program Specification

TrackCSPC integrates mobile and web apps, GPS trackers, and Firebase to provide secure, near real-time vehicle tracking and reservations, with synchronized data, live maps, role-based access, and efficient performance for a reliable and user-friendly platform. This integration helps create a smooth experience for users, where information is updated quickly and services feel consistent across all platforms.

3.3.4 Programming Environment

The TrackCSPC system uses a combination of mobile, web, backend, and embedded technologies for real-time operation. The mobile app (Flutter/Dart) and web dashboard (HTML/CSS/JS) provide responsive interfaces, Firebase manages data and authentication, and an ESP32 with Arduino IDE handles GPS tracking. Development and collaboration are supported through Visual Studio Code and GitHub. This environment allows the development team to work efficiently, collaborate easily, and maintain code quality throughout the system's lifecycle.

3.3.5 Front End

The TrackCSPC frontend uses Flutter and Dart for the mobile app, offering responsive layouts and high-performance Android interfaces, while the web admin dashboard is built with HTML, CSS, and JavaScript for cross-browser compatibility and user-friendly interaction.

3.3.6 Back End

The TrackCSPC backend uses Firebase, with Realtime Database enabling instant data synchronization for vehicle locations and reservations, and Firebase Authentication securing user access. Firebase SDKs and REST APIs allow smooth communication between the mobile and web apps and the database.

3.3.7 Embedded System

The TrackCSPC embedded system relies on an ESP32 microcontroller, which collects GPS data and sends it to Firebase for real-time vehicle tracking. It is programmed via the Arduino IDE, enabling efficient control of the GPS module and seamless transmission of location data.

4. TESTING

This section summarizes the testing approach for TrackCSPC, covering Unit, Load, Functional, and User Acceptance Testing (UAT) to ensure accurate tracking, database synchronization, responsive interfaces, and overall system reliability across mobile and web platforms.

4.1 Load Testing

The purpose of the Load Testing was to determine the ability of the TrackCSPC system to handle multiple users accessing the platform simultaneously. The inputs were simulated by users from different groups, including students, drivers, GSU staff, and an IT expert.

Table 1. Load Testing

USER GROUP	NO. OF THREADS (USERS)	AVERAGE RESPONSE TIME (MS)	THROUGHPUT (REQ/SEC)	ERROR%	SCORE
Students	20	1845	1.25	0.00%	0.90
Drivers	5	2110	0.85	0.00%	0.85
GSU Staff	5	2320	0.72	0.00%	0.80
Faculty	0	N/A	N/A	N/A	N/A
IT Expert	1	1280	0.30	0.00%	1.00
Overall Result	31	1889	0.78	0.00%	0.89

4.2 Functional Testing

The purpose of the Functional Testing was to verify whether each module of the TrackCSPC system operated according to its defined specifications.

Table 2. Functionality Testing Results

MODULE	RESULT	INTERPRETATION
User Authentication	1.00	Passed
Real - time GPS Tracking	1.00	Passed
Route and Schedule Display	1.00	Passed
Admin Dashboard	1.00	Passed
Vehicle Reservation	1.00	Passed
Parental Monitoring	1.00	Passed
Arithmetic Mean (Total)	1.00	Passed

4.3 User Acceptance Testing (UAT)

User Acceptance Testing assessed the usability and acceptability of TrackCSPC across students, drivers, GSU staff, faculty, and an IT expert. The system was tested in a real environment, with participants providing feedback via questionnaires and checklists.

Table 3. User Acceptance Testing (UAT)

USER GROUP	RESULTS	INTERPRETATION
Student	1.00	Acceptable
Driver	1.00	Acceptable
GSU Staff	1.00	Acceptable
Faculty	0.00	Unacceptable
IT Expert	1.00	Acceptable
Arithmetic Mean (Total)	1.00	Acceptable

5. CONCLUSIONS

The researchers found that TrackCSPC: A Smart Campus Vehicle Tracking and Reservation Platform effectively addresses the challenges of the manual vehicle reservation and tracking system at Camarines Sur Polytechnic Colleges' General Services Unit (GSU). The system provides essential features for vehicle reservation, real-time GPS tracking, live map visualization, and role-based access for administrators, drivers, and users. TrackCSPC reduces delays, minimizes scheduling conflicts, and improves operational accuracy. While the system demonstrates strong

performance and security, future enhancements could focus on scalability and long-term optimization to further support campus transportation management.

6. RECOMMENDATION

1. Provide regular training and orientation programs for students, faculty, drivers, GSU staff, and administrators to ensure proper use of the system, familiarize users with new features, and maximize the platform's benefits for campus transportation.
2. Implement continuous system maintenance and scheduled updates to ensure security, reliability, and optimal performance. This includes resolving bugs, improving system responsiveness, and upgrading database structure and security protocols.
3. Enhance mobile compatibility across major device ecosystems, ensuring smooth operation on Huawei, iPhone (iOS), and Xiaomi (Android) devices. This will allow seamless access regardless of brand or operating system. Improve the system's communication features so faculty and administrators can interact more easily, such as through built-in messaging or a request clarification option.
4. Explore future improvements such as adding online payment options, offline mobile access, AI-based trip scheduling, and better reporting tools to make the system more efficient and easier to use.

ACKNOWLEDGEMENT

The researchers sincerely thank Rosel O. Onesa, MIT, the Capstone Adviser, for her invaluable guidance, expertise, and continuous support throughout this study. Special appreciation is extended Gratitude is given to the Camarines Sur Polytechnic Colleges' College of Computer Studies (CCS) for their cooperation as the client and for providing valuable feedback during the system development phase. Special thanks are also extended to General Services Unit (GSU), the Client for their dedication in reviewing and refining the system to ensure clarity and correctness.

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